Introduction to Time Series

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Contents

1 Basics **7**
- 1.1 Shocks and lag operator . . . 7
- 1.2 Moments and stationarity: the practical view . . . 9
- 1.3 Central Limit Theorem (CLT) for persistent processes . . . 11
- 1.4 Exercises from past exams . . . 13

2 Univariate processes **17**
- 2.1 Moving Average (MA) processes . . . 17
- 2.2 Auto-Regressive (AR) processes . . . 20
- 2.3 AR-MA processes . . . 23
- 2.4 PACF approach to identify AR/MA processes . . . 24
- 2.5 Wold decomposition . . . 25
- 2.6 Impulse Response Functions (IRFs) in ARMA models . . . 26
- 2.7 ARMA processes with exogenous variables (ARMA-X) . . . 28
- 2.8 Maximum Likelihood Estimation (MLE) of ARMA processes . . . 34
- 2.9 Specification choice . . . 40
- 2.10 Exercises . . . 42

3 Multivariate models **53**
- 3.1 Definition of VARs (and SVARMA) models . . . 53
- 3.2 IRFs in SVARMA . . . 54
- 3.3 Covariance-stationary VARMA models . . . 56
- 3.4 VAR estimation . . . 58
- 3.5 Block exogeneity and Granger causality . . . 62
- 3.6 Standard identification techniques . . . 63
- 3.7 Exercise . . . 69

4 Forecasting **73**
- 4.1 Exercise . . . 78

5 Non-stationary processes **81**
- 5.1 Issues when working with nonstationary time series . . . 81
- 5.2 Nonstationarity tests . . . 83
- 5.3 Stationarity test: KPSS . . . 87
- 5.4 Exercises . . . 88

6 Introduction to cointegration **91**
- 6.1 Intuition . . . 91
- 6.2 The bivariate case . . . 91
- 6.3 Multivariate case and VECM . . . 93
- 6.4 Cointegration in practice . . . 97

6.5 Exercise 98

7 ARCH and GARCH models 101
7.1 Conditional heteroskedasticity 101
7.2 The ARCH model 102
7.3 The GARCH model 104
7.4 ARCH-in-mean 105

8 Appendix 109
8.1 Principal component analysis (PCA) 109
8.2 Linear algebra: definitions and results 110
8.3 Statistical analysis: definitions and results 114
8.4 Proofs 119
8.5 Inference 122
8.6 Statistical Tables 124

Introduction to Time Series

Time series constitute a prevalent data type in several disciplines, notably macroeconomics and finance. The modeling of time series is crucial for many purposes, including forecasting, understanding macroeconomic mechanisms, and risk assessment. This course proposes an introduction to time series analysis. It has been developed by Jean-Paul Renne and Alain Monfort.

Codes associated with this course are part of the `IdSS` package, which is available on GitHub. To load a package from GitHub, you need to use function `install_github` from the `devtools` package:

Useful (R) links:

- Download R:
  - R software: https://cran.r-project.org (the basic R software)
  - RStudio: https://www.rstudio.com (a convenient R editor)
- Tutorials:
  - Rstudio: https://dss.princeton.edu/training/RStudio101.pdf (by Oscar Torres-Reyna)
  - R: https://cran.r-project.org/doc/contrib/Paradis-rdebuts_en.pdf (by Emmanuel Paradis)
  - My own tutorial: https://jrenne.shinyapps.io/Rtuto_publiShiny/

Chapter 1

Basics

1.1 Shocks and lag operator

A time series is an infinite sequence of random variables indexed by time: $\{y_t\}_{t=-\infty}^{+\infty} = \{\dots, y_{-2}, y_{-1}, y_0, y_1, \dots, y_t, \dots\}$, $y_i \in \mathbb{R}^k$. In practice, we only observe samples, typically: $\{y_1, \dots, y_T\}$.

Standard time series models are built using **shocks** that we will often denote by ε_t. Typically, $\mathbb{E}(\varepsilon_t) = 0$. In many models, the shocks are supposed to be i.i.d., but there exist other (less restrictive) notions of shocks. In particular, the definition of many processes is based on white noises:

Definition 1.1 (White noise)**.** The process $\{\varepsilon_t\}_{t\in]-\infty,+\infty[}$ is a white noise if, for all t:

a. $\mathbb{E}(\varepsilon_t) = 0$,
b. $\mathbb{E}(\varepsilon_t^2) = \sigma^2 < \infty$,
c. for all $s \neq t$, $\mathbb{E}(\varepsilon_t \varepsilon_s) = 0$.

Another type of shocks that are commonly used are Martingale Difference Sequences:

Definition 1.2 (Martingale Difference Sequence)**.** The process $\{\varepsilon_t\}_{t=-\infty}^{+\infty}$ is a martingale difference sequence (MDS) if $\mathbb{E}(|\varepsilon_t|) < \infty$ and if, for all t,

$$\underbrace{\mathbb{E}_{t-1}(\varepsilon_t)}_{\text{conditional on the past}} = 0.$$

By definition, if y_t is a martingale, then $y_t - y_{t-1}$ is a MDS.

Example 1.1 (ARCH process)**.** The Autoregressive conditional heteroskedasticity process—studied in Section 7—is an example of shock that satisfies the white noise and MDS definitions, but that is not i.i.d.:

$$\varepsilon_t = \sigma_t \times z_t,$$

where $z_t \sim i.i.d.\, \mathcal{N}(0,1)$ and $\sigma_t^2 = w + \alpha \varepsilon_{t-1}^2$ (with $w > 0$ and $0 \leq \alpha < 1$).

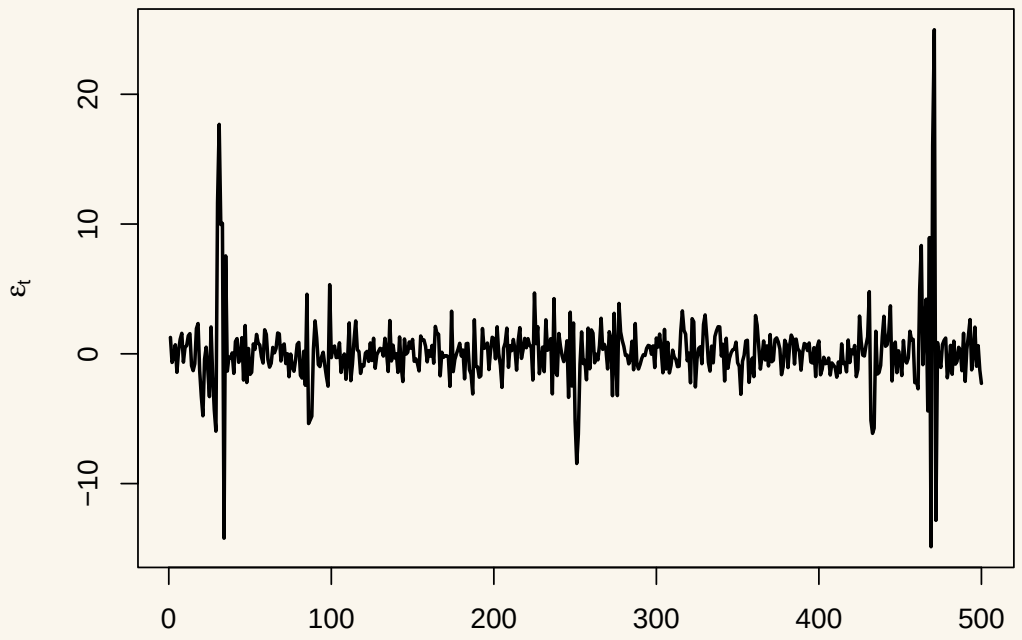


Figure 1.1: Simulated ARCH(1) innovation $\varepsilon_t = \sigma_t z_t$ with $z_t \sim i.i.d.\,\mathcal{N}(0,1)$ and $\sigma_t^2 = 1 + 0.98\varepsilon_{t-1}^2$. The sample contains 500 observations and illustrates volatility clustering despite zero serial correlation in levels.

Example 1.2. A white noise process is not necessarily a MDS. This is for instance the case for following process:

$$\varepsilon_t = z_t + z_{t-1}z_{t-2},$$

where $z_t \sim i.i.d.\mathcal{N}(0,1)$.

Indeed, it can be shown that the ε_t's are not correlated through time, we have $\mathbb{E}_{t-1}(\varepsilon_t) = z_{t-1}z_{t-2} \neq 0$.

In the following, to simplify the exposition, we will essentially consider **strong white noises**. A strong white noise is a particular case of white noise where the ε_t's are serially independent. Using the law of iterated expectation, it can be shown that a strong white noise is a martingale difference sequence. Indeed, when the ε_t's are serially independent, we have that $\mathbb{E}(\varepsilon_t|\varepsilon_{t-1}, \varepsilon_{t-2}, \dots) = \mathbb{E}(\varepsilon_t) = 0$, as $f_{\mathcal{E}_t}(\varepsilon_t|\varepsilon_{t-1}, \varepsilon_{t-2}, \dots) = f_{\mathcal{E}_t}(\varepsilon_t)$.

Let us now introduce the lag operator. The lag operator, denoted by L, is defined on the time series space and is defined by:

$$L : \{y_t\}_{t=-\infty}^{+\infty} \to \{w_t\}_{t=-\infty}^{+\infty} \quad \text{with} \quad w_t = y_{t-1}. \tag{1.1}$$

It is easily seen that we have $L^2 y_t = L(Ly_t) = y_{t-2}$ and, more generally, $L^k y_t = y_{t-k}$.

Consider a process y_t whose law of motion is $y_t = \mu + \phi y_{t-1} + \varepsilon_t$, where the ε_t's are i.i.d. $\mathcal{N}(0, \sigma^2)$ (this is an AR(1) process, as we will see in Section 2). Using the lag operator, the dynamics of y_t can be expressed as follows:

$$(1 - \phi L)y_t = \mu + \varepsilon_t.$$

1.2 Moments and stationarity: the practical view

Moments are numerical summaries of a time series. In practice, we will use them to answer three questions:

1. What is the typical level of the series?
2. How uncertain or volatile is it?
3. How strongly is its current value related to its past values?

Let $\mathcal{J}_t$ denote the information available at date t, including the current and past observations of the series. We distinguish two perspectives:

Perspective	Relevant moments	Interpretation
Unconditional	$\mathbb{E}(y_t)$ and $\mathbb{V}ar(y_t)$	The typical level and dispersion across all possible histories
Conditional on $\mathcal{J}_t$	$\mathbb{E}_t(y_{t+h}) := \mathbb{E}(y_{t+h} \mid \mathcal{J}_t)$ and $\mathbb{V}ar_t(y_{t+h}) := \mathbb{V}ar(y_{t+h} \mid \mathcal{J}_t)$	The forecast at date t and the uncertainty surrounding it

There is no need to write these moments as integrals for the methods developed in this course. What matters is the distinction between a **long-run description** of the process and a **forecast based on today's information**.

For example, consider the stationary AR(1) process

$$y_{t+1} = (1-\phi)m + \phi y_t + \varepsilon_{t+1}, \qquad |\phi| < 1,$$

where ε_{t+1} has mean zero and variance σ^2. Its unconditional moments are

$$\mathbb{E}(y_t) = m \qquad \text{and} \qquad \mathbb{V}ar(y_t) = \frac{\sigma^2}{1-\phi^2}.$$

By contrast, its one-step-ahead conditional moments are

$$\mathbb{E}_t(y_{t+1}) = (1-\phi)m + \phi y_t \qquad \text{and} \qquad \mathbb{V}ar_t(y_{t+1}) = \sigma^2.$$

Thus, a stationary process can have a constant unconditional mean even though its forecast changes with the latest observation.

Definition 1.3 (Autocovariance)**.** The jth autocovariance of y_t measures the linear association between observations separated by j periods:

$$\gamma_{j,t} := \mathbb{C}ov(y_t, y_{t-j}) = \mathbb{E}([y_t - \mathbb{E}(y_t)][y_{t-j} - \mathbb{E}(y_{t-j})]).$$

A positive value means that observations j periods apart tend to deviate from their means in the same direction; a negative value means that they tend to deviate in opposite directions.

In particular, note that $\gamma_{0,t} = \mathbb{V}ar(y_t)$.

Definition 1.4 (Covariance stationarity)**.** The process y_t is covariance stationary —or weakly stationary— if its mean is finite and constant and if its autocovariances depend on the lag j, but not on the calendar date t. That is, for all t and j,

$$\mathbb{E}(y_t) = \mu \quad \text{and} \quad \mathbb{E}\{(y_t - \mu)(y_{t-j} - \mu)\} = \gamma_j.$$

For a covariance-stationary process, $\mathbb{V}ar(y_t) = \gamma_0$ at every date. Notice that stationarity does **not** imply that conditional moments or forecasts must be constant.

Figure 1.2 displays the simulation of a process that is not covariance stationary. This process follows $y_t = 0.1t + \varepsilon_t$, where $\varepsilon_t \sim \ i.i.d.\, \mathcal{N}(0,1)$. Indeed, for such a process, we have: $\mathbb{E}(y_t) = 0.1t$, which depends on t.

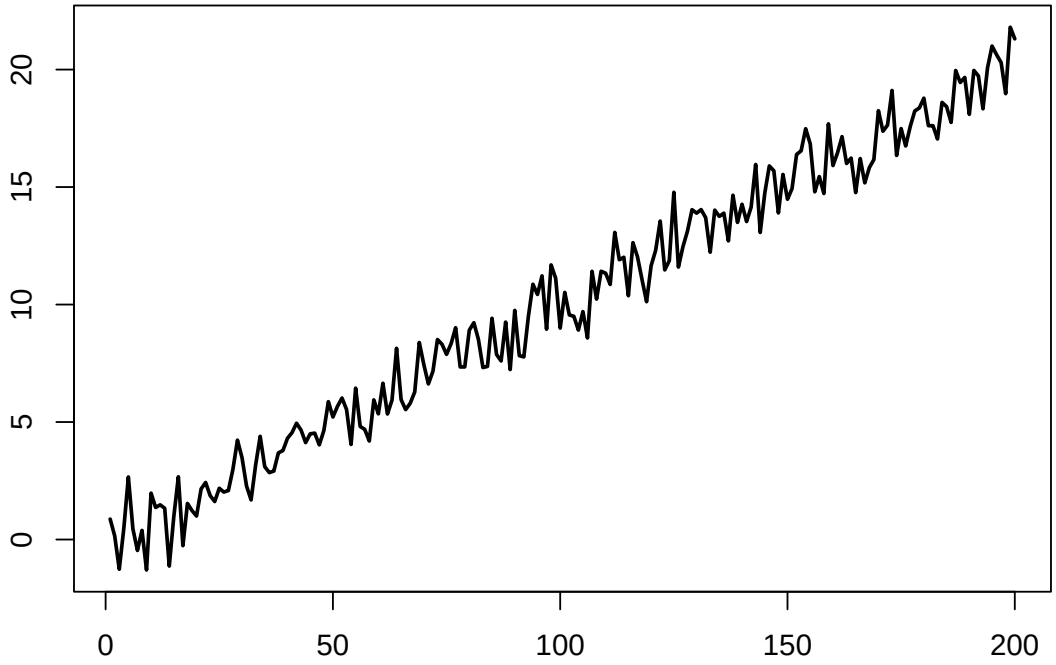


Figure 1.2: Simulated nonstationary process $y_t = 0.1t + \varepsilon_t$, where $\varepsilon_t \sim i.i.d.\, \mathcal{N}(0,1)$. Its unconditional mean increases linearly with time, so the process is not covariance stationary.

Remark. A stronger concept, **strict stationarity**, requires the entire joint distribution of the process —not only its first two moments— to be invariant over time. In this course, *stationary* will generally mean *covariance stationary* unless explicitly stated otherwise.

Proposition 1.1. *If y_t is covariance stationary, then $\gamma_j = \gamma_{-j}$.*

Proof. Since y_t is covariance stationary, the covariance between y_t and y_{t-j} (i.e., γ_j) is the same as that between y_{t+j} and y_{t+j-j} (i.e. γ_{-j}). □

Definition 1.5 (Auto-correlation)**.** The jth auto-correlation of a covariance-stationary process is:

$$\rho_j = \frac{\gamma_j}{\gamma_0}.$$

Consider a long historical time series of the Swiss GDP growth, taken from the Jordà et al. (2017) dataset.[1]

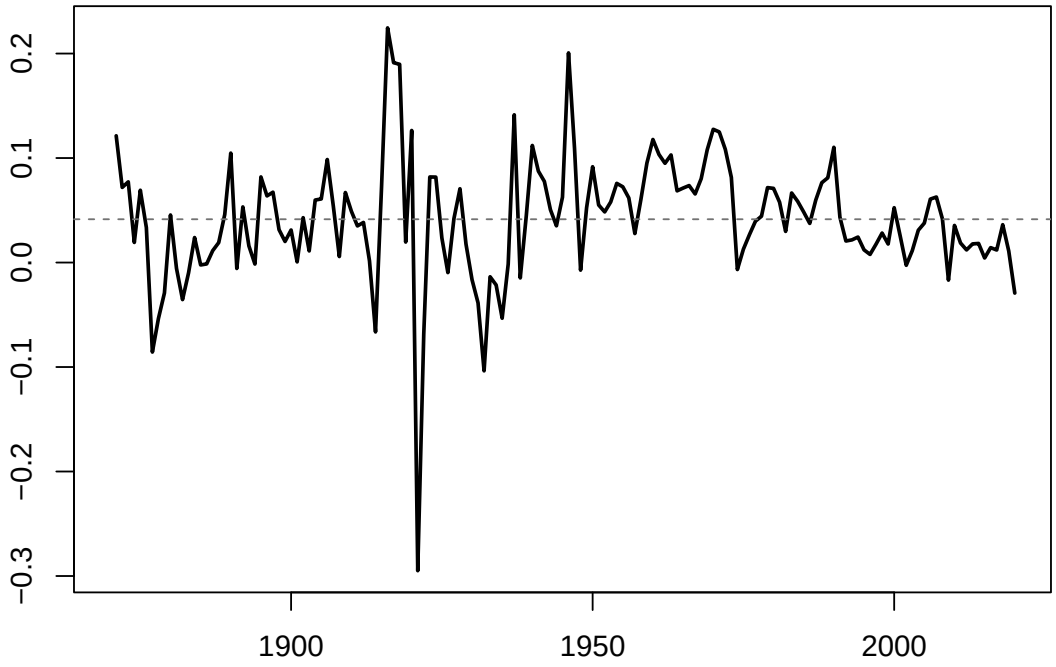


Figure 1.3: Annual Swiss real-GDP growth, computed as the log difference of real GDP. The dashed grey line indicates the sample mean. Source: Jordà-Schularick-Taylor Macrohistory Database, release 6.

Figure 1.4 shows two scatter plots. In the first one, the coordinates of each point are of the form (y_{t-1}, y_t). Hence, the slope of the OLS regression line is $\widehat{\mathbb{C}ov}(y_t, y_{t-1})/\widehat{\mathbb{V}ar}(y_{t-1})$ (where the hats indicate that these moments are the sample ones). Assuming that the process is covariance stationary, the slope therefore is an estimate of the auto-correlation of order one. By the same logic, the slope of the line in the second plot is an estimate of the auto-correlation of order three.

1.3 Central Limit Theorem (CLT) for persistent processes

This subsection shows that the Central Limit Theorem (CLT, see Theorem 8.5) can be extended to cases where the observations are auto-correlated.

[1] Version 6 of the dataset, available on this website.

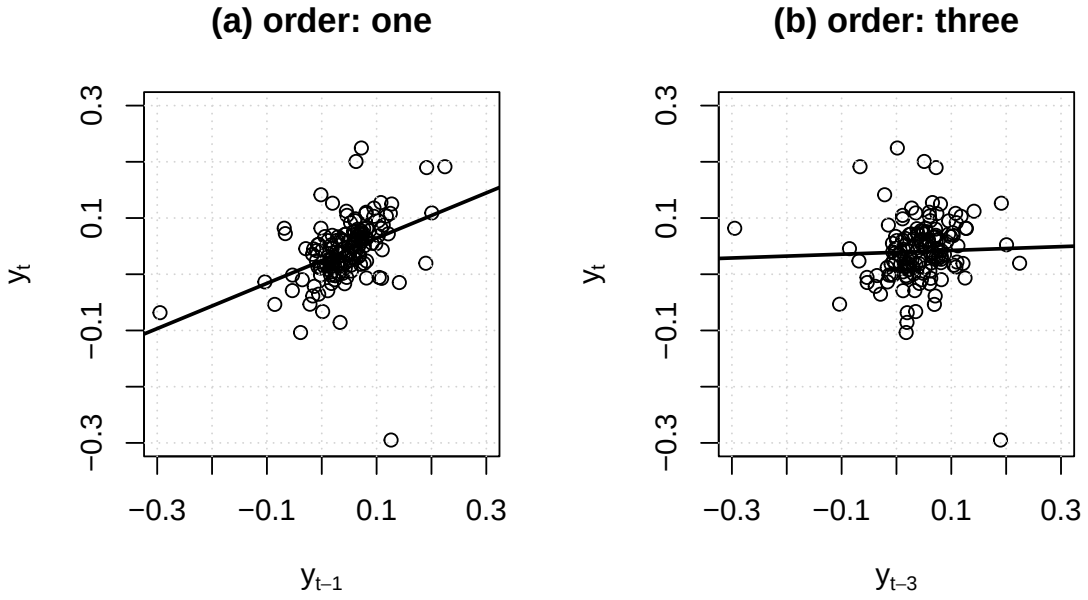


Figure 1.4: Swiss real-GDP growth plotted against its own lags: (a) first lag and (b) third lag. The solid black lines are OLS fits. For lag j, the fitted slope is approximately $\hat{\gamma}_j / \widehat{\mathbb{V}ar}(y_t)$, where hats denote sample moments.

Theorem 1.1 (Central Limit Theorem for covariance-stationary processes)**.** *If process y_t is covariance stationary and if the series of autocovariances is absolutely summable ($\sum_{j=-\infty}^{+\infty} |\gamma_j| < \infty$), then:*

$$\bar{y}_T \overset{m.s.}{\to} \mu = \mathbb{E}(y_t) \tag{1.2}$$

$$lim_{T\to+\infty} T\mathbb{E}\left[(\bar{y}_T - \mu)^2\right] = \sum_{j=-\infty}^{+\infty} \gamma_j \tag{1.3}$$

$$\sqrt{T}(\bar{y}_T - \mu) \overset{d}{\to} \mathcal{N}\left(0, \sum_{j=-\infty}^{+\infty} \gamma_j\right). \tag{1.4}$$

[Mean square (m.s.) and distribution (d.) convergences: see Definitions 8.16 and 8.14.]

Proof. The absolute value of the series of the autocovariances is finite as it is lower than the sum of the absolute values of the autocovariances which is finite by the absolute summability assumption. As a result, there exists a finite $M \in \mathbb{R}$ such that $\sum_{j=-\infty}^{+\infty} \gamma_j = M$ and Eq. (1.3) implies Eq. (1.2). For Eq. (1.3), see Appendix 8.4. For Eq. (1.4), see Anderson (1971), p. 429. □

Definition 1.6 (Long-run variance)**.** Under the assumptions of Theorem 1.1, the limit appearing in Eq. (1.3) exists and is called **long-run variance**. It is denoted by S, i.e.:

$$S = \Sigma_{j=-\infty}^{+\infty} \gamma_j = \lim_{T \to +\infty} T \mathbb{E}[(\bar{y}_T - \mu)^2].$$

If y_t is ergodic for second moments (see Def. 8.8), a natural estimator of S is:

$$\hat{\gamma}_0 + 2 \sum_{\nu=1}^{q} \hat{\gamma}_\nu, \tag{1.5}$$

where $\hat{\gamma}_\nu = \frac{1}{T} \sum_{\nu+1}^{T} (y_t - \bar{y})(y_{t-\nu} - \bar{y})$.

However, for small samples, Eq. (1.5) does not necessarily result in a positive definite matrix. Newey and West (1987) have proposed an estimator that does not have this defect. Their estimator is given by:

$$S^{NW} = \hat{\gamma}_0 + 2 \sum_{\nu=1}^{q} \left(1 - \frac{\nu}{q+1}\right) \hat{\gamma}_\nu. \tag{1.6}$$

Loosely speaking, Theorem 1.1 says that, for a given sample size, the higher the "persistency" of a process, the lower the accuracy of the sample mean as an estimate of the population mean. To illustrate, consider three processes that feature the same marginal variance (equal to one, say), but different autocorrelations: 0%, 70%, and 99.9%. Figure 1.5 displays simulated paths of such three processes. It indeed appears that, the larger the autocorrelation of the process, the further the sample mean (dashed line) from the population mean (solid line).

The same type of simulations can be performed using this web interface (use panel "AR(1)").

1.4 Exercises from past exams

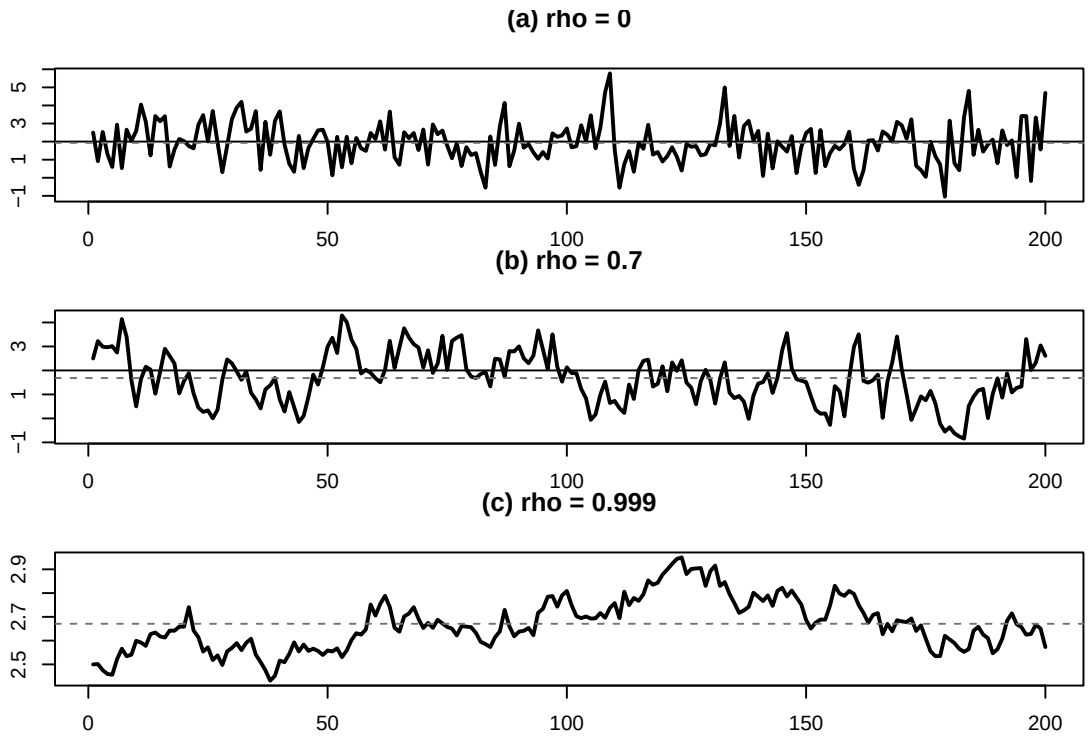


Figure 1.5: Simulated AR(1) processes $x_t = \mu + \rho(x_{t-1} - \mu) + \sqrt{1-\rho^2}\varepsilon_t$, with $\varepsilon_t \sim \mathcal{N}(0,1)$, $\mu = 2$, and $\mathbb{V}ar(x_t) = 1$: (a) $\rho = 0$; (b) $\rho = 0.7$; and (c) $\rho = 0.999$. In each panel, the solid black line marks the population mean and the dashed grey line the sample mean.

Exercise 1.1 (Autocovariances and the long-run variance)**. EXAM-2024 | Exam | True/false**
Review Section 1.

A covariance-stationary process has $\gamma_0 = 4$, $\gamma_1 = 3$, $\gamma_2 = 1$, and $\sum_{h=3}^{\infty} \gamma_h = 2$. Let $\bar{y}_T = T^{-1} \sum_{t=1}^{T} y_t$ and $\mathbb{E}(y_t) = \mu$.

State whether each assertion is true or false and justify your answer.

1. $\mathbb{V}ar(y_t + y_{t-1}) = 14$.
2. $\mathbb{V}ar(y_{t-1}) = \gamma_1$.
3. For large T, $\sqrt{T}(\bar{y}_T - \mu)$ is approximately $\mathcal{N}(0, 16)$.
4. For large T, $\mathbb{V}ar[\sqrt{T}(\bar{y}_T - \mu)]$ is approximately 4.
5. If $T = 1600$ and $\bar{y}_T = 1$, then 0.70 lies in an approximate 95% confidence interval for μ.

Exercise 1.2 (Autocovariances, correlations, and the time-series CLT)**. EXAM-2025 | Exam | True/false**
Review Section 1.

A covariance-stationary process y_t has autocovariances

$$\gamma_0 = 3, \quad \gamma_1 = 1.5, \quad \gamma_2 = 1, \quad \gamma_3 = 0.5, \quad \text{and} \quad \gamma_h = 0 \text{ for } h \geq 4.$$

Let $\bar{y}_T = T^{-1} \sum_{t=1}^{T} y_t$, where T is large.

State whether each assertion is true or false and justify your answer.

1. The process can be a nondegenerate AR(4).
2. $\mathbb{V}ar(\bar{y}_T) = 3$.
3. The correlation between y_{t-3} and y_{t-5} is zero.
4. If $\bar{y}_T = 1$ and $T = 900$, an approximate 95% confidence interval for $\mathbb{E}(y_t)$ does not include zero.
5. $\mathbb{E}(y_{t+1}) = \mathbb{E}(y_{t-1})$.

Chapter 2

Univariate processes

2.1 Moving Average (MA) processes

Moving average (MA) processes are important basic processes. The moving-average process of order one (MA(1)) is defined as follows:

Definition 2.1 (Moving average process of order one)**.** Process y_t is a first-order moving average process if, for all t:

$$y_t = \mu + \varepsilon_t + \theta\varepsilon_{t-1}, \tag{2.1}$$

where $\{\varepsilon_t\}_{t=-\infty}^{+\infty}$ is a white noise process (see Def. 1.1).

If $\mathbb{E}(\varepsilon_t^2) = \sigma^2$, it is easily obtained that the unconditional mean and variance of y_t are:

$$\mathbb{E}(y_t) = \mu, \quad \mathbb{V}ar(y_t) = (1+\theta^2)\sigma^2.$$

The first auto-covariance is:

$$\gamma_1 = \mathbb{E}\{(y_t - \mu)(y_{t-1} - \mu)\} = \mathbb{E}\{(\varepsilon_t + \theta\varepsilon_{t-1})(\varepsilon_{t-1} + \theta\varepsilon_{t-2})\} = \theta\sigma^2.$$

It is easily seen that higher-order auto-covariances are zero ($\gamma_j = 0$ for $j > 1$). Therefore: An MA(1) process is covariance-stationary (Def. 1.4).

From what precedes, the autocorrelation of order j (see Def. 1.5) of an MA(1) process is given by:

$$\rho_j = \begin{cases} 1 & \text{if} \quad j = 0, \\ \theta/(1+\theta^2) & \text{if} \quad j = 1 \\ 0 & \text{if} \quad j > 1. \end{cases}$$

Notice that process y_t defined through Eq. (2.1), with $\mathbb{V}ar(\varepsilon_t) = \sigma^2$, has the same mean and autocovariances as

$$y_t = \mu + \varepsilon_t^* + \frac{1}{\theta}\varepsilon_{t-1}^*,$$

where $\mathbb{V}ar(\varepsilon_t^*) = \theta^2\sigma^2$. That is, knowing the mean and auto-covariances of an MA(1) process is not sufficient to identify the process, since two different processes possess the same moments. Only one of these two specifications is said to be *fundamental*, that is the one that satisfies $|\theta_1| < 1$ (see Eq. (2.27)).

Definition 2.2 (MA(q) process)**.** A qth-order Moving Average process $\{y_t\}$ is defined through:

$$y_t = \mu + \varepsilon_t + \theta_1 \varepsilon_{t-1} + \cdots + \theta_q \varepsilon_{t-q},$$

where $\{\varepsilon_t\}_{t=-\infty}^{+\infty}$ is a white noise process (Def. 1.1).

Proposition 2.1 (Covariance-stationarity of an MA(q) process)**.** *Finite-order Moving Average processes are covariance-stationary.*

Moreover, the autocovariances of an MA(q) process (as defined in Def. 2.2) are given by:

$$\gamma_j = \begin{cases} \sigma^2(\theta_j\theta_0 + \theta_{j+1}\theta_1 + \cdots + \theta_q\theta_{q-j}) & \text{for} \quad j \in \{0, \dots, q\} \\ 0 & \text{for} \quad j > q, \end{cases} \tag{2.2}$$

where we use the notation $\theta_0 = 1$*, and* $\mathbb{V}ar(\varepsilon_t) = \sigma^2$*.*

Proof. The unconditional expectation of y_t does not depend on time, since $\mathbb{E}(y_t) = \mu$. Turning to the autocovariances, we can extend the series of θ_j's by setting $\theta_j = 0$ for $j > q$. We then have:

$$\begin{aligned} \mathbb{E}((y_t - \mu)(y_{t-j} - \mu)) \quad = \quad & \mathbb{E}\left[(\theta_0\varepsilon_t + \theta_1\varepsilon_{t-1} + \cdots + \theta_j\varepsilon_{t-j} + \theta_{j+1}\varepsilon_{t-j-1} + \dots) \times \right. \\ & \left. (\theta_0\varepsilon_{t-j} + \theta_1\varepsilon_{t-j-1} + \dots)\right]. \end{aligned}$$

Using the fact that $\mathbb{E}(\varepsilon_t\varepsilon_s) = 0$ if $t \neq s$ (because $\{\varepsilon_t\}_{t=-\infty}^{+\infty}$ is a white noise process) leads to the result. □

Figure 2.1 displays simulated paths of two MA processes (an MA(1) and an MA(4)). Such simulations can also be produced by using panel "ARMA(p,q)" of this web interface.

What if the order q of an MA(q) process gets infinite? The notion of **infinite-order Moving Average process** exists and is important in time series analysis, as it relates to impulse response functions (as illustrated in Section 2.6). The (infinite) sequence of θ_j has to satisfy some conditions for such a process to be well-defined (see Theorem 2.1 below). These conditions relate to the "summability" of the components of the sequence $\{\theta_i\}_{i\in\mathbb{N}}$:

Definition 2.3 (Absolute and square summability)**.** The sequence $\{\theta_i\}_{i\in\mathbb{N}}$ is absolutely summable if $\sum_{i=0}^{\infty} |\theta_i| < +\infty$, and it is square summable if $\sum_{i=0}^{\infty} \theta_i^2 < +\infty$.

According to Prop. 8.8 (in the appendix), absolute summability implies square summability.

Theorem 2.1 (Existence condition for an infinite MA process)**.** *If* $\{\theta_i\}_{i\in\mathbb{N}}$ *is square summable (see Def. 2.3) and if* $\{\varepsilon_t\}_{t=-\infty}^{+\infty}$ *is a white noise process (see Def. 1.1), then*

$$\mu + \sum_{i=0}^{+\infty} \theta_i \varepsilon_{t-i}$$

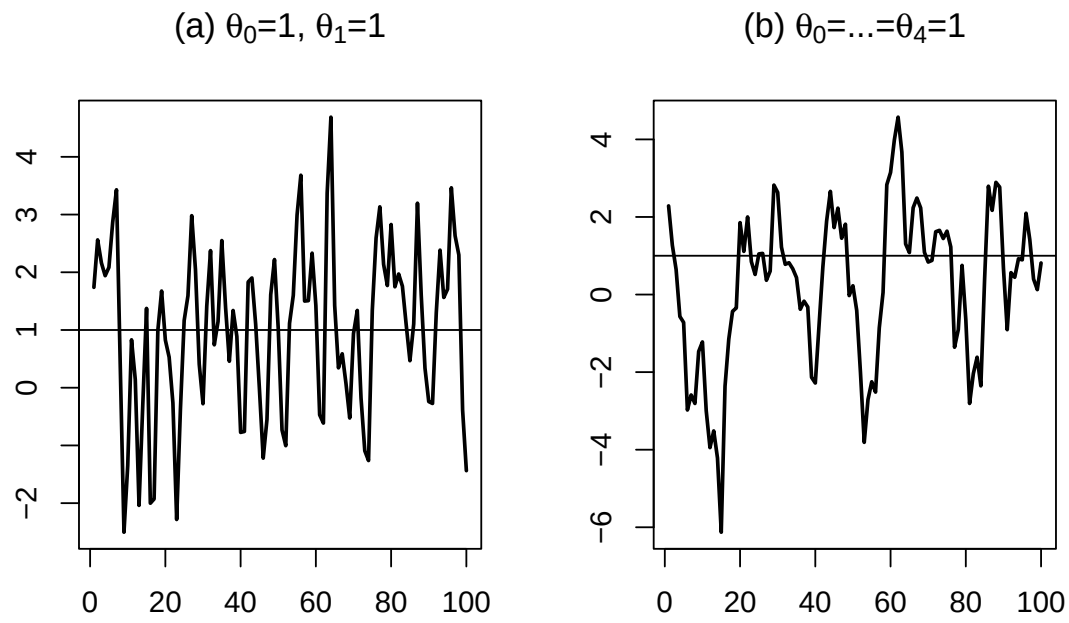


Figure 2.1: Simulated moving-average processes with unit innovation variance and intercept equal to one: (a) MA(1) process with $\theta_0 = \theta_1 = 1$; (b) MA(4) process with $\theta_0 = \cdots = \theta_4 = 1$. The horizontal line marks the intercept.

defines a well-behaved [covariance-stationary] process, called infinite-order MA process (MA(∞)).

Proof. See Appendix 3.A in Hamilton. "Well behaved" means that $\Sigma_{i=0}^{T}\theta_{t-i}\varepsilon_{t-i}$ converges in mean square (Def. 8.14) to some random variable Z_t. The proof makes use of the fact that:

$$\mathbb{E}\left[\left(\sum_{i=N}^{M}\theta_i\varepsilon_{t-i}\right)^2\right] = \sum_{i=N}^{M}|\theta_i|^2\sigma^2,$$

and that, when $\{\theta_i\}$ is square summable, $\forall\eta > 0$, $\exists N$ s.t. the right-hand-side term in the last equation is lower than η for all $M \geq N$ (static Cauchy criterion, Theorem 8.2). This implies that $\Sigma_{i=0}^{T}\theta_i\varepsilon_{t-i}$ converges in mean square (stochastic Cauchy criterion, see Theorem 8.3). □

Proposition 2.2 (First two moments of an infinite MA process)**.** *If $\{\theta_i\}_{i\in\mathbb{N}}$ is absolutely summable, i.e., if $\sum_{i=0}^{\infty}|\theta_i| < +\infty$, then*

i. $y_t = \mu + \sum_{i=0}^{+\infty}\theta_i\varepsilon_{t-i}$ *exists (Theorem 2.1) and is such that:*

$$\begin{aligned}\mathbb{E}(y_t) &= \mu\\ \gamma_0 = \mathbb{E}([y_t-\mu]^2) &= \sigma^2(\theta_0^2+\theta_1^2+\dots)\\ \gamma_j = \mathbb{E}([y_t-\mu][y_{t-j}-\mu]) &= \sigma^2(\theta_0\theta_j+\theta_1\theta_{j+1}+\dots).\end{aligned}$$

ii. Process y_t has absolutely summable auto-covariances, which implies that the results of Theorem 1.1 (Central Limit) apply.

Proof. The absolute summability of $\{\theta_i\}$ and the fact that $\mathbb{E}(\varepsilon^2) < \infty$ imply that the order of integration and summation is interchangeable (see Hamilton, 1994, Footnote p. 52), which proves (i). For (ii), see end of Appendix 3.A in Hamilton (1994). □

2.2 Auto-Regressive (AR) processes

Definition 2.4 (First-order AR process (AR(1))). Process y_t is an AR(1) process if its dynamics is defined by the following difference equation:

$$y_t = c + \phi y_{t-1} + \varepsilon_t,$$

where $\{\varepsilon_t\}_{t=-\infty}^{+\infty}$ is a white noise process (see Def. 1.1).

If $|\phi| \geq 1$, y_t is not stationary. Indeed, we have:

$$\begin{aligned} y_{t+k} &= c + \varepsilon_{t+k} + \phi(c + \varepsilon_{t+k-1}) + \phi^2(c + \varepsilon_{t+k-2}) + \cdots + \\ &\quad \phi^{k-1}(c + \varepsilon_{t+1}) + \phi^k y_t. \end{aligned}$$

Therefore, the conditional variance

$$\mathbb{V}ar_t(y_{t+k}) = \sigma^2(1 + \phi^2 + \phi^4 + \cdots + \phi^{2(k-1)})$$

(where σ^2 is the variance of ε_t) does not converge when k gets infinitely large. This implies that $\mathbb{V}ar(y_t)$ does not exist.[1] By contrast, if $|\phi| < 1$, we have:

$$y_t = c + \varepsilon_t + \phi(c + \varepsilon_{t-1}) + \phi^2(c + \varepsilon_{t-2}) + \cdots + \phi^k(c + \varepsilon_{t-k}) + \ldots$$

Hence, if $|\phi| < 1$, the unconditional mean and variance of y_t are:

$$\mathbb{E}(y_t) = \frac{c}{1-\phi} =: \mu \quad \text{and} \quad \mathbb{V}ar(y_t) = \frac{\sigma^2}{1-\phi^2}.$$

Let us compute the jth autocovariance of the AR(1) process:

$$\begin{aligned} \mathbb{E}([y_t - \mu][y_{t-j} - \mu]) &= \mathbb{E}([\varepsilon_t + \phi\varepsilon_{t-1} + \phi^2\varepsilon_{t-2} + \cdots + \phi^j\varepsilon_{t-j} + \phi^{j+1}\varepsilon_{t-j-1} \ldots] \times \\ &\quad [\varepsilon_{t-j} + \phi\varepsilon_{t-j-1} + \phi^2\varepsilon_{t-j-2} + \cdots + \phi^k\varepsilon_{t-j-k} + \ldots]) \\ &= \mathbb{E}(\phi^j\varepsilon_{t-j}^2 + \phi^{j+2}\varepsilon_{t-j-1}^2 + \phi^{j+4}\varepsilon_{t-j-2}^2 + \ldots) \\ &= \frac{\phi^j\sigma^2}{1-\phi^2}. \end{aligned}$$

Therefore, the auto-correlation is given by $\rho_j = \phi^j$.

By what precedes, we have:

[1] Indeed, by the law of total variance $\mathbb{V}ar(y_t) = \mathbb{V}ar(\mathbb{E}_{t-k}(y_t)) + \mathbb{E}(\mathbb{V}ar_{t-k}(y_t)) >$. Since we have $\mathbb{V}ar(\mathbb{E}_{t-k}(y_t)) \geq 0$ and $\mathbb{E}(\mathbb{V}ar_{t-k}(y_t)) = \mathbb{V}ar_{t-k}(y_t)$ that goes to $+\infty$ (by what precedes), it comes that $\mathbb{V}ar(y_t)$ is not finite.

Proposition 2.3 (Covariance-stationarity of an AR(1) process)**.** *The AR(1) process, as defined in Def. 2.4, is covariance-stationary iff* $|\phi| < 1$.

Definition 2.5 (AR(p) process)**.** Process y_t is a pth-order autoregressive process (AR(p)) if its dynamics is defined by the following difference equation (with $\phi_p \neq 0$):

$$y_t = c + \phi_1 y_{t-1} + \phi_2 y_{t-2} + \cdots + \phi_p y_{t-p} + \varepsilon_t, \tag{2.3}$$

where $\{\varepsilon_t\}_{t=-\infty}^{+\infty}$ is a white noise process (see Def. 1.1).

As we will see, the covariance-stationarity of process y_t hinges on the eigenvalues of matrix F, defined as:

$$F = \begin{bmatrix} \phi_1 & \phi_2 & \dots & \phi_p \\ 1 & 0 & \dots & 0 \\ 0 & 1 & \dots & 0 \\ \vdots & & \ddots & \vdots \\ 0 & 0 & \dots & 1 & 0 \end{bmatrix}. \tag{2.4}$$

Note that this matrix F is such that if y_t follows Eq. (2.3), then process $\mathbf{y}_t$ follows:

$$\mathbf{y}_t = \mathbf{c} + F\mathbf{y}_{t-1} + \xi_t$$

with

$$\mathbf{c} = \begin{bmatrix} c \\ 0 \\ \vdots \\ 0 \end{bmatrix}, \quad \xi_t = \begin{bmatrix} \varepsilon_t \\ 0 \\ \vdots \\ 0 \end{bmatrix}, \quad \mathbf{y}_t = \begin{bmatrix} y_t \\ y_{t-1} \\ \vdots \\ y_{t-p+1} \end{bmatrix}.$$

Proposition 2.4 (The eigenvalues of matrix F)**.** *The eigenvalues of* F *(defined by Eq.* (2.4)*) are the solutions of:*

$$\lambda^p - \phi_1 \lambda^{p-1} - \cdots - \phi_{p-1}\lambda - \phi_p = 0. \tag{2.5}$$

Proof. See Appendix 1.A in Hamilton (1994) □

Proposition 2.5 (Covariance-stationarity of an AR(p) process)**.** *These four statements are equivalent:*

i. Process $\{y_t\}$*, defined in Def. 2.5, is covariance-stationary.*
ii. The eigenvalues of F *(as defined Eq.* (2.4)*) lie strictly within the unit circle.*
iii. The roots of Eq. (2.6) *(below) lie strictly outside the unit circle.*

$$1 - \phi_1 z - \cdots - \phi_{p-1} z^{p-1} - \phi_p z^p = 0. \tag{2.6}$$

iv. The roots of Eq. (2.7) *(below) lie strictly inside the unit circle.*

$$\lambda^p - \phi_1 \lambda^{p-1} - \cdots - \phi_{p-1}\lambda - \phi_p = 0. \tag{2.7}$$

Proof. We consider the case where the eigenvalues of F are distinct.[2] When the eigenvalues of F are distinct, F admits the following spectral decomposition: $F = PDP^{-1}$, where D is diagonal. Using the notations introduced in Eq. (2.4), we have:

$$\mathbf{y}_t = \mathbf{c} + F\mathbf{y}_{t-1} + \xi_t.$$

Let's introduce $\mathbf{d} = P^{-1}\mathbf{c}$, $\mathbf{z}_t = P^{-1}\mathbf{y}_t$ and $\eta_t = P^{-1}\xi_t$. We have:

$$\mathbf{z}_t = \mathbf{d} + D\mathbf{z}_{t-1} + \eta_t.$$

Because D is diagonal, the different component of $\mathbf{z}_t$, denoted by $z_{i,t}$, follow AR(1) processes. The (scalar) autoregressive parameters of these AR(1) processes are the diagonal entries of D—which also are the eigenvalues of F—that we denote by λ_i.

Process y_t is covariance-stationary iff $\mathbf{y}_t$ also is covariance-stationary, which is the case iff all $z_{i,t}$, $i \in \{1, ..., p\}$, are covariance-stationary. By Prop. 2.3, process $z_{i,t}$ is covariance-stationary iff $|\lambda_i| < 1$. This proves that (i) is equivalent to (ii). Prop. 2.4 further proves that (ii) is equivalent to (iv). Finally, it is easily seen that (iii) is equivalent to (iv) (as long as $\phi_p \neq 0$[3]). □

Using the lag operator (see Eq (1.1)), if y_t is a covariance-stationary AR(p) process (Def. 2.5), we can write:

$$y_t = \mu + \psi(L)\varepsilon_t,$$

where

$$\psi(L) = (1 - \phi_1 L - \cdots - \phi_p L^p)^{-1}, \tag{2.8}$$

and

$$\mu = \mathbb{E}(y_t) = \frac{c}{1 - \phi_1 - \cdots - \phi_p}. \tag{2.9}$$

In the following lines of codes, we compute the eigenvalues of the F matrices associated with the following processes (where ε_t is a white noise):

$$\begin{aligned} x_t &= 0.9x_{t-1} - 0.2x_{t-2} + \varepsilon_t \\ y_t &= 1.1y_{t-1} - 0.3y_{t-2} + \varepsilon_t \\ w_t &= 1.4w_{t-1} - 0.7w_{t-2} + \varepsilon_t \\ z_t &= 0.9z_{t-1} + 0.2z_{t-2} + \varepsilon_t. \end{aligned}$$

```
##                              [,1]                  [,2]
## lambda_x 0.500000+0.0000000i  0.4000000+0.0000000i
## lambda_y 0.600000+0.0000000i  0.5000000+0.0000000i
## lambda_w 0.700000+0.4582576i  0.7000000-0.4582576i
## lambda_z 1.084429+0.0000000i -0.1844289+0.0000000i
```

The absolute values of the eigenvalues associated with process w_t are both equal to 0.837. Therefore, according to Proposition 2.5, processes x_t, y_t, and w_t are covariance-stationary, but not z_t (because, for the latter process, the absolute value of one of the eigenvalues of matrix F is larger than 1).

[2] The Jordan matrix decomposition can be used in the general case.

[3] If $\phi_p \neq 0$ and λ solves Eq. (2.7) with $|\lambda| < 1$, then $\lambda \neq 0$. If we define $z = \frac{1}{\lambda}$, then z solves Eq. (2.6) and $|z| > 1$; Conversely, if $\phi_p \neq 0$ and z solves Eq. (2.6) with $|z| > 1$, then $z \neq 0$ .If we define $\lambda = \frac{1}{z}$, then λ solves Eq. (2.7) and $|\lambda| < 1$

The computation of the autocovariances of y_t is based on the so-called **Yule-Walker equations** (Eq. (2.10)). Let's rewrite Eq. (2.3):

$$(y_t - \mu) = \phi_1(y_{t-1} - \mu) + \phi_2(y_{t-2} - \mu) + \cdots + \phi_p(y_{t-p} - \mu) + \varepsilon_t.$$

Multiplying both sides by $y_{t-j} - \mu$ and taking expectations leads to the (Yule-Walker) equations:

$$\gamma_j = \begin{cases} \phi_1\gamma_{j-1} + \phi_2\gamma_{j-2} + \cdots + \phi_p\gamma_{j-p} & if \quad j > 0 \\ \phi_1\gamma_1 + \phi_2\gamma_2 + \cdots + \phi_p\gamma_p + \sigma^2 & for \quad j = 0, \end{cases} \tag{2.10}$$

where σ^2 is the variance of ε_t.

Using $\gamma_j = \gamma_{-j}$ (Prop. 1.1), one can express $(\gamma_0, \gamma_1, \dots, \gamma_p)$ as functions of $(\sigma^2, \phi_1, \dots, \phi_p)$. Indeed, we have:

$$\begin{bmatrix} \gamma_0 \\ \gamma_1 \\ \gamma_2 \\ \vdots \\ \gamma_p \end{bmatrix} = \underbrace{\begin{bmatrix} 0 & \phi_1 & \phi_2 & \cdots & & \phi_p & \\ \phi_1 & \phi_2 & \cdots & & \phi_p & 0 & \\ \phi_2 & (\phi_1 + \phi_3) & \phi_4 & \cdots & \phi_p & 0 & 0 \\ \vdots & & & & & & \\ \phi_p & \phi_{p-1} & \cdots & & \phi_2 & \phi_1 & 0 \end{bmatrix}}_{=H} \begin{bmatrix} \gamma_0 \\ \gamma_1 \\ \gamma_2 \\ \vdots \\ \gamma_p \end{bmatrix} + \begin{bmatrix} \sigma^2 \\ 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix},$$

which is easily solved by inverting matrix $Id - H$.

2.3 AR-MA processes

Definition 2.6 (ARMA(p,q) process)**.** $\{y_t\}$ is an ARMA(p,q) process if its dynamics is described by the following equation:

$$y_t = c + \underbrace{\phi_1 y_{t-1} + \cdots + \phi_p y_{t-p}}_{\text{AR part}} + \underbrace{\varepsilon_t + \theta_1\varepsilon_{t-1} + \cdots + \theta_q\varepsilon_{t-q}}_{\text{MA part}}, \tag{2.11}$$

where $\{\varepsilon_t\}_{t\in]-\infty,+\infty[}$, of variance σ^2 (say), is a white noise process (see Def. 1.1).

Proposition 2.6 (Stationarity of an ARMA(p,q) process)**.** *The ARMA(p,q) process defined in 2.6 is covariance stationary iff the roots of*

$$1 - \phi_1 z - \cdots - \phi_p z^p = 0$$

lie strictly outside the unit circle or, equivalently, iff those of

$$\lambda^p - \phi_1\lambda^{p-1} - \cdots - \phi_p = 0$$

lie strictly within the unit circle.

Proof. The proof of Prop. 2.5 can be adapted to the present case. The MA part of the ARMA process is not problematic for this process to be covariance-stationary. Only the AR part matters and needs to be checked (i.e. check if all the eigenvalues of the underlying F matrix are stictly lower than 1 in absolute value) to make sure that this process is stationary. □

We can write:

$$(1 - \phi_1 L - \cdots - \phi_p L^p) y_t = c + (1 + \theta_1 L + \cdots + \theta_q L^q)\varepsilon_t.$$

If the roots of $1 - \phi_1 z - \cdots - \phi_p z^p = 0$ lie outside the unit circle, we have:

$$y_t = \mu + \psi(L)\varepsilon_t, \tag{2.12}$$

where

$$\psi(L) = \frac{1 + \theta_1 L + \cdots + \theta_q L^q}{1 - \phi_1 L - \cdots - \phi_p L^p} \quad and \quad \mu = \frac{c}{1 - \phi_1 - \cdots - \phi_p}.$$

Eq. (2.12) is the **Wold representation** of this ARMA process (see Theorem 2.2 below). In Section 2.6 (and more precisely in Proposition 2.7), we will see how to obtain the first h terms of the infinite polynomial $\Psi(L)$.

Importantly, note that the stationarity of the process depends only on the AR specification (or on the eigenvalues of matrix F, exactly as in Prop. 2.5). When an ARMA process is stationary, the weights in $\psi(L)$ decay at a geometric rate.

2.4 PACF approach to identify AR/MA processes

We have seen that the kth-order auto-correlation of an MA(q) process is null if $k > q$. This is exploited, in practice, to determine the order of an MA process. Moreover, since this is not the case for an AR process, this can be used to distinguish an AR from an MA process.

There exists an equivalent condition that is satisfied by AR processes, and that can be used to determine whether a time series process can be modeled as an AR process. This condition relates to partial auto-correlations:

Definition 2.7 (Partial auto-correlation)**.** The partial auto-correlation ($\phi_{h,h}$) of process $\{y_t\}$ is defined as the partial correlation of y_{t+h} and y_t given $y_{t+h-1}, \ldots, y_{t+1}$. (see Def. 8.5 for the definition of partial correlation.)

If $h > p$, the regression of y_{t+h} on $y_{t+h-1}, \ldots, y_{t+1}$ is:

$$y_{t+h} = c + \phi_1 y_{t+h-1} + \cdots + \phi_p y_{t+h-p} + \underbrace{0 \times y_{t+h-p-1} + \cdots + 0 \times y_{t+1}}_{=0} + \varepsilon_{t+h}.$$

The residuals of the latter regressions (ε_{t+h}) are uncorrelated to y_t. Then the partial autocorrelation is zero for $h > p$.

Besides, it can be shown that $\phi_{p,p} = \phi_p$. Hence $\phi_{p,p} = \phi_p$ but $\phi_{h,h} = 0$ for $h > p$. This can be used to determine the order of an AR process. By contrast (importantly) if y_t follows an MA(q) process, then $\phi_{k,k}$ asymptotically approaches zero instead of cutting off abruptly.

As illustrated below, the functions `acf` and `pacf` of R allow us to conveniently implement the (P)ACF approach. (In these lines of codes, note also the use of function `sim.arma` to simulate ARMA processes.)

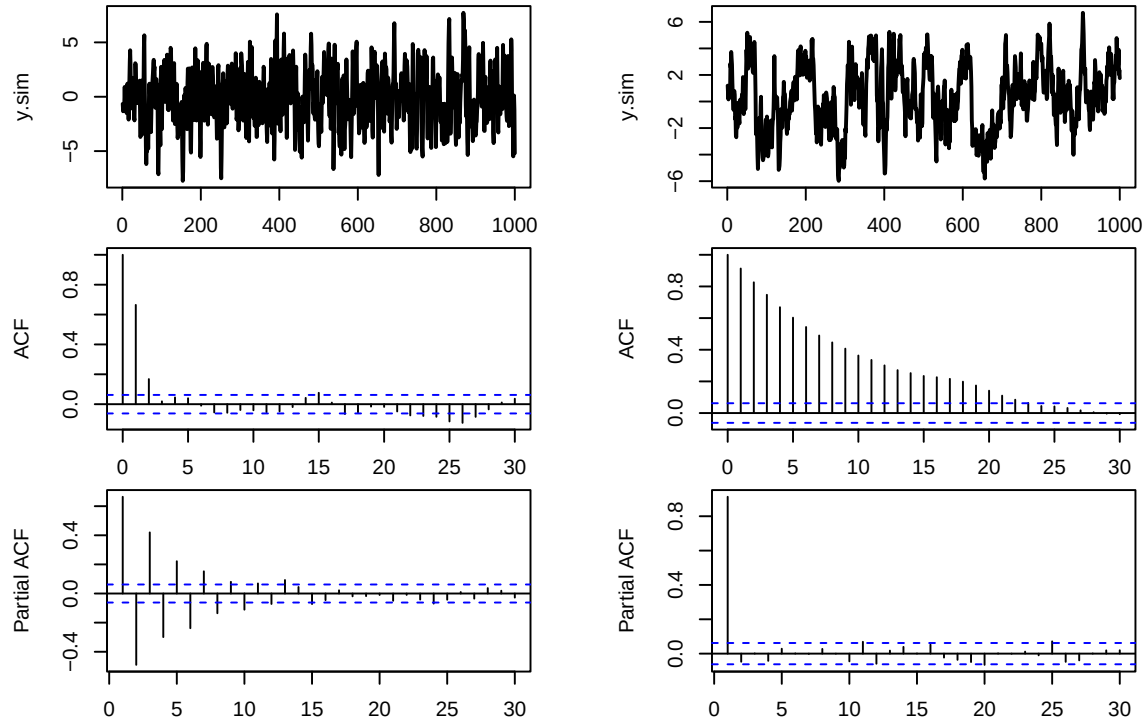


Figure 2.2: Simulated samples and correlograms for two processes, each with 1,000 observations. Left column: (a) simulated MA(2) process, (c) its sample autocorrelation function, and (e) its sample partial autocorrelation function. Right column: (b) simulated AR(1) process, (d) its sample autocorrelation function, and (f) its sample partial autocorrelation function.

2.5 Wold decomposition

The Wold decomposition is an important result in time series analysis:

Theorem 2.2 (Wold decomposition)**.** *Any covariance-stationary process admits the following representation:*

$$y_t = \mu + \sum_{0}^{+\infty} \psi_i \varepsilon_{t-i} + \kappa_t,$$

where

- $\psi_0 = 1$, $\sum_{i=0}^{\infty} \psi_i^2 < +\infty$ *(square summability, see Def. 2.3).*
- $\{\varepsilon_t\}$ *is a white noise (see Def. 1.1);* ε_t *is the error made when forecasting* y_t *based on a linear combination of lagged* y_t*'s (*$\varepsilon_t = y_t - \hat{\mathbb{E}}[y_t | y_{t-1}, y_{t-2}, \dots]$*).*
- *For any* $j \geq 1$*,* κ_t *is not correlated with* ε_{t-j}*; but* κ_t *can be perfectly forecasted based on a linear combination of lagged* y_t*'s (i.e.* $\kappa_t = \hat{\mathbb{E}}(\kappa_t | y_{t-1}, y_{t-2}, \dots)$*).* κ_t *is called the* ***deterministic component*** *of* y_t*.*

Proof. See Anderson (1971). Partial proof in L. Christiano's lecture notes. □

For an ARMA process, the Wold representation is given by Eq. (2.12). As detailed in Prop. 2.7, it can be computed by recursively replacing the lagged y_t's in Eq. (2.11). In this case, the deterministic component (κ) is null.

2.6 Impulse Response Functions (IRFs) in ARMA models

Consider the ARMA(p,q) process defined in Def. 2.6. Let us construct a novel (counterfactual) sequence of shocks $\{\tilde{\varepsilon}_t^{(s)}\}$:

$$\tilde{\varepsilon}_t^{(s)} = \begin{cases} \varepsilon_t & if \quad t \neq s, \\ \varepsilon_t + 1 & if \quad t = s. \end{cases}$$

Hence, the only difference between processes $\{\varepsilon_t^{(s)}\}$ and $\{\tilde{\varepsilon}_t^{(s)}\}$ pertains to date s, where ε_s is replaced with $\varepsilon_s + 1$ in $\{\tilde{\varepsilon}_t^{(s)}\}$.

We denote by $\{\tilde{y}_t^{(s)}\}$ the process following Eq. (2.11) where $\{\varepsilon_t\}$ is replaced with $\{\tilde{\varepsilon}_t^{(s)}\}$. The time series $\{\tilde{y}_t^{(s)}\}$ is the counterfactual series $\{y_t\}$ that would have prevailed if ε_t had been shifted by one unit on date s (and that would be the only change).

The relationship between $\{y_t\}$ and $\{\tilde{y}_t^{(s)}\}$ defines the **dynamic multipliers** of $\{y_t\}$. The dynamic multiplier $\frac{\partial y_t}{\partial \varepsilon_s}$ corresponds to the impact on y_t of a unit increase in ε_s (on date s). Using the notation introduced before for $\tilde{y}_t^{(s)}$, we have:

$$\tilde{y}_t^{(s)} = y_t + \frac{\partial y_t}{\partial \varepsilon_s}.$$

Let us show that the dynamic multipliers are closely related to the infinite MA representation (or **Wold decomposition**, Theorem 2.2) of y_t:

$$y_t = \mu + \sum_{i=0}^{+\infty} \psi_i \varepsilon_{t-i}.$$

For $t < s$, we have $y_t = \tilde{y}_t^{(s)}$ (because $\tilde{\varepsilon}_{t-i} = \varepsilon_{t-i}$ for all $i \geq 0$ if $t < s$).

For $t \geq s$:

$$\tilde{y}_t^{(s)} = \mu + \left(\sum_{i=0}^{t-s-1} \psi_i \varepsilon_{t-i} \right) + \psi_{t-s}(\varepsilon_s + 1) + \left(\sum_{i=t-s+1}^{+\infty} \psi_i \varepsilon_{t-i} \right) = y_t + \frac{\partial y_t}{\partial \varepsilon_s}.$$

Therefore, it comes that the only difference between $\tilde{y}_t^{(s)}$ and y_t is ψ_{t-s}. As a result, for $t \geq s$, we have:

$$\boxed{\frac{\partial y_t}{\partial \varepsilon_s} = \psi_{t-s}.}$$

That is, $\{y_t\}$'s dynamic multiplier of order k is the same object as the kth loading ψ_k in the Wold decomposition of $\{y_t\}$. The sequence $\left\{ \frac{\partial y_{t+h}}{\partial \varepsilon_t} \right\}_{h \geq 0} \equiv \{\psi_h\}_{h \geq 0}$ defines the **impulse response function (IRF)** of y_t to the shock ε_t.

For ARMA processes, one can compute the IRFs (or the Wold decomposition) by using a simple recursive algorithm:

Proposition 2.7 (IRF of an ARMA(p,q) process)**.** *The coefficients ψ_h, that define the IRF of process y_t to ε_t, can be computed recursively as follows:*

1. *Set $\psi_{-1} = \cdots = \psi_{-p} = 0$.*
2. *For $h \geq 0$, (recursively) apply:*

$$\psi_h = \phi_1 \psi_{h-1} + \cdots + \phi_p \psi_{h-p} + \theta_h,$$

where $\theta_h = 0$ for $h > q$.

Proof. This is obtained by applying the operator $\frac{\partial}{\partial \varepsilon_t}$ on both sides of Eq. (2.11):

$$y_{t+h} = c + \phi_1 y_{t+h-1} + \cdots + \phi_p y_{t+h-p} + \varepsilon_{t+h} + \theta_1 \varepsilon_{t+h-1} + \cdots + \theta_q \varepsilon_{t+h-q}.$$

□

Note that Proposition 2.7 constitutes a simple way to compute the MA(∞) representation (or Wold representation) of an ARMA process.

One can use function `sim.arma` of package `IdSS` to compute ARMA's IRFs (with the argument `make.IRF = 1`):

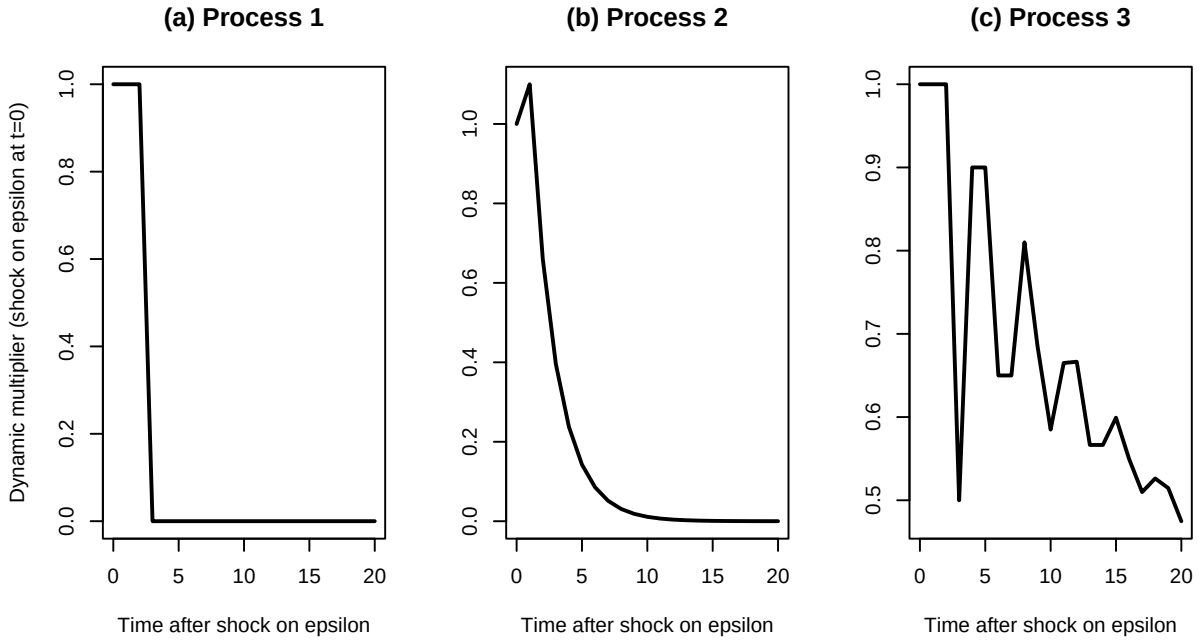


Figure 2.3: IRFs associated with the three processes. Process 1 (MA(2)): $y_t = \varepsilon_t + \varepsilon_{t-1} + \varepsilon_{t-2}$. Process 2 (ARMA(1,1)): $y_t = 0.6y_{t-1} + \varepsilon_t + 0.5\varepsilon_{t-1}$. Process 3 (ARMA(4,2)): $y_t = 0.5y_{t-3} + 0.4y_{t-4} + \varepsilon_t + \varepsilon_{t-1} + \varepsilon_{t-2}$.

Consider the annual Swiss GDP growth from the JST macro-history database. Let us first determine relevant orders for AR and MA processes using the (P)ACF approach.

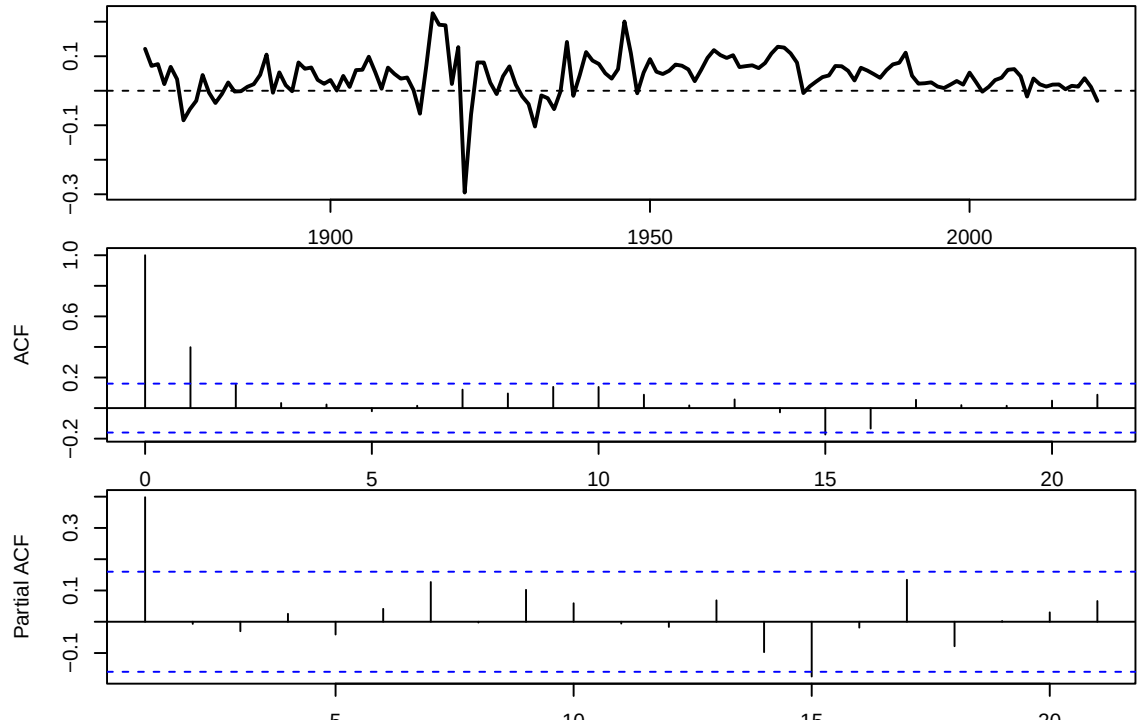


Figure 2.4: Serial-dependence diagnostics for annual Swiss real-GDP growth: (a) growth series; (b) sample autocorrelation function; and (c) sample partial autocorrelation function. Source: Jordà-Schularick-Taylor Macrohistory Database, release 6.

The two bottom plots of Figure 2.4 suggest that either an MA(2)[4] or an AR(1)[5]could be used to model the GDP growth rate series. Figure 2.5 shows the IRFs based on these two respective specifications.

The same kind of algorithm (as in Prop. 2.7) can be used to compute the impact of an increase in an exogenous variable x_t within an ARMAX(p,q,r) model (see next section).

2.7 ARMA processes with exogenous variables (ARMA-X)

2.7.1 Definitions (ARMA-X)

ARMA processes do not allow to investigate the influence of an exogenous variable (say x_t) on the variable of interest (say y_t). When x_t and y_t have reciprocal influences, the Vector Autoregressive (VAR) model may be used (this tools will be studied later, in Section 3). However, when one suspects that x_t has an "exogenous" influence on y_t, then a simple extension of the ARMA processes may be considered. Loosely speaking, x_t has an "exogenous" influence on y_t if y_t does not affect x_t. This extension is referred to as ARMA-X.

To begin with, let us formalize this notion of exogeneity. Consider a white noise sequence $\{\varepsilon_t\}$ (Def. 1.1). This white noise will enter the dynamics of y_t, alongside with x_t; but x_t will be exogenous to ε_t. (We will also say that x_t is exogenous to y_t.)

[4]Indeed, the sample autocorrelation coefficients, (ρ_j), are statistically significant for $j \leq 2$ where ρ_2 is barely significant

[5]Indeed, the sample partial autocorrelation coefficients, $\rho_{j,j}$, are statistically significant for $j = 1$ only

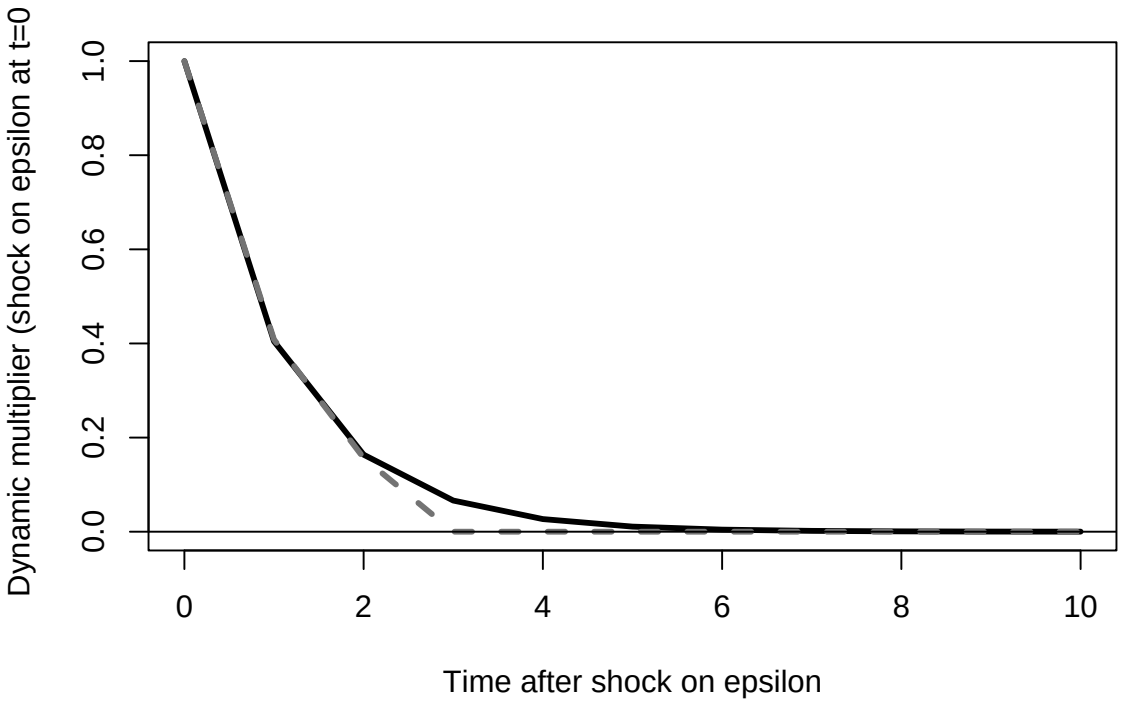


Figure 2.5: Dynamic response of Swiss annual growth to a shock on the innovation ε_t at date $t = 0$. The solid line corresponds to an AR(1) specification; the dashed line corresponds to a MA(2) specification.

Definition 2.8 (Exogeneity)**.** We say that x_t is (strictly) exogenous to $\{\varepsilon_t\}$ if

$$\mathbb{E}(\varepsilon_t | \underbrace{\ldots, x_{t+1}}_{\text{future}}, \underbrace{x_t, x_{t-1}, \ldots}_{\text{present and past}}) = 0.$$

Hence, if $\{x_t\}$ is strictly exogenous to ε_t, then past, present and future values of x_t do not allow to predict the ε_t's.

In the following, we assume that $\{x_t\}$ is a covariance stationary process.

Definition 2.9 (ARMA-X(p,q,r) model)**.** The process $\{y_t\}$ is an ARMA-X(p,q,r) if it follows a difference equation of the form:

$$\begin{aligned} y_t = & \underbrace{c + \phi_1 y_{t-1} + \cdots + \phi_p y_{t-p}}_{\text{AR(p) part}} + \underbrace{\beta_0 x_t + \cdots + \beta_r x_{t-r}}_{\text{X(r) part}} + \\ & \underbrace{\varepsilon_t + \theta_1 \varepsilon_{t-1} + \cdots + \theta_q \varepsilon_{t-q}}_{\text{MA(q) part}}, \end{aligned} \tag{2.13}$$

where $\{\varepsilon_t\}$ is an i.i.d. white noise sequence and $\{x_t\}$ is exogenous to y_t.

2.7.2 Dynamic multipliers in ARMA-X models

What is the effect of a one-unit increase in x_t on y_t? To address this question, this notion of "effect" has to be formalized. Let us introduce two related sequences of values for $\{x\}$.

Denote the first by $\{a\}$ and the second by $\{\tilde{a}^t\}$. Further, we posit $a_s = \tilde{a}^t_s$ for all $s \neq t$, and $\tilde{a}^t_t = a_t + 1$.

With these notations, we define $\frac{\partial y_{t+h}}{\partial x_t}$ as follows:

$$\frac{\partial y_{t+h}}{\partial x_t} := \mathbb{E}_{t-1}(y_{t+h}|\{x\} = \{\tilde{a}^t\}) - \mathbb{E}_{t-1}(y_{t+h}|\{x\} = \{a\}). \tag{2.14}$$

Under the exogeneity assumption, it is easily seen that

$$\frac{\partial y_t}{\partial x_t} = \beta_0.$$

Now, since

$$\begin{aligned} y_{t+1} &= c + \phi_1 y_t + \cdots + \phi_p y_{t+1-p} + \beta_0 x_{t+1} + \cdots + \beta_r x_{t+1-r} + \\ &\quad \varepsilon_{t+1} + \theta_1 \varepsilon_t + \cdots + \theta_q \varepsilon_{t+1-q}, \end{aligned}$$

and using the exogeneity assumption, we obtain:

$$\frac{\partial y_{t+1}}{\partial x_t} := \phi_1 \frac{\partial y_t}{\partial x_t} + \beta_1 = \phi_1 \beta_0 + \beta_1.$$

This can be applied recursively to give $\frac{\partial y_{t+h}}{\partial x_t}$ for any $h \geq 0$:

Proposition 2.8 (Dynamic multipliers in ARMA-X models)**.** *One can recursively compute the dynamic multipliers $\frac{\partial y_{t+h}}{\partial x_t}$ as follows:*

i. Initialization: $\frac{\partial y_{t+h}}{\partial x_t} = 0$ for $h < 0$.

ii. For $h \geq 0$ and assuming that the first $h-1$ multipliers have been computed, we have:

$$\frac{\partial y_{t+h}}{\partial x_t} = \phi_1 \frac{\partial y_{t+h-1}}{\partial x_t} + \cdots + \phi_p \frac{\partial y_{t+h-p}}{\partial x_t} + \beta_h, \tag{2.15}$$

where we use the notation $\beta_h = 0$ if $h > r$.

Remark that the resulting dynamic multipliers are the same as those obtained for an ARMA(p,r) model where the θ_i's are replaced with β_i's (see Proposition 2.7 in Section 2.7).

It has to be stressed that the definition of the dynamic multipliers (Eq. (2.14)) does not reflect a potential persistency of the shock occuring on date t in process $\{x\}$ itself. Going in this direction would necessitate to model the joint dynamics of x_t (for instance using a VAR model, see Section 3).

Example 2.1 (Influence of the number of freezing days on the price of orange juice)**.** This example is based on data used in Stock and Watson (2003) (Chapter 16). The objective is to study the influence of the number of freezing days on the price of orange juice. Let us first estimate a ARMA-X(0,0,3) model:

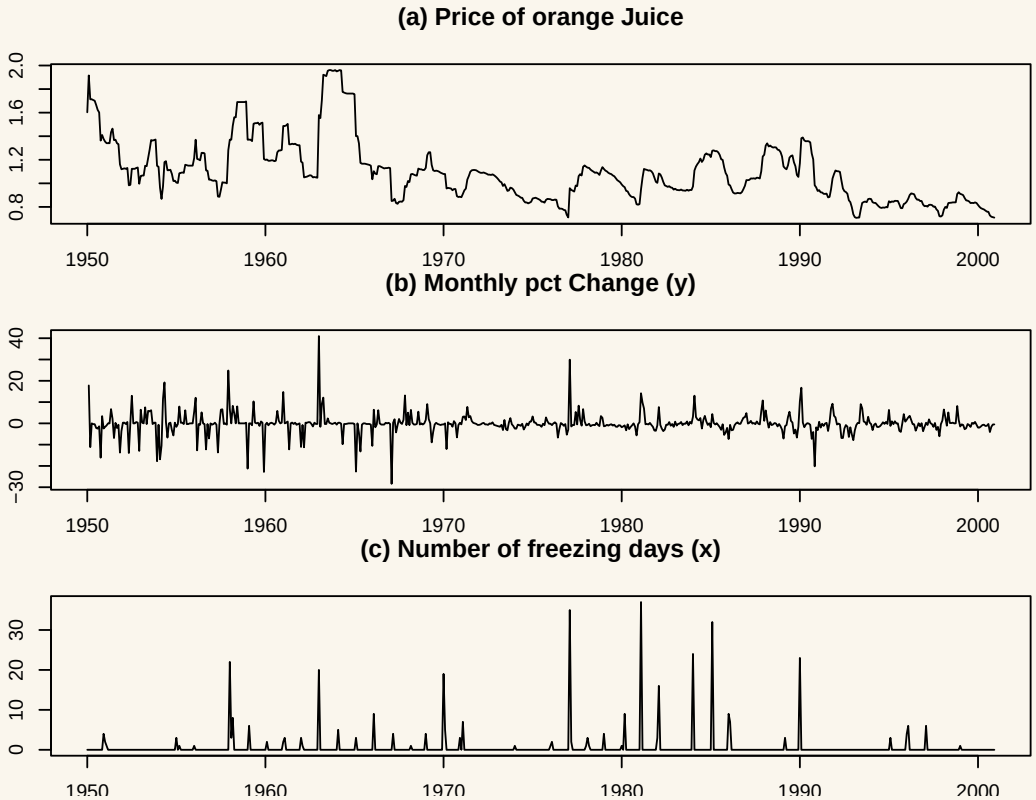


Figure 2.6: Orange-juice data used in the ARMAX example: (a) real orange-juice price; (b) monthly log price change, in percent; and (c) number of freezing-degree days. Source: FrozenJuice data distributed with the AER package, based on Stock and Watson (2003).

```
##
## ========================================
##                  Dependent variable:
##              ---------------------------
##                         dprice
##                 (no HAC)        (HAC)
##                   (1)             (2)
## ----------------------------------------
## FDD Lag 0       0.467***       0.467***
##                 (0.057)         (0.135)
## FDD Lag 1       0.140**         0.140*
##                 (0.057)         (0.083)
## FDD Lag 2        0.055           0.055
##                 (0.057)         (0.056)
## FDD Lag 3        0.073           0.073
##                 (0.057)         (0.047)
```

```
## Constant       -0.599***      -0.599***
##                 (0.204)        (0.213)
## ---------------------------------------
## Observations      609            609
## =======================================
## Note:         *p<0.1; **p<0.05; ***p<0.01
```

Let us now use function `estim.armax`, from package `IdSS` to fit an ARMA-X(3,0,1) model:

Figure 2.7 shows the IRF associated with each of the two models.

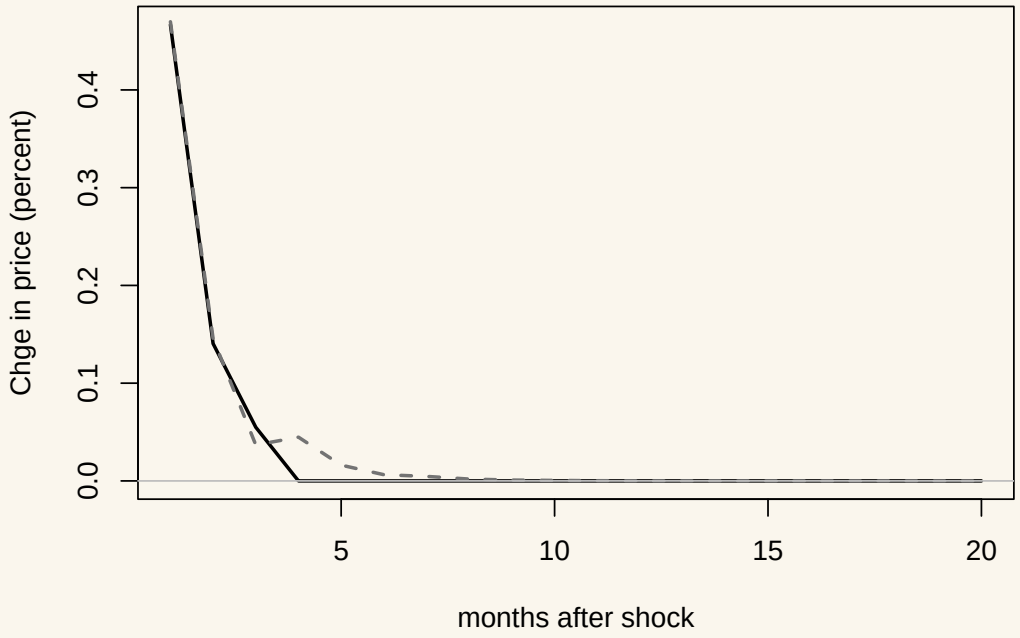


Figure 2.7: Response of changes in orange juice price (in percent) to the number of freezing days. The solid (respectively dashed) line corresponds to the ARMAX(0,0,3) (resp. ARMAX(3,0,1)) model. The first model is estimated by OLS (see above), the second by MLE.

Example 2.2 (Real effect of a monetary policy shock)**.** In this example, we use the monthly monetary-policy shock developed by Bu, Rogers, and Wu. The series is distributed by the Board of Governors of the Federal Reserve System and is included in the redistributable `USmonthly` dataset of `IdSS`.

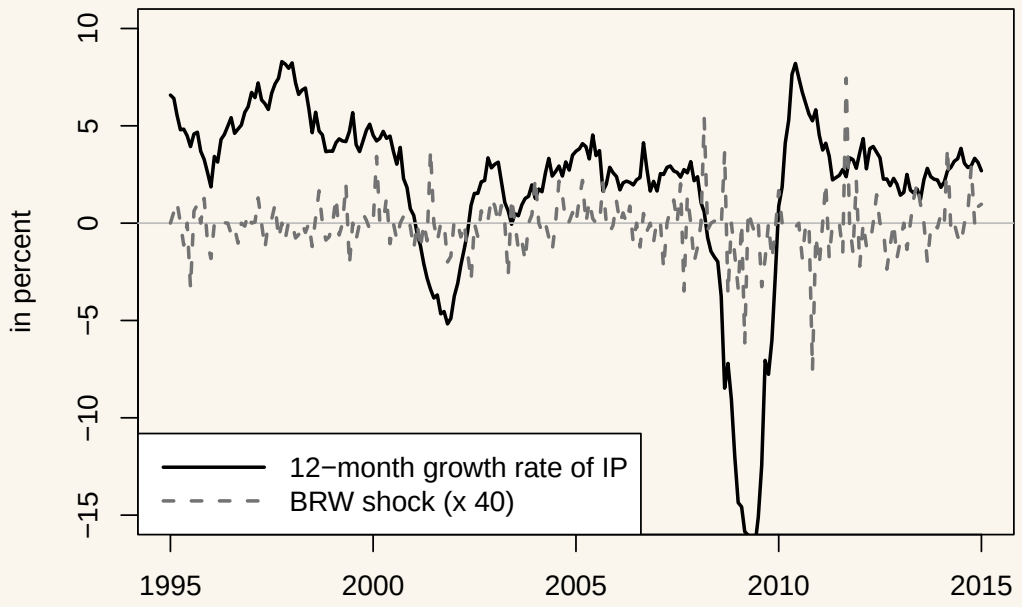


Figure 2.8: U.S. industrial production and monetary-policy shocks. The solid black line is the year-on-year growth rate of industrial production; the dashed grey line is the monthly BRW monetary-policy shock multiplied by 40 for readability. The common sample begins in 1994. Sources: U.S. government macroeconomic series via FRED and Bu, Rogers, and Wu shock series via the Board of Governors.

Figure 2.9 displays the resulting IRF, with a 95% confidence band. The code used to produce the confidence bands (i.e., to compute the standard deviation of the dynamic multipliers for the different horizons) is based on the Delta method (see Appendix 8.5.2).[a]

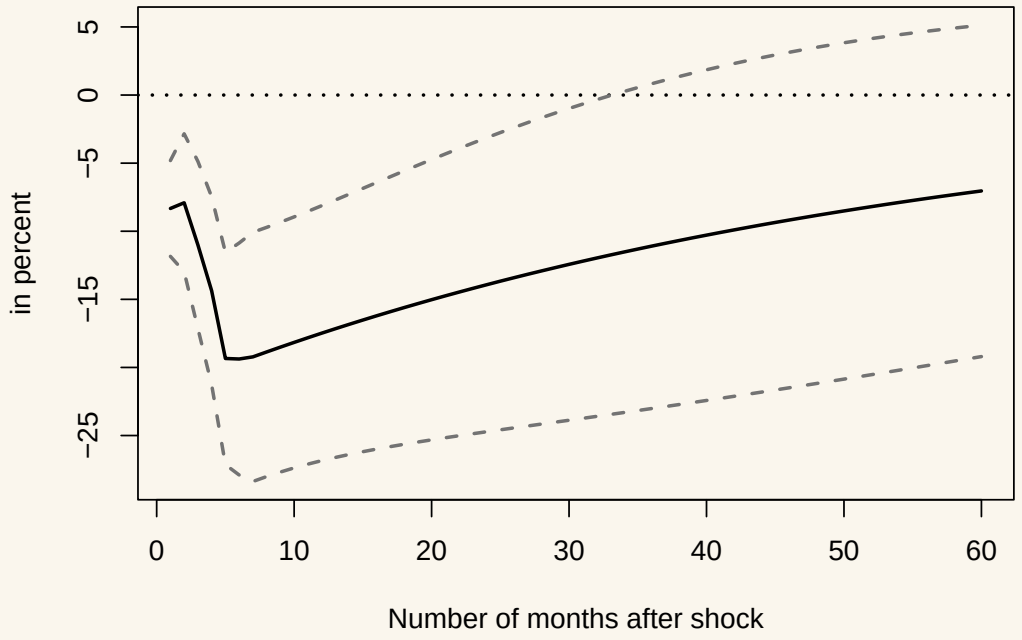


Figure 2.9: Estimated response of U.S. industrial-production growth to a BRW monetary-policy shock. The solid black line is the point estimate; the dashed grey lines are pointwise bands equal to the estimate plus or minus two standard errors, computed with the delta method.

[a]This method consists in approximating $\mathbb{V}ar(f(\theta))$ with $\frac{\partial f(\theta)}{\partial \theta}' \mathbb{V}ar(\theta) \frac{\partial f(\theta)}{\partial \theta}$.

2.8 Maximum Likelihood Estimation (MLE) of ARMA processes

2.8.1 Generalities

Consider the general case (of any time series); assume we observe a sample $\mathbf{y} = [y_1, \dots, y_T]'$. In order to implement ML techniques, we need to evaluate the joint p.d.f. (or "likelihood") of $\mathbf{y}$, i.e., $\mathcal{L}(\theta; \mathbf{y})$, where θ is a vector of parameters that characterizes the dynamics of y_t. The Maximum Likelihood (ML) estimate of θ is then given by:

$$\boxed{\theta_{MLE} = \arg\max_{\theta} \mathcal{L}(\theta; \mathbf{y}) = \arg\max_{\theta} \log \mathcal{L}(\theta; \mathbf{y}).}$$

It is useful to remember Bayes' formula:

$$\mathbb{P}(X_2 = x, X_1 = y) = \mathbb{P}(X_2 = x | X_1 = y)\mathbb{P}(X_1 = y).$$

Using it leads to the following decomposition of our likelihood function (by considering that $X_2 = Y_T$ and $X_1 = (Y_{T-1}, Y_{T-2}, \dots, Y_1)$):

$$\begin{aligned} f_{Y_T,\dots,Y_1}(y_T, \dots, y_1; \theta) &= f_{\underbrace{Y_T}_{=X_2} | \underbrace{Y_{T-1},\dots,Y_1}_{=X_1}}(y_T, \dots, y_1; \theta) \times \\ & \quad f_{\underbrace{Y_{T-1},\dots,Y_1}_{=X_1}}(y_{T-1}, \dots, y_1; \theta). \end{aligned}$$

Using the previous expression recursively, one obtains that for any $1 \leq p \leq T$:

$$f_{Y_T,\dots,Y_1}(y_T, \dots, y_1; \theta) = f_{Y_p,\dots,Y_1}(y_p, \dots, y_1; \theta) \times \prod_{t=p+1}^{T} f_{Y_t|Y_{t-1},\dots,Y_1}(y_t, \dots, y_1; \theta). \tag{2.16}$$

In the time series context, if process y_t is Markovian, then there exists a useful way to rewrite the likelihood $\mathcal{L}(\theta; \mathbf{y})$. Let us first recall the definition of a Markovian process:

Definition 2.10 (Markovian process)**.** Process y_t is Markovian of order one if $f_{Y_t|Y_{t-1},Y_{t-2},\dots} = f_{Y_t|Y_{t-1}}$. More generally, it is Markovian of order k if $f_{Y_t|Y_{t-1},Y_{t-2},\dots} = f_{Y_t|Y_{t-1},\dots,Y_{t-k}}$.

2.8.2 Maximum Likelihood Estimation of AR processes

2.8.2.1 Maximum Likelihood Estimation of an AR(1) process

Let us start with the Gaussian AR(1) process (which is Markovian of order one):

$$y_t = c + \phi_1 y_{t-1} + \varepsilon_t, \quad \varepsilon_t \sim i.i.d.\, \mathcal{N}(0, \sigma^2).$$

For $t > 1$:

$$f_{Y_t|Y_{t-1},\dots,Y_1}(y_t, \dots, y_1; \theta) = f_{Y_t|Y_{t-1}}(y_t, y_{t-1}; \theta)$$

and

$$f_{Y_t|Y_{t-1}}(y_t, y_{t-1}; \theta) = \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{(y_t - c - \phi_1 y_{t-1})^2}{2\sigma^2}\right).$$

These expressions can be plugged into Eq. (2.16) that ultimately takes the follwing form with $p = 1$

$$f_{Y_T,\dots,Y_1}(y_T, \dots, y_1; \theta) = f_{Y_1}(y_1; \theta) \times \prod_{t=2}^{T} f_{Y_t|Y_{t-1}}(y_t, y_{t-1}; \theta)$$

But what about $f_{Y_1}(y_1; \theta)$? There exist two possibilities:

1. **Case 1**: We use the marginal distribution: $y_1 \sim \mathcal{N}\left(\frac{c}{1-\phi_1}, \frac{\sigma^2}{1-\phi_1^2}\right)$.
2. **Case 2**: y_1 is considered to be deterministic. In a way, that means that the first observation is "sacrificed".

For a Gaussian AR(1) process, we have:

1. **Case 1**: The (exact) log-likelihood is:

$$\begin{aligned} \log \mathcal{L}(\theta; \mathbf{y}) &= -\frac{T}{2}\log(2\pi) - T\log(\sigma) + \frac{1}{2}\log(1-\phi_1^2) \\ &\quad -\frac{(y_1 - c/(1-\phi_1))^2}{2\sigma^2/(1-\phi_1^2)} - \sum_{t=2}^{T}\left[\frac{(y_t - c - \phi_1 y_{t-1})^2}{2\sigma^2}\right]. \end{aligned} \tag{2.17}$$

   The Maximum Likelihood Estimator of $\theta = [c, \phi_1, \sigma^2]$ is obtained by numerical optimization.

2. **Case 2**: The (conditional) log-likelihood (denoted with a $*$) is:

$$\begin{aligned} \log \mathcal{L}^*(\theta; \mathbf{y}) &= -\frac{T-1}{2}\log(2\pi) - (T-1)\log(\sigma) \\ &\quad -\sum_{t=2}^{T}\left[\frac{(y_t - c - \phi_1 y_{t-1})^2}{2\sigma^2}\right]. \end{aligned} \tag{2.18}$$

Exact MLE and conditional MLE have the same asymptotic (i.e. large-sample) distribution. Indeed, when the process is stationary, $f_{Y_1}(y_1;\theta)$ makes a relatively negligible contribution to $\log \mathcal{L}(\theta;\mathbf{y})$.

The conditional MLE has a substantial advantage: in the Gaussian case, the conditional MLE is simply obtained by OLS. Indeed, let us introduce the notations:

$$Y = \begin{bmatrix} y_2 \\ \vdots \\ y_T \end{bmatrix} \quad and \quad X = \begin{bmatrix} 1 & y_1 \\ \vdots & \vdots \\ 1 & y_{T-1} \end{bmatrix}.$$

Eq. (2.18) then rewrites:

$$\begin{aligned} \log \mathcal{L}^*(\theta;\mathbf{y}) &= -\frac{T-1}{2}\log(2\pi) - (T-1)\log(\sigma) \\ &\quad -\frac{1}{2\sigma^2}\underbrace{(Y - X[c,\phi_1]')'(Y - X[c,\phi_1]')}_{=\sum_{t=2}^{T}(y_t - c - \phi_1 y_{t-1})^2}, \end{aligned} \tag{2.19}$$

which is maximised for[6]:

$$[\hat{c},\hat{\phi}_1]' = (X'X)^{-1}X'Y \tag{2.20}$$

$$\begin{aligned} \hat{\sigma^2} &= \frac{1}{T-1}\sum_{t=2}^{T}(y_t - \hat{c} - \hat{\phi}_1 y_{t-1})^2 \\ &= \frac{1}{T-1}Y'(I - X(X'X)^{-1}X')Y. \end{aligned} \tag{2.21}$$

2.8.2.2 Maximum Likelihood Estimation of an AR(p) process

Let us turn to the case of an AR(p) process which is Markovian of order p. We have:

$$\begin{aligned} \log \mathcal{L}(\theta;\mathbf{y}) &= \log f_{Y_p,\dots,Y_1}(y_p,\dots,y_1;\theta) + \\ &\quad \underbrace{\sum_{t=p+1}^{T} \log f_{Y_t|Y_{t-1},\dots,Y_{t-p}}(y_t,\dots,y_{t-p};\theta)}_{\log \mathcal{L}^*(\theta;\mathbf{y})}. \end{aligned}$$

where $f_{Y_p,\dots,Y_1}(y_p,\dots,y_1;\theta)$ is the marginal distribution of $\mathbf{y}_{1:p} := [y_p,\dots,y_1]'$. The marginal distribution of $\mathbf{y}_{1:p}$ is Gaussian; it is therefore fully characterised by its mean and covariance matrix:

$$\begin{aligned} \mathbb{E}(\mathbf{y}_{1:p}) &= \frac{c}{1-\phi_1-\dots-\phi_p}\mathbf{1}_{p\times 1} \\ \mathbb{V}ar(\mathbf{y}_{1:p}) &= \begin{bmatrix} \gamma_0 & \gamma_1 & \cdots & \gamma_{p-1} \\ \gamma_1 & \gamma_0 & \cdots & \gamma_{p-2} \\ \vdots & & \ddots & \vdots \\ \gamma_{p-1} & \gamma_{p-2} & \cdots & \gamma_0 \end{bmatrix}, \end{aligned}$$

[6] Indeed, $(Y - X[c,\phi_1]')'(Y - X[c,\phi_1]') = Y'Y - [c,\phi_1]X'Y - Y'X[c,\phi_1]' + [c,\phi_1]X'X[c,\phi_1]'$. If we use the notation $\beta = [c,\phi_1]'$, then $(Y - X[c,\phi_1]')'(Y - X[c,\phi_1]') = Y'Y - \beta'X'Y - Y'X\beta + \beta'X'X\beta$ whose derivative with respect to β corresponds to $-2X'Y + 2X'X\beta$. If $\hat{\beta}$ maximizes the log-likelihood, then $X'X\hat{\beta} = X'Y \Leftrightarrow \hat{\beta} = [\hat{c},\hat{\phi}_1]' = (X'X)^{-1}X'Y$

where the γ_i's are computed using the Yule-Walker equations (Eq. (2.10)). Note that they depend, in a non-linear way, on the model parameters. Hence, the maximization of the exact log-likelihood necessitates numerical optimization procedures. By contrast, the maximization of the conditional log-likelihood $\log \mathcal{L}^*(\theta; \mathbf{y})$ only requires OLS, using Eqs. (2.20) and (2.21), with:

$$Y = \begin{bmatrix} y_{p+1} \\ \vdots \\ y_T \end{bmatrix} \quad and \quad X = \begin{bmatrix} 1 & y_p & \dots & y_1 \\ \vdots & \vdots & & \vdots \\ 1 & y_{T-1} & \dots & y_{T-p} \end{bmatrix}.$$

Again, for stationary processes, conditional and exact MLE have the same asymptotic (large-sample) distribution. In small samples, the OLS formula is however biased. Indeed, consider the regression (where y_t follows an AR(p) process):

$$y_t = \mathbf{x}_t \beta + \varepsilon_t, \tag{2.22}$$

with $\mathbf{x}_t = [1, y_{t-1}, \dots, y_{t-p}]$ of dimensions $1 \times (p+1)$ and $\beta = [c, \phi_1, \dots, \phi_p]'$ of dimensions $(p+1) \times 1$.

The bias results from the fact that $\mathbf{x}_t$ correlates to the ε_s's for $s < t$. To be sure:

$$\mathbf{b} = \beta + (X'X)^{-1} X' \varepsilon, \tag{2.23}$$

and because of the specific form of X, we have non-zero correlation[7] between $\mathbf{x}_t$ and ε_s for $s < t$, therefore $\mathbb{E}[(X'X)^{-1}X'\varepsilon] \neq 0$. Again, asymptotically, the previous expectation goes to zero, and we have:

Proposition 2.9 (Large-sample properties of the OLS estimator of AR(p) models)**.** *Assume $\{y_t\}$ follows the AR(p) process:*

$$y_t = c + \phi_1 y_{t-1} + \cdots + \phi_p y_{t-p} + \varepsilon_t$$

where $\{\varepsilon_t\}$ is an i.i.d. white noise process. If $\mathbf{b}$ *is the OLS estimator of β (Eq.* (2.22)*), we have:*

$$\sqrt{T}(\mathbf{b} - \beta) = \underbrace{\left[\frac{1}{T}\sum_{t=p}^{T} \mathbf{x}_t \mathbf{x}_t'\right]^{-1}}_{\xrightarrow{p} \mathbf{Q}^{-1}} \underbrace{\sqrt{T}\left[\frac{1}{T}\sum_{t=1}^{T} \mathbf{x}_t \varepsilon_t\right]}_{\xrightarrow{d} \mathcal{N}(0, \sigma^2 \mathbf{Q})},$$

where $\mathbf{Q} = plim\ \frac{1}{T}\sum_{t=p}^{T} \mathbf{x}_t \mathbf{x}_t' = plim\ \frac{1}{T}\sum_{t=1}^{T} \mathbf{x}_t \mathbf{x}_t'$ *is given by:*

$$\mathbf{Q} = \begin{bmatrix} 1 & \mu & \mu & \dots & \mu \\ \mu & \gamma_0 + \mu^2 & \gamma_1 + \mu^2 & \dots & \gamma_{p-1} + \mu^2 \\ \mu & \gamma_1 + \mu^2 & \gamma_0 + \mu^2 & \dots & \gamma_{p-2} + \mu^2 \\ \vdots & \vdots & \vdots & \dots & \vdots \\ \mu & \gamma_{p-1} + \mu^2 & \gamma_{p-2} + \mu^2 & \dots & \gamma_0 + \mu^2 \end{bmatrix}. \tag{2.24}$$

Proof. Rearranging Eq. (2.23), we have:

$$\sqrt{T}(\mathbf{b} - \beta) = (X'X/T)^{-1} \sqrt{T}(X'\varepsilon/T).$$

[7] The Gauss-Markov theorem's strict exogeneity assumption is therefore not satisfied.

Let us consider the autocovariances of $\mathbf{v}_t = \mathbf{x}_t\varepsilon_t$, denoted by γ_j^v. Using the fact that $\mathbf{x}_t$ is a linear combination of past ε_t's and that ε_t is a white noise, we get that $\mathbb{E}(\varepsilon_t\mathbf{x}_t) = \mathbb{E}(\mathbb{E}_{t-1}(\varepsilon_t\mathbf{x}_t)) = \mathbb{E}(\underbrace{\mathbb{E}_{t-1}(\varepsilon_t|\mathbf{x}_t)}_{=0}\mathbf{x}_t) = 0$. Therefore

$$\gamma_j^v = \mathbb{E}(\varepsilon_t\varepsilon_{t-j}\mathbf{x}_t\mathbf{x}_{t-j}').$$

If $j > 0$, we have

$$\begin{aligned}\mathbb{E}(\varepsilon_t\varepsilon_{t-j}\mathbf{x}_t\mathbf{x}_{t-j}') &= \mathbb{E}(\mathbb{E}[\varepsilon_t\varepsilon_{t-j}\mathbf{x}_t\mathbf{x}_{t-j}'|\varepsilon_{t-j},\mathbf{x}_t,\mathbf{x}_{t-j}]) \\ &= \mathbb{E}(\varepsilon_{t-j}\mathbf{x}_t\mathbf{x}_{t-j}'\mathbb{E}[\varepsilon_t|\varepsilon_{t-j},\mathbf{x}_t,\mathbf{x}_{t-j}]) = 0.\end{aligned}$$

Note that, for $j > 0$, we have $\mathbb{E}[\varepsilon_t|\varepsilon_{t-j},\mathbf{x}_t,\mathbf{x}_{t-j}] = 0$ because $\{\varepsilon_t\}$ is an i.i.d. white noise sequence. If $j = 0$, we have:

$$\gamma_0^v = \mathbb{E}(\varepsilon_t^2\mathbf{x}_t\mathbf{x}_t') = \mathbb{E}(\varepsilon_t^2)\mathbb{E}(\mathbf{x}_t\mathbf{x}_t') = \sigma^2\mathbf{Q}.$$

The convergence in distribution of $\sqrt{T}(X'\varepsilon/T) = \sqrt{T}\frac{1}{T}\sum_{t=1}^T v_t$ results from Theorem 1.1 (applied on $\mathbf{v}_t = \mathbf{x}_t\varepsilon_t$), using the γ_j^v computed above. □

These two cases (exact or conditional log-likelihoods) can be implemented when asking R to fit an AR process by means of function `arima`. Let us for instance use the output gap of the `US3var` dataset (US quarterly data, covering the period 1959:2 to 2015:1, used in Gouriéroux et al. (2017)).

```
##           ar1         ar2        ar3  intercept
## [1,] 1.191267 -0.08934705 -0.1781163 -0.9226007
## [2,] 1.192003 -0.08811150 -0.1787662 -1.0341696
```

The two sets of estimated coefficients appear to be very close to each other.

2.8.3 Maximum Likelihood Estimation of MA processes

2.8.3.1 Maximum Likelihood Estimation of an MA(1) process

Let us now turn to Moving-Average processes. Start with the MA(1):

$$y_t = \mu + \varepsilon_t + \theta_1\varepsilon_{t-1}, \quad \varepsilon_t \sim i.i.d.\mathcal{N}(0,\sigma^2).$$

The ε_t's are easily computed recursively, starting with $\varepsilon_t = y_t - \mu - \theta_1\varepsilon_{t-1}$. We obtain:

$$\varepsilon_t = y_t - \mu - \theta_1(y_{t-1}-\mu) + \theta_1^2(y_{t-2}-\mu) + \cdots + (-1)^{t-1}\theta_1^{t-1}(y_1-\mu) + (-1)^t\theta_1^t\varepsilon_0.$$

Assume that one wants to recover the sequence of $\{\varepsilon_t\}$'s based on observed values of y_t (from date 1 to date t). One can use the previous expression, but what value should be used for ε_0? If one does not use the true value of ε_0 but 0 (say), one does not obtain ε_t, but only an estimate of it ($\widehat{\varepsilon}_t$, say), with:

$$\widehat{\varepsilon}_t = \varepsilon_t - (-1)^t\theta_1^t\varepsilon_0.$$

Clearly, if $|\theta_1| < 1$, then the error becomes small for large t. Formally, when $|\theta_1| < 1$, we have:

$$\widehat{\varepsilon}_t \xrightarrow{p} \varepsilon_t.$$

Hence, when $|\theta_1| < 1$, a consistent estimate of the conditional log-likelihood is given by:

$$\log \widehat{\mathcal{L}}^*(\theta; \mathbf{y}) = -\frac{T}{2}\log(2\pi) - \frac{T}{2}\log(\sigma^2) - \sum_{t=1}^{T}\frac{\widehat{\varepsilon}_t^2}{2\sigma^2}. \tag{2.25}$$

Loosely speaking, if $|\theta_1| < 1$ and if T is sufficiently large:

$$\text{approximate conditional MLE} \approx \text{exact MLE}.$$

Note that $\widehat{\mathcal{L}}^*(\theta; \mathbf{y})$ is a complicated nonlinear function of μ and θ. Its maximization therefore has to be based on numerical optimization procedures.

2.8.3.2 Maximum Likelihood Estimation of MA(q) processes

Let us not consider the case of a Gaussian MA(q) process:

$$y_t = \mu + \varepsilon_t + \theta_1\varepsilon_{t-1} + \cdots + \theta_q\varepsilon_{t-q}, \quad \varepsilon_t \sim i.i.d.\mathcal{N}(0, \sigma^2). \tag{2.26}$$

Let us assume that this process is an **invertible MA process**. That is, assume that the roots of:

$$\lambda^q + \theta_1\lambda^{q-1} + \cdots + \theta_{q-1}\lambda + \theta_q = 0 \tag{2.27}$$

lie strictly inside of the unit circle. In this case, the polynomial $\Theta(L) = 1 + \theta_1 L + \cdots + \theta_q L^q$ is *invertible* and Eq. (2.26) writes:

$$\varepsilon_t = \Theta(L)^{-1}(y_t - \mu),$$

which implies that, if we knew all past values of y_t, we would also know ε_t. In this case, we can consistently estimate the ε_t's by recursively computing the $\widehat{\varepsilon}_t$'s as follows (for $t > 0$):

$$\widehat{\varepsilon}_t = y_t - \mu - \theta_1\widehat{\varepsilon}_{t-1} - \cdots - \theta_q\widehat{\varepsilon}_{t-q}, \tag{2.28}$$

with

$$\widehat{\varepsilon}_0 = \cdots = \widehat{\varepsilon}_{-q+1} = 0. \tag{2.29}$$

In this context, a consistent estimate of the conditional log-likelihood is still given by Eq. (2.25), using Eqs. (2.28) and (2.29) to recursively compute the $\widehat{\varepsilon}_t$'s.

Note that we could determine the exact likelihood of an MA process. Indeed, vector $\mathbf{y} = [y_1, \ldots, y_T]'$ is a Gaussian-distributed vector of mean $\mu = [\mu, \ldots, \mu]'$ and of variance:

$$\Omega = \begin{bmatrix} \gamma_0 & \gamma_1 & \cdots & \gamma_q & 0 & \cdots & 0 \\ \gamma_1 & \gamma_0 & \gamma_1 & & \ddots & \ddots & \vdots \\ \vdots & \gamma_1 & \ddots & \ddots & & \ddots & 0 \\ \gamma_q & & \ddots & & & & \gamma_q \\ 0 & & & \ddots & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & & \gamma_1 & \gamma_0 & \gamma_1 \\ 0 & \cdots & 0 & \gamma_q & \cdots & \gamma_1 & \gamma_0 \end{bmatrix},$$

where the γ_j's are given by Eq. (2.2). The p.d.f. of $\mathbf{y}$ is then given by (see Prop. 8.10):

$$(2\pi)^{-T/2}|\Omega|^{-1/2}\exp\left(-\frac{1}{2}(\mathbf{y}-\mu)'\Omega^{-1}(\mathbf{y}-\mu)\right).$$

For large samples, the computation of this likelihood however becomes numerically demanding.

2.8.4 Maximum Likelihood Estimation of an ARMA(p,q) process

Finally, let us consider the MLE of an ARMA(p,q) processes:

$$y_t = c + \phi_1 y_{t-1} + \cdots + \phi_p y_{t-p} + \varepsilon_t + \theta_1 \varepsilon_{t-1} + \cdots + \theta_q \varepsilon_{t-q},\ \varepsilon_t \sim i.i.d.\ \mathcal{N}(0, \sigma^2).$$

If the MA part of this process is invertible, the log-likelihood function can be consistently approximated by its conditional counterpart (of the form of Eq. (2.25)), using consistent estimates $\widehat{\varepsilon}_t$ of the ε_t. The $\widehat{\varepsilon}_t$'s are computed recursively as:

$$\widehat{\varepsilon}_t = y_t - c - \phi_1 y_{t-1} - \cdots - \phi_p y_{t-p} - \theta_1 \widehat{\varepsilon}_{t-1} - \cdots - \theta_q \widehat{\varepsilon}_{t-q}, \tag{2.30}$$

given some initial conditions, for instance:

a. $\widehat{\varepsilon}_0 = \cdots = \widehat{\varepsilon}_{-q+1} = 0$ and $y_0 = \cdots = y_{-p+1} = \mathbb{E}(y_i) = \mu$. (Recursions in Eq. (2.30) then start for $t = 1$.)
b. $\widehat{\varepsilon}_p = \cdots = \widehat{\varepsilon}_{p-q+1} = 0$ and actual values of the y_i's for $i \in [1, p]$. In that case, the first p observations of y_t will not be used. Recursions in Eq. (2.30) then start for $t = p + 1$.

2.9 Specification choice

The previous section explains how to fit a given ARMA specification. But how to choose an appropriate specification? A possibility is to employ the (P)ACF approach (see Figure 2.2). However, the previous approach leads to either an AR or a MA process (and not an ARMA process). If one wants to consider various ARMA(p,q) specifications, for $p \in \{1, \dots, P\}$ and $q \in \{1, \dots, Q\}$, say, then one can resort to **information criteria**.

In general, when choosing a specification, one faces the following dilemma:

a. Too rich a specification may lead to "overfitting"/misspecification, implying additional estimation errors (in out-of-sample forecasts).
b. Too simple a specification may lead to potential omission of valuable information (e.g., contained in older lags).

The lag selection approach based on the so-called **information criteria** consists in maximizing the fit of the data, but adding a penalty for the "richness" of the model. More precisely, using this approach amounts to minimizing a loss function that (a) negatively depends on the fitting errors and (b) positively depends on the number of parameters in the model.

Definition 2.11 (Information Criteria)**.** The Akaike (AIC), Hannan-Quinn (HQ) and Schwarz information (BIC) criteria are of the form

$$c^{(i)}(k) = \underbrace{\frac{-2 \log \mathcal{L}(\widehat{\theta}_T(k); \mathbf{y})}{T}}_{\text{decreases w.r.t. } k} + \underbrace{\frac{k \phi^{(i)}(T)}{T}}_{\text{increases w.r.t. } k},$$

with $(i) \in \{AIC, HQ, BIC\}$ and where $\widehat{\theta}_T(k)$ denotes the ML estimate of $\theta_0(k)$, which is a vector of parameters of length k.

Table 2.1: (#tabcriteria) This table shows ϕ functions used by the different criteria.

Criterion		$\phi^{(i)}(T)$
Akaike	AIC	2
Hannan-Quinn	HQ	$2\log(\log(T))$
Schwarz	BIC	$\log(T)$

The lag suggested by criterion (i) is then given by:

$$\boxed{\hat{k}^{(i)} = \underset{k}{\text{argmin}} \quad c^{(i)}(k).}$$

In the case of an ARMA(p,q) process, $k = 2 + p + q$.

Proposition 2.10 (Consistency of the criteria-based lag selection)**.** *The lag selection procedure is consistent if*

$$\lim_{T\to\infty} \phi(T) = \infty \quad and \quad \lim_{T\to\infty} \phi(T)/T = 0.$$

This is notably the case of the HQ and the BIC criteria.

Proof. The true number of lags is denoted by k_0. We will show that $\lim_{T\to\infty} \mathbb{P}(\hat{k}_T \neq k_0) = 0$.

- Case $k < k_0$: The model with k parameter is misspecified, therefore:

  $$\text{plim}_{T\to\infty} \log \mathcal{L}(\hat{\theta}_T(k); \mathbf{y})/T < \text{plim}_{T\to\infty} \log \mathcal{L}(\hat{\theta}_T(k_0); \mathbf{y})/T.$$

  Hence, if $\lim_{T\to\infty} \phi(T)/T = 0$, we have: $\lim_{T\to\infty} \mathbb{P}(c(k_0) \geq c(k)) \to 0$ and

  $$\lim_{T\to\infty} \mathbb{P}(\hat{k} < k_0) \leq \lim_{T\to\infty} \mathbb{P}\left\{c(k_0) \geq c(k) \text{ for some } k < k_0\right\} = 0.$$

- Case $k > k_0$: under the null hypothesis, the likelihood ratio (LR) test statistic satisfies:

  $$2\left(\log \mathcal{L}(\hat{\theta}_T(k); \mathbf{y}) - \log \mathcal{L}(\hat{\theta}_T(k_0); \mathbf{y})\right) \sim \chi^2(k - k_0).$$

  If $\lim_{T\to\infty} \phi(T) = \infty$, we have: $\text{plim}_{T\to\infty} -2\left(\log \mathcal{L}(\hat{\theta}_T(k); \mathbf{y}) - \log \mathcal{L}(\hat{\theta}_T(k_0); \mathbf{y})\right)/\phi(T) = 0$. Hence $\text{plim}_{T\to\infty} T[c(k_0) - c(k)]/\phi(T) \leq -1$ and $\lim_{T\to\infty} \mathbb{P}(c(k_0) \geq c(k)) \to 0$, which implies, in the same spirit as before, that $\lim_{T\to\infty} \mathbb{P}(\hat{k} > k_0) = 0$.

Therefore, $\lim_{T\to\infty} \mathbb{P}(\hat{k} = k_0) = 1$. □

Example 2.3 (Linear regression)**.** Consider a linear regression with normal disturbances:

$$y_t = \mathbf{x}_t'\beta + \varepsilon_t, \quad \varepsilon_t \sim i.i.d. \mathcal{N}(0, \sigma^2).$$

The associated log-likelihood is of the form of Eq. (2.25). In that case, we have:

$$\begin{aligned} c^{(i)}(k) &= \frac{-2 \log \mathcal{L}(\hat{\theta}_T(k); \mathbf{y})}{T} + \frac{k\phi^{(i)}(T)}{T} \\ &\approx \log(2\pi) + \log(\widehat{\sigma^2}) + \frac{1}{T}\sum_{t=1}^{T} \frac{\varepsilon_t^2}{\widehat{\sigma^2}} + \frac{k\phi^{(i)}(T)}{T}. \end{aligned}$$

For a large T, for all consistent estimation scheme, we have:

$$\widehat{\sigma^2} \approx \frac{1}{T}\sum_{t=1}^{T} \varepsilon_t^2 = SSR/T.$$

Hence $\widehat{k}^{(i)} \approx \underset{k}{\text{argmin}} \quad \log(SSR/T) + \frac{k\phi^{(i)}(T)}{T}.$

Example 2.4 (Swiss GDP growth)**.** Consider a long historical time series of the Swiss GDP growth (see Figure 1.3), taken from the Jordà et al. (2017) dataset. Let us look for the best ARMA specification using the AIC criteria:

```
## [1] 1 0
```

The best specification therefore is an AR(1) model. That is, although an AR(2) (say) would result in a better fit of the data, the fit improvement is not be large enough to compensate for the additional AIC cost associated with an additional parameter.

2.10 Exercises

The following exercises consolidate stationarity, Wold representations, likelihood-based estimation, model identification, and impulse-response analysis. They come from PS1, PS2, and PS3.

Exercise 2.1 (Stationarity of ARMA processes)**. PS1 | Core | Analytical**
Review Section 2.2.

For each process below, determine whether it is covariance stationary. When it is, identify an ARMA representation and justify your answer using the roots of the autoregressive polynomial.

1. $y_t = y_{t-1} + \varepsilon_t$.
2. $y_t = 0.9y_{t-1} + \varepsilon_t + \varepsilon_{t-1}$.
3. $y_t = \varepsilon_t + \varepsilon_{t-1} + 2\varepsilon_{t-2}$.
4. $y_t = 0.9y_{t-1} + 0.2y_{t-2} + \varepsilon_t$.
5. $y_t = 0.9y_{t-1} - 0.2y_{t-2} + \varepsilon_t$.

Assume throughout that $\{\varepsilon_t\}$ is white noise with finite variance.

Exercise 2.2 (A stationary transformation)**. PS1 | Core | Analytical**
Review Section 2.

Let $y_t = y_{t-1} + \varepsilon_t$, where $\{\varepsilon_t\}$ is white noise, and define $x_t = y_t - y_{t-1}$.

1. Is y_t covariance stationary?
2. Express x_t in terms of the innovation and determine whether x_t is covariance stationary.
3. State the order of integration of y_t.

Exercise 2.3 (Autocovariances of an ARMA(2,1))**. PS1 | Advanced | Analytical**
Review Section 2.

Consider

$$y_t = \phi_1 y_{t-1} + \phi_2 y_{t-2} + \varepsilon_t + \varepsilon_{t-1},$$

where $\{\varepsilon_t\}$ is white noise with variance σ^2 and the AR polynomial is stable. Let $\gamma_h = \mathbb{C}ov(y_t, y_{t-h})$.

1. Compute $\mathbb{C}ov(\varepsilon_t, y_t)$ and $\mathbb{C}ov(\varepsilon_{t-1}, y_t)$.
2. Derive the equations linking γ_0, γ_1, and γ_2.
3. Show that $\gamma_h = \phi_1 \gamma_{h-1} + \phi_2 \gamma_{h-2}$ for $h \geq 2$.
4. Solve the three-equation system for γ_0, γ_1, and γ_2 (a matrix expression is acceptable).

Exercise 2.4 (Wold representations and forecasts)**. PS1 | Advanced | Analytical**
Review Section 2.6.

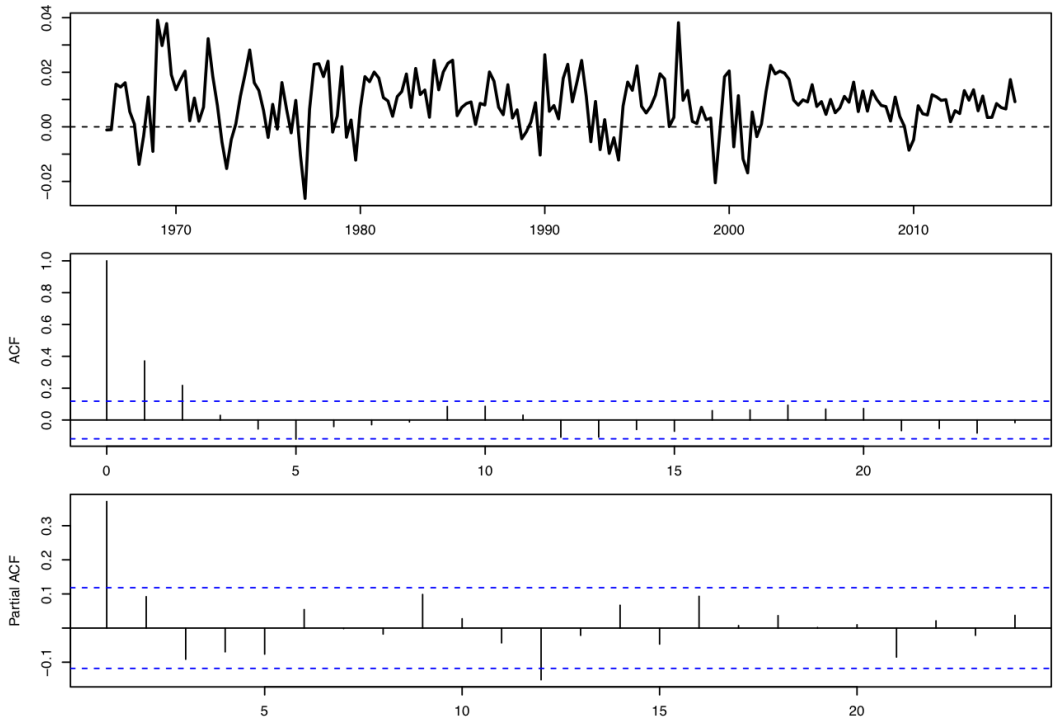
For each stationary process in Exercise 2.1:

1. Derive the first four Wold coefficients $\psi_0, \ldots, \psi_3$ in $y_t - \mu = \sum_{j=0}^{\infty} \psi_j \varepsilon_{t-j}$.
2. Compute the three-step-ahead conditional forecast $\mathbb{E}_t(y_{t+3})$.
3. Compute the corresponding forecast-error variance.

Take the innovations to have variance σ^2 and state clearly which information set is used.

Exercise 2.5 (Reading an ACF and a PACF)**. PS1 | Core | Empirical interpretation** Review Section 2.4.

The figure shows a stationary-looking macroeconomic series, together with its sample autocorrelation function (ACF) and partial autocorrelation function (PACF).



1. What low-order AR or MA specifications are suggested by these diagnostics?
2. Can the plots alone establish strict stationarity?
3. Explain how you would choose between the candidate models in practice.

Exercise 2.6 (Maximum likelihood for an exponential duration)**. PS2 | Core | Analytical**
Review Section 2.8.1.

Customer inter-arrival times $y_1, \dots, y_T$ are i.i.d. exponential with **scale** parameter $\lambda > 0$:

$$f(y; \lambda) = \lambda^{-1} \exp(-y/\lambda), \qquad y \geq 0.$$

1. Write the log-likelihood.
2. Derive the maximum-likelihood estimator of λ.
3. Compute the Fisher information and the estimator's asymptotic standard error.

Exercise 2.7 (A periodically centered AR(1))**. PS2 | Advanced | Likelihood and testing**
Review Section 2.8.2.1.

Consider

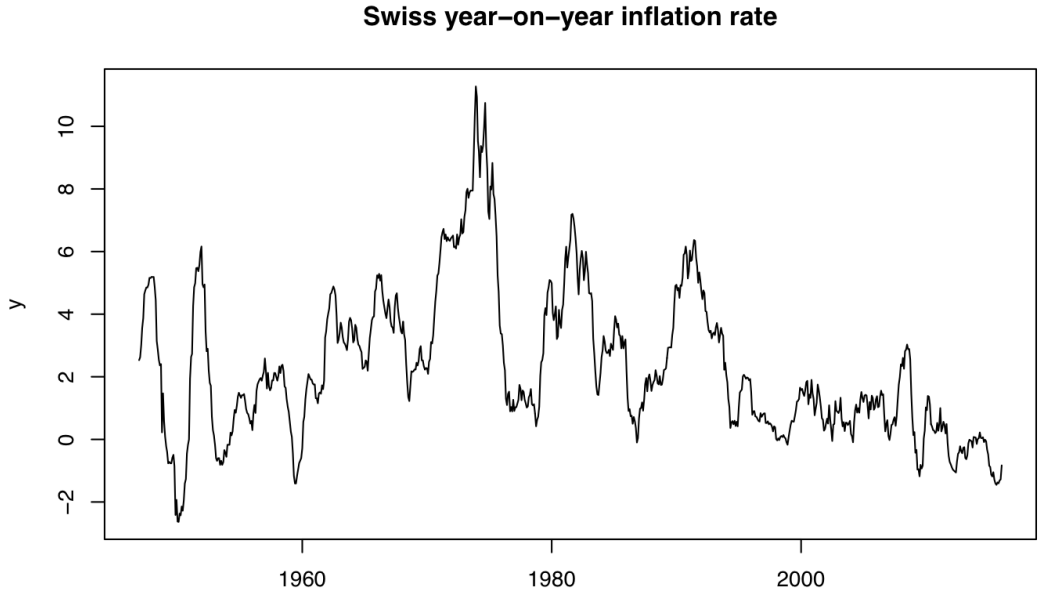
$$y_t = \cos(\psi t) + \rho\{y_{t-1} - \cos[\psi(t-1)]\} + \varepsilon_t, \qquad \varepsilon_t \overset{i.i.d.}{\sim} \mathcal{N}(0, \sigma^2),$$

with $|\rho| < 1$.

1. Identify the process followed by $x_t = y_t - \cos(\psi t)$.
2. Derive the unconditional mean and variance of y_t.
3. Derive the conditional Gaussian log-likelihood for $\theta = (\psi, \rho, \sigma^2)'$.
4. Explain how to estimate θ and its asymptotic covariance matrix.
5. For monthly data, interpret $\psi = 2\pi/12$.
6. Describe Wald and likelihood-ratio tests of $H_0 : \psi = 2\pi/12$ and $\rho = 0.9$. Give the 5% and 1% critical values.

Exercise 2.8 (Estimating an AR(p) model for Swiss inflation)**. PS2 | Core | Estimation and diagnostics**
Review Section 2.8.2.2.

The figure shows Swiss year-on-year inflation.



Model $y_t = c + \phi_1 y_{t-1} + \cdots + \phi_p y_{t-p} + \varepsilon_t$, with $\varepsilon_t \overset{i.i.d.}{\sim} \mathcal{N}(0, \sigma^2)$. Let $Y = (y_{p+1}, \ldots, y_T)'$ and let X contain an intercept and the p lags.

1. Give the conditional maximum-likelihood estimators of $\beta = (c, \phi_1, \ldots, \phi_p)'$ and σ^2.
2. For $p = 1$, $\hat{c} = 0.025$ (s.e. 0.019) and $\hat{\phi}_1 = 0.987$ (s.e. 0.006). Is the fitted process stationary? Comment on persistence.
3. Regressing the AR(1) residual on its lag gives 0.207 (s.e. 0.034). What does this imply about specification?
4. For $p = 2$, the estimates are $\hat{c} = 0.033$, $\hat{\phi}_1 = 1.194$, and $\hat{\phi}_2 = -0.210$. The lagged-residual coefficient is -0.027 (s.e. 0.035). Check stationarity from the companion roots and assess the new specification.

Exercise 2.9 (Impulse responses and forecast uncertainty)**. PS3 | Core | Analytical**
Review Section 2.6.

Let $\varepsilon_t \overset{i.i.d.}{\sim} \mathcal{N}(0, \sigma^2)$ and consider:

1. $y_t = \varepsilon_t + 0.5\varepsilon_{t-1} + 0.2\varepsilon_{t-2}$;
2. $y_t = 1 + 0.9y_{t-1} + \varepsilon_t$;
3. $y_t = 1 + 0.5y_{t-1} + \varepsilon_t + 0.5\varepsilon_{t-1}$.

For each process:

1. Compute the impulse responses at horizons 0 through 4.
2. State the corresponding first five Wold coefficients.
3. Compute the mean squared error of the four-step-ahead forecast.
4. Explain how parameter estimation changes forecast uncertainty.

Finally, consider $y_t = 1 + 0.5y_{t-1} + x_t + 0.5x_{t-1} + \varepsilon_t + 0.8\varepsilon_{t-1}$, with x_t strictly exogenous. Compute the dynamic multipliers of x_t at horizons 0 through 4.

2.10.1 Exercises from the 2024 exam

Exercise 2.10 (Two AR(1) processes driven by a common shock)**. EXAM-2024 | Exam | True/false**
Review Section 2.2.

Let $x_t = \rho_x x_{t-1} + \varepsilon_t$ and $y_t = \rho_y y_{t-1} + \varepsilon_t$, where $\varepsilon_t \overset{i.i.d.}{\sim} \mathcal{N}(0, \sigma^2)$, $\sigma^2 > 0$, and $0 < \rho_x < \rho_y < 1$. Conditional moments at date t use the history of both processes.

State whether each assertion is true or false and justify your answer.

1. The contemporaneous correlation between x_t and y_t is zero.
2. $\mathbb{V}ar(y_t) = \mathbb{V}ar(x_t)$.
3. $\mathbb{E}_t(y_{t+1}) = \mathbb{E}_t(x_{t+1})$.
4. $\mathbb{E}_t(y_{t+h}) = \rho_y^h y_t$ for every $h \geq 1$.
5. $\mathbb{C}ov_t(x_{t+1}, y_{t+1}) = \sigma^2$.

Exercise 2.11 (Interpreting an autocovariance sequence)**. EXAM-2024 | Exam | True/false**
Review Section 2.

A covariance-stationary process has $\gamma_0 = 5$, $\gamma_1 = 4$, $\gamma_2 = 3$, $\gamma_3 = 2$, $\gamma_4 = 1$, and $\gamma_h = 0$ for $h > 4$.

State whether each assertion is true or false and justify your answer.

1. The process can be a nondegenerate AR(4).
2. It could be $y_t = \varepsilon_t + \varepsilon_{t-1} + \varepsilon_{t-2} + \varepsilon_{t-3} + \varepsilon_{t-4}$ with $\mathbb{V}ar(\varepsilon_t) = 1$.
3. $\mathbb{V}ar(y_t) = \sum_{h=0}^{4} \gamma_h$.
4. Corr$(y_{t-1}, y_{t-4}) = 2/5$.
5. Necessarily, $\mathbb{E}(y_t) = \gamma_0$.

Exercise 2.12 (ARMA roots and moments)**. EXAM-2024 | Exam | True/false**
Review Section 2.2.

Let $\varepsilon_t \overset{i.i.d.}{\sim} \mathcal{N}(0, \sigma^2)$ and

$$(1 - 0.5L^2)x_t = 0.1 + 0.5x_{t-1} + \varepsilon_t,$$
$$(1 + 1.1L)(1 - 0.5L)(1 + 0.4L)y_t = \varepsilon_t,$$
$$z_t + 0.8z_{t-1} = (1 + L + L^2)\varepsilon_t.$$

State whether each assertion is true or false and justify your answer.

1. x_t is covariance stationary.
2. y_t is covariance stationary.
3. z_t is covariance stationary.
4. $\gamma_1(z) = -0.8\gamma_0(z)$.
5. $\mathbb{V}ar(z_t + 0.8z_{t-1}) = 3\sigma^2$.

Exercise 2.13 (A one-date exogenous intervention)**. EXAM-2024 | Exam | True/false**
Review Section 2.7.2.

For $t \geq 0$, let

$$y_t = 0.5y_{t-1} + x_t + \varepsilon_t, \qquad \tilde{y}_t = 0.5\tilde{y}_{t-1} + \tilde{x}_t + \varepsilon_t,$$

with $y_0 = \tilde{y}_0 = 0$. The inputs are deterministic, $\tilde{x}_t = x_t$ except at one date $s > 0$, and $\tilde{x}_s = x_s + 1$.

State whether each assertion is true or false and justify your answer.

1. For $t < s$, $\tilde{y}_t = y_t$.
2. For all $t \geq s$, $\tilde{y}_t = y_t$.
3. For every $h \geq 0$, $\tilde{y}_{s+h} - y_{s+h} = 1$.
4. If x_t is constant, y_t is an AR(1) with an intercept.
5. $\mathbb{V}ar(y_t) = \mathbb{V}ar(\tilde{y}_t)$.

2.10.2 Exercises from the 2025 exam

Exercise 2.14 (Conditional moments of two AR(1) processes)**. EXAM-2025 | Exam | True/false**
Review Section 2.2.

Consider

$$x_t = \rho_x x_{t-1} + \varepsilon_t, \qquad y_t = \rho_y y_{t-1} + \varepsilon_t,$$

where $\varepsilon_t \overset{i.i.d.}{\sim} \mathcal{N}(0, \sigma^2)$, $\sigma^2 > 0$, and $0 < \rho_x < \rho_y < 1$. Conditional moments at date t use the history of both processes.

State whether each assertion is true or false and justify your answer.

1. $\mathbb{V}ar(y_t) > \mathbb{V}ar(x_t)$.
2. $\mathbb{V}ar_t(y_{t+2}) = \mathbb{V}ar_t(x_{t+2})$.
3. $\mathbb{E}(y_t) > \mathbb{E}(x_t)$.
4. $\mathbb{E}_{t+1}(y_{t+h}) = \rho_y^h y_{t+1}$ for $h \geq 2$.
5. $\mathbb{C}ov_t(y_{t+1}, y_t) = 0$.

Exercise 2.15 (Effects and forecasts in an ARMAX model)**. EXAM-2025 | Exam | True/false**
Review Section 2.7.2.

Consider

$$y_t = \phi_1 y_{t-1} + \beta_0 x_t + \beta_1 x_{t-1} + \varepsilon_t,$$

where $\varepsilon_t \overset{i.i.d.}{\sim} \mathcal{N}(0, \sigma^2)$ and $x_t \overset{i.i.d.}{\sim} \mathcal{N}(\mu_x, \sigma_x^2)$. The two innovation sequences are independent. Conditional moments at date t use the history of x_t, y_t, and ε_t.

State whether each assertion is true or false and justify your answer.

1. Everything else equal, a unit increase in x_t increases y_{t+1} by β_1.
2. $\mathbb{E}_t(x_{t+1}) = \mu_x$.
3. $\mathbb{E}_t(y_{t+1}) = \phi_1 y_t + \beta_0 \mu_x + \beta_1 x_t$.
4. $\mathbb{V}ar_t(y_{t+1}) = \sigma^2$.
5. Everything else equal, a unit increase in ε_t increases y_{t+1} by $\phi_1 + \beta_0$.

Chapter 3

Multivariate models

This section presents Vector Auto-Regressive Moving-Average (SVARMA) models. These models are widely used in macroeconomic analysis. While simple and easy to estimate, they make it possible to conveniently capture the dynamics of complex multivariate systems. VAR popularity is notably due to Sims (1980)'s influential work. A comprehensive survey is proposed by Stock and Watson (2016).

In economics, VAR models are often employed in order to identify *structural* shocks, that are independent primitive exogenous forces that drive economic variables (Ramey (2016)). They are often given a specific economic meaning (e.g., demand and supply shocks). Once structural shocks have been identified, researchers use the resulting Structural VAR model to produce Impulse Response Functions that describe the dynamic reactions of endogenous variables to the shocks.

Working with these models (VAR and VARMA models) often involves two steps: in a first step, the **reduced-form** version of the model is estimated; in a second step, **structural shocks** are identified and IRFs are produced.

3.1 Definition of VARs (and SVARMA) models

Definition 3.1 ((S)VAR model)**.** Let y_t denote a $n \times 1$ vector of (endogenous) random variables. Process y_t follows a pth-order (S)VAR if, for all t, we have

$$\begin{array}{rrcl} VAR: & y_t & = & c + \Phi_1 y_{t-1} + \cdots + \Phi_p y_{t-p} + \varepsilon_t, \\ SVAR: & y_t & = & c + \Phi_1 y_{t-1} + \cdots + \Phi_p y_{t-p} + B\eta_t, \end{array} \tag{3.1}$$

with $\varepsilon_t = B\eta_t$, where $\{\eta_t\}$ is a white noise sequence whose components are mutually and serially independent.

The first line of Eq. (3.1) corresponds to the **reduced-form** of the VAR model (**structural form** for the second line). While the **structural shocks** (the components of η_t) are mutually uncorrelated, this is not the case of the **innovations**, that are the components of ε_t. However, in boths cases, vectors η_t and ε_t are serially correlated (through time).

As is the case for univariate models, VARs can be extended with MA terms in η_t, giving rise to VARMA models:

Definition 3.2 ((S)VARMA model)**.** Let y_t denote a $n \times 1$ vector of random variables. Process y_t follows a VARMA model of order (p,q) if, for all t, we have

$$\begin{array}{rrcl} VARMA: & y_t & = & c + \Phi_1 y_{t-1} + \cdots + \Phi_p y_{t-p} + \\ & & & \varepsilon_t + \Theta_1 \varepsilon_{t-1} + \cdots + \Theta_q \varepsilon_{t-q}, \\ SVARMA: & y_t & = & c + \Phi_1 y_{t-1} + \cdots + \Phi_p y_{t-p} + \\ & & & B_0 \eta_t + B_1 \eta_{t-1} + \cdots + B_q \eta_{t-q}, \end{array} \tag{3.2}$$

with $\varepsilon_t = B_0 \eta_t$, and $B_j = \Theta_j B_0$, for $j \geq 0$ (with $\Theta_0 = Id$), where $\{\eta_t\}$ is a white noise sequence whose components are are mutually and serially independent.

3.2 IRFs in SVARMA

One of the main objectives of macro-econometrics is to derive IRFs, that represent the dynamic effects of structural shocks (components of η_t) though the system of variables y_t.

Formally, an IRF is a difference in conditional expectations:

$$\boxed{\Psi_{i,j,h} = \mathbb{E}(y_{i,t+h} | \eta_{j,t} = 1) - \mathbb{E}(y_{i,t+h})} \tag{3.3}$$

(effect on $y_{i,t+h}$ of a one-unit shock on $\eta_{j,t}$).

IRFs closely relate to the Wold decomposition of y_t. Indeed, if the dynamics of process y_t can be described as a VARMA model, and if y_t is covariance stationary (see Def. 1.4), then y_t admits the following infinite MA representation (or Wold decomposition):

$$y_t = \mu + \sum_{h=0}^{\infty} \Psi_h \eta_{t-h}. \tag{3.4}$$

With these notations, we get $\mathbb{E}(y_{i,t+h} | \eta_{j,t} = 1) = \mu_i + \Psi_{i,j,h}$, where $\Psi_{i,j,h}$ is the component (i,j) of matrix Ψ_h and μ_i is the ith entry of vector μ. Since we also have $\mathbb{E}(y_{i,t+h}) = \mu_i$, we obtain Eq. (3.3).

Hence, estimating IRFs amounts to estimating the Ψ_h's. In general, there exist three main approaches for that:

- Calibrate and solve a (purely structural) Dynamic Stochastic General Equilibrium (DSGE) model at the first order (linearization). The solution takes the form of Eq. (3.4).
- Directly estimate the Ψ_h based on *projection approaches* à la Jordà (2005) (see this web-page).
- Approximate the infinite MA representation by estimating a parsimonious type of model, e.g. **VAR(MA) models** (see Section 3.4). Once a (Structural) VARMA representation is obtained, Eq. (3.4) is easily deduced using the following proposition:

Proposition 3.1 (IRF of a VARMA(p,q) process). *If y_t follows the VARMA model described in Def. 3.2, then the matrices Ψ_h appearing in Eq.* (3.4) *can be computed recursively as follows:*

1. *Set $\Psi_{-1} = \cdots = \Psi_{-p} = 0$.*
2. *For $h \geq 0$, (recursively) apply:*

$$\Psi_h = \Phi_1 \Psi_{h-1} + \cdots + \Phi_p \Psi_{h-p} + \Theta_h B_0,$$

with $\Theta_0 = Id$ and $\Theta_h = 0$ for $h > q$.

Proof. This is obtained by applying the operator $\frac{\partial}{\partial \varepsilon_t}$ on both sides of Eq. (3.2). □

Typically, consider the VAR(2) case. The first steps of the algorithm mentioned in the last bullet point are as follows:

$$\begin{aligned}
y_t &= \Phi_1 y_{t-1} + \Phi_2 y_{t-2} + B\eta_t \\
&= \Phi_1(\Phi_1 y_{t-2} + \Phi_2 y_{t-3} + B\eta_{t-1}) + \Phi_2 y_{t-2} + B\eta_t \\
&= B\eta_t + \Phi_1 B\eta_{t-1} + (\Phi_2 + \Phi_1^2) y_{t-2} + \Phi_1\Phi_2 y_{t-3} \\
&= B\eta_t + \Phi_1 B\eta_{t-1} + (\Phi_2 + \Phi_1^2)(\Phi_1 y_{t-3} + \Phi_2 y_{t-4} + B\eta_{t-2}) + \Phi_1\Phi_2 y_{t-3} \\
&= \underbrace{B}_{=\Psi_0}\eta_t + \underbrace{\Phi_1 B}_{=\Psi_1}\eta_{t-1} + \underbrace{(\Phi_2 + \Phi_1^2)B}_{=\Psi_2}\eta_{t-2} + f(y_{t-3}, y_{t-4}).
\end{aligned}$$

In particular, we have $B = \Psi_0$. Matrix B indeed captures the contemporaneous impact of η_t on y_t. That is why matrix B is sometimes called **impulse matrix**.

Example 3.1 (IRFs of an SVARMA model). Consider the following VARMA(1,1) model:

$$y_t = \underbrace{\begin{bmatrix} 0.5 & 0.3 \\ -0.4 & 0.7 \end{bmatrix}}_{\Phi_1} y_{t-1} + \underbrace{\begin{bmatrix} 1 & 2 \\ -1 & 1 \end{bmatrix}}_{B} \eta_t - \underbrace{\begin{bmatrix} -0.4 & 0 \\ 1 & 0.5 \end{bmatrix}}_{\Theta_1} \underbrace{\begin{bmatrix} 1 & 2 \\ -1 & 1 \end{bmatrix}}_{B} \eta_{t-1} \quad (3.5)$$

We can use function `simul.VARMA` of package `IdSS` to produce IRFs (using `indic.IRF=1` in the list of arguments):

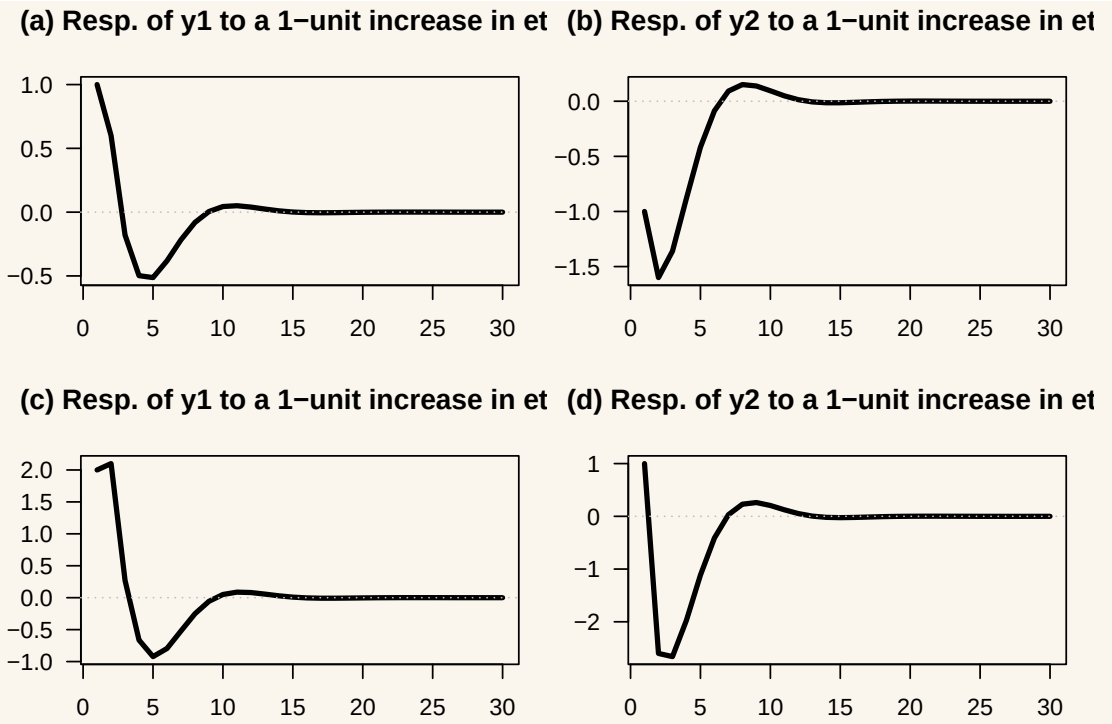


Figure 3.1: Impulse response functions of the bivariate SVARMA(1,1) model specified in Equation eqrefeq:VARMA111: (a) response of $y_{1,t}$ to $\eta_{1,t}$; (b) response of $y_{2,t}$ to $\eta_{1,t}$; (c) response of $y_{1,t}$ to $\eta_{2,t}$; and (d) response of $y_{2,t}$ to $\eta_{2,t}$. Each structural innovation is increased by one unit on impact.

This web interface allows to explore IRFs in the context of a simple VAR(1) model with $n = 2$. (Select *VAR(1) Simulation* and tick *IRFs*.)

3.3 Covariance-stationary VARMA models

Let's come back to the infinite MA case (Eq. (3.4)):

$$y_t = \mu + \sum_{h=0}^{\infty} \Psi_h \eta_{t-h}.$$

For y_t to be covariance-stationary (and ergodic for the mean), it has to be the case that

$$\sum_{i=0}^{\infty} \|\Psi_i\| < \infty, \tag{3.6}$$

where $\|A\|$ denotes a norm of the matrix A (e.g. $\|A\| = \sqrt{tr(AA')}$). This notably implies that if y_t is stationary (and ergodic for the mean), then $\|\Psi_h\| \to 0$ when h gets large.

What should be satisfied by Φ_k's and Θ_k's for a VARMA-based process (Eq. (3.2)) to be stationary? The conditions will be similar to that we have in the univariate case. Let us introduce

the following notations:

$$\begin{aligned} y_t &= c + \underbrace{\Phi_1 y_{t-1} + \cdots + \Phi_p y_{t-p}}_{\text{AR component}} + \\ &\quad \underbrace{B\eta_t - \Theta_1 B\eta_{t-1} - \cdots - \Theta_q B\eta_{t-q}}_{\text{MA component}} \\ &\Leftrightarrow \underbrace{(I - \Phi_1 L - \cdots - \Phi_p L^p)}_{=\Phi(L)} y_t = c + \underbrace{(I - \Theta_1 L - \ldots - \Theta_q L^q)}_{=\Theta(L)} B\eta_t. \end{aligned} \tag{3.7}$$

Process y_t is stationary iff the roots of $\det(\Phi(z)) = 0$ are strictly outside the unit circle or, equivalently, iff the eigenvalues of

$$\Phi = \begin{bmatrix} \Phi_1 & \Phi_2 & \cdots & \Phi_p \\ I & 0 & \cdots & 0 \\ 0 & \ddots & 0 & 0 \\ 0 & 0 & I & 0 \end{bmatrix} \tag{3.8}$$

lie strictly within the unit circle. Hence, as is the case for univariate processes, the covariance-stationarity of a VARMA model depends only on the specification of its AR part.

Let's derive the first two unconditional moments of a (covariance-stationary) VARMA process.

Eq. (3.7) gives $\mathbb{E}(\Phi(L)y_t) = c$, therefore $\Phi(1)\mathbb{E}(y_t) = c$, or

$$\mathbb{E}(y_t) = (I - \Phi_1 - \cdots - \Phi_p)^{-1} c.$$

The autocovariances of y_t can be deduced from the infinite MA representation (Eq. (3.4)). We have:

$$\gamma_j \equiv \mathbb{C}ov(y_t, y_{t-j}) = \sum_{i=j}^{\infty} \Psi_i \Psi'_{i-j}.$$

(This infinite sum exists as soon as Eq. (3.6) is satisfied.)

Conditional means and autocovariances can also be deduced from Eq. (3.4). For $0 \leq h$ and $0 \leq h_1 \leq h_2$:

$$\begin{aligned} \mathbb{E}_t(y_{t+h}) &= \mu + \sum_{k=0}^{\infty} \Psi_{k+h} \eta_{t-k} \\ \mathbb{C}ov_t(y_{t+1+h_1}, y_{t+1+h_2}) &= \sum_{k=0}^{h_1} \Psi_k \Psi'_{k+h_2-h_1}. \end{aligned}$$

The previous formula implies in particular that the forecasting error $y_{t+h} - \mathbb{E}_t(y_{t+h})$ has a variance equal to:

$$\mathbb{V}ar_t(y_{t+1+h}) = \sum_{k=0}^{h} \Psi_k \Psi'_k.$$

Because the η_t are mutually and serially independent (and therefore uncorrelated), we have:

$$\mathbb{V}ar(\Psi_k \eta_{t-k}) = \mathbb{V}ar\left(\sum_{i=1}^{n} \psi_{k,i} \eta_{i,t-k}\right) = \sum_{i=1}^{n} \psi_{k,i} \psi'_{k,i},$$

where $\psi_{k,i}$ denotes the ith column of Ψ_k. This suggests the following decomposition of the variance of the forecast error (called **variance decomposition**):

$$\mathbb{V}ar_t(y_{t+1+h}) = \sum_{i=1}^{n} \underbrace{\sum_{k=0}^{h} \psi_{k,i}\psi_{k,i}'}_{\text{Contribution of } \eta_{i,t}}.$$

Let us now turn to the estimation of VAR models. Note that if there is an MA component (i.e., if we consider a VARMA model), then OLS regressions yield biased estimates (even for asymptotically large samples). Assume for instance that y_t follows a VARMA(1,1) model:

$$y_{i,t} = \phi_i y_{t-1} + \varepsilon_{i,t},$$

where ϕ_i is the ith row of Φ_1, and where $\varepsilon_{i,t}$ is a linear combination of η_t and η_{t-1}. Since y_{t-1} (the regressor) is correlated to η_{t-1}, it is also correlated to $\varepsilon_{i,t}$. The OLS regression of $y_{i,t}$ on y_{t-1} yields a biased estimator of ϕ_i (see Figure 3.2). Hence, SVARMA models cannot be consistently estimated by simple OLS regressions (contrary to VAR models, as we will see in the next section); instrumental-variable approaches can be employed to estimate SVARMA models (using past values of y_t as instruments, see, e.g., Gouriéroux et al. (2020)).

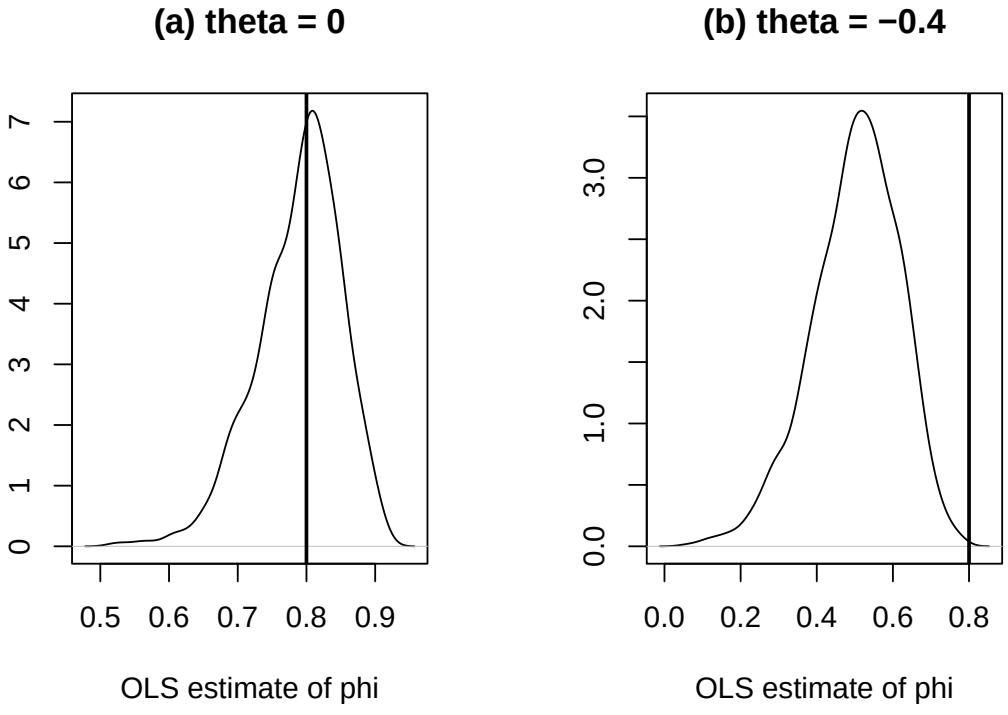


Figure 3.2: Monte Carlo distributions of the OLS estimator of the autoregressive coefficient in an ARMA(1,1) model, based on 1,000 samples of length 100: (a) $\theta = 0$ and (b) $\theta = -0.4$. The true autoregressive coefficient is $\phi = 0.8$, marked by the vertical black line.

3.4 VAR estimation

This section discusses the estimation of VAR models. Eq. (3.1) can be written:

$$y_t = c + \Phi(L)y_{t-1} + \varepsilon_t,$$

with $\Phi(L) = \Phi_1 + \Phi_2 L + \cdots + \Phi_p L^{p-1}$.

Consequently:

$$y_t \mid y_{t-1}, y_{t-2}, \dots, y_{-p+1} \sim \mathcal{N}(c + \Phi_1 y_{t-1} + \dots \Phi_p y_{t-p}, \Omega).$$

Using Hamilton (1994)'s notations, denote with Π the matrix $\begin{bmatrix} c & \Phi_1 & \Phi_2 & \dots & \Phi_p \end{bmatrix}'$ and with x_t the vector $\begin{bmatrix} 1 & y'_{t-1} & y'_{t-2} & \dots & y'_{t-p} \end{bmatrix}'$, we have:

$$y_t = \Pi' x_t + \varepsilon_t. \tag{3.9}$$

The previous representation is convenient to discuss the estimation of the VAR model, as parameters are gathered in two matrices only: Π and Ω.

Let us start with the case where the shocks are Gaussian.

Proposition 3.2 (MLE of a Gaussian VAR)**.** *If y_t follows a VAR(p) (see Definition 3.1), and if $\varepsilon_t \sim i.i.d.\ \mathcal{N}(0, \Omega)$, then the ML estimate of Π, denoted by $\hat{\Pi}$ (see Eq.* (3.9)*), is given by*

$$\hat{\Pi} = \left[\sum_{t=1}^{T} x_t x_t'\right]^{-1} \left[\sum_{t=1}^{T} y_t' x_t\right] = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y}, \tag{3.10}$$

where **X** *is the $T \times (1 + np)$ matrix whose t^{th} row is x_t and where* **y** *is the $T \times n$ matrix whose t^{th} row is y_t'.*

That is, the ith column of $\hat{\Pi}$ (b_i, say) is the OLS estimate of β_i, where:

$$y_{i,t} = \beta_i' x_t + \varepsilon_{i,t}, \tag{3.11}$$

(i.e., $\beta_i' = [c_i, \phi_{i,1}', \dots, \phi_{i,p}']'$).

The ML estimate of Ω, denoted by $\hat{\Omega}$, coincides with the sample covariance matrix of the n series of the OLS residuals in Eq. (3.11)*, i.e.:*

$$\hat{\Omega} = \frac{1}{T}\sum_{i=1}^{T} \hat{\varepsilon}_t \hat{\varepsilon}_t', \quad \text{with } \hat{\varepsilon}_t = y_t - \hat{\Pi}' x_t. \tag{3.12}$$

The asymptotic distributions of these estimators are the ones resulting from standard OLS formula.

Proof. See Appendix 8.4. □

As stated by Proposition 3.3, when the shocks are not Gaussian, then the OLS regressions still provide consistent estimates of the model parameters. However, since x_t correlates to ε_s for $s < t$, the OLS estimator $\mathbf{b}_i$ of β_i is biased in small sample. (That is also the case for the ML estimator.) Indeed, denoting by ε_i the $T \times 1$ vector of $\varepsilon_{i,t}$'s, and using the notations of b_i and β_i introduced in Proposition 3.2, we have:

$$\mathbf{b}_i = \beta_i + (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\varepsilon_i. \tag{3.13}$$

We have non-zero correlation between x_t and $\varepsilon_{i,s}$ for $s < t$ and, therefore, $\mathbb{E}[(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\varepsilon_i] \neq 0$.

However, when y_t is covariance stationary, then $\frac{1}{n}\mathbf{X}'\mathbf{X}$ converges to a positive definite matrix $\mathbf{Q}$, and $\frac{1}{n}X'\varepsilon_i$ converges to 0. Hence $\mathbf{b}_i \xrightarrow{p} \beta_i$. More precisely:

Proposition 3.3 (Asymptotic distribution of the OLS estimate)**.** *If y_t follows a VAR model, as defined in Definition 3.1, we have:*

$$\sqrt{T}(\mathbf{b}_i - \beta_i) = \underbrace{\left[\frac{1}{T}\sum_{t=p}^{T} x_t x_t'\right]^{-1}}_{\xrightarrow{p}\mathbf{Q}^{-1}} \underbrace{\sqrt{T}\left[\frac{1}{T}\sum_{t=1}^{T} x_t \varepsilon_{i,t}\right]}_{\xrightarrow{d}\mathcal{N}(0,\sigma_i^2\mathbf{Q})},$$

where $\sigma_i = \mathbb{V}ar(\varepsilon_{i,t})$ and where $\mathbf{Q} = plim\ \frac{1}{T}\sum_{t=p}^{T} x_t x_t'$ is given by:

$$\mathbf{Q} = \begin{bmatrix} 1 & \mu' & \mu' & \dots & \mu' \\ \mu & \gamma_0 + \mu\mu' & \gamma_1 + \mu\mu' & \dots & \gamma_{p-1} + \mu\mu' \\ \mu & \gamma_1 + \mu\mu' & \gamma_0 + \mu\mu' & \dots & \gamma_{p-2} + \mu\mu' \\ \vdots & \vdots & \vdots & \dots & \vdots \\ \mu & \gamma_{p-1} + \mu\mu' & \gamma_{p-2} + \mu\mu' & \dots & \gamma_0 + \mu\mu' \end{bmatrix}. \tag{3.14}$$

Proof. See Appendix 8.4. □

The following proposition extends the previous proposition and includes covariances between different β_i's as well as the asymptotic distribution of the ML estimates of Ω.

Proposition 3.4 (Asymptotic distribution of the OLS estimates)**.** *If y_t follows a VAR model, as defined in Definition 3.1, we have:*

$$\sqrt{T}\begin{bmatrix} vec(\hat{\Pi} - \Pi) \\ vec(\hat{\Omega} - \Omega) \end{bmatrix} \sim \mathcal{N}\left(0, \begin{bmatrix} \Omega \otimes \mathbf{Q}^{-1} & 0 \\ 0 & \Sigma_{22} \end{bmatrix}\right), \tag{3.15}$$

where the component of Σ_{22} corresponding to the covariance between $\hat{\sigma}_{i,j}$ and $\hat{\sigma}_{k,l}$ (for $i, j, l, m \in \{1, \dots, n\}^4$) is equal to $\sigma_{i,l}\sigma_{j,m} + \sigma_{i,m}\sigma_{j,l}$.

Proof. See Hamilton (1994), Appendix of Chapter 11. □

In practice, to use the previous proposition (for instance to implement Monte-Carlo simulations, see Section 8.5.1), Ω is replaced with $\hat{\Omega}$, $\mathbf{Q}$ is replaced with $\hat{\mathbf{Q}} = \frac{1}{T}\sum_{t=p}^{T} x_t x_t'$ and Σ with the matrix whose components are of the form $\hat{\sigma}_{i,l}\hat{\sigma}_{j,m} + \hat{\sigma}_{i,m}\hat{\sigma}_{j,l}$, where the $\hat{\sigma}_{i,l}$'s are the components of $\hat{\Omega}$.

The simplicity of the VAR framework and the tractability of its MLE open the way to convenient

econometric testing. Let's illustrate this with the likelihood ratio test.[1] The maximum value achieved by the MLE is

$$\log \mathcal{L}(Y_T; \widehat{\Pi}, \widehat{\Omega}) = -\frac{Tn}{2}\log(2\pi) + \frac{T}{2}\log\left|\widehat{\Omega}^{-1}\right| - \frac{1}{2}\sum_{t=1}^{T}\left[\widehat{\varepsilon}_t'\widehat{\Omega}^{-1}\widehat{\varepsilon}_t\right].$$

The last term is:

$$\begin{aligned}\sum_{t=1}^{T}\widehat{\varepsilon}_t'\widehat{\Omega}^{-1}\widehat{\varepsilon}_t &= \text{Tr}\left[\sum_{t=1}^{T}\widehat{\varepsilon}_t'\widehat{\Omega}^{-1}\widehat{\varepsilon}_t\right] = \text{Tr}\left[\sum_{t=1}^{T}\widehat{\Omega}^{-1}\widehat{\varepsilon}_t\widehat{\varepsilon}_t'\right] \\ &= \text{Tr}\left[\widehat{\Omega}^{-1}\sum_{t=1}^{T}\widehat{\varepsilon}_t\widehat{\varepsilon}_t'\right] = \text{Tr}\left[\widehat{\Omega}^{-1}\left(T\widehat{\Omega}\right)\right] = Tn.\end{aligned}$$

Therefore, the optimized log-likelihood is simply obtained by:

$$\log \mathcal{L}(Y_T; \widehat{\Pi}, \widehat{\Omega}) = -(Tn/2)\log(2\pi) + (T/2)\log\left|\widehat{\Omega}^{-1}\right| - Tn/2. \tag{3.16}$$

Assume that we want to test the null hypothesis that a set of variables follows a VAR(p_0) against the alternative specification of p_1 ($> p_0$). Let us denote by $\hat{L}_0$ and $\hat{L}_1$ the maximum log-likelihoods obtained with p_0 and p_1 lags, respectively. Under the null hypothesis (H_0: $p = p_0$), we have:

$$2\left(\hat{L}_1 - \hat{L}_0\right) = T\left(\log\left|\hat{\Omega}_1^{-1}\right| - \log\left|\hat{\Omega}_0^{-1}\right|\right) \sim \chi^2(n^2(p_1 - p_0)).$$

What precedes can be used to help determine the appropriate number of lags to use in the specification. In a VAR, using too many lags consumes numerous degrees of freedom: with p lags, each of the n equations in the VAR contains $n \times p$ coefficients plus the intercept term. Adding lags improve in-sample fit, but is likely to result in over-parameterization and affect the **out-of-sample** prediction performance.

To select appropriate lag length, **selection criteria** are often used. In the context of VAR models, using Eq. (3.16) (Gaussian case), we have for instance:

$$\begin{aligned}AIC &= cst + \log\left|\widehat{\Omega}\right| + \frac{2}{T}N \\ BIC &= cst + \log\left|\widehat{\Omega}\right| + \frac{\log T}{T}N,\end{aligned}$$

where $N = p \times n^2$.

```
## $selection
## AIC(n)  HQ(n)  SC(n) FPE(n)
##      3      3      2      3
##
```

[1] We implement a likelihood ratio test when we want to test whether a restricted version of a wider model is acceptable. For that, we maximize the restricted and unrestricted log-likelihood functions. Assume the respective maximum values are $\log \mathcal{L}_0$ (for the restricted model, called Model 0) and $\log \mathcal{L}_1$ (unrestricted model, Model 1). Under the null hypothesis (the restricted model is the true one), we have $2(\log \mathcal{L}_1 - \log \mathcal{L}_0) \sim \chi^2(r)$, where r is the number of restrictions imposed on Model 0. (Note that we necessarily have $\log \mathcal{L}_0 < \log \mathcal{L}_1$.)

```
## $criteria
##                 1          2          3          4          5           6
## AIC(n) -0.3394120 -0.4835525 -0.5328327 -0.5210835 -0.5141079 -0.49112812
## HQ(n)  -0.3017869 -0.4208439 -0.4450407 -0.4082080 -0.3761491 -0.32808581
## SC(n)  -0.2462608 -0.3283005 -0.3154798 -0.2416298 -0.1725534 -0.08747275
## FPE(n)  0.7121914  0.6165990  0.5869659  0.5939325  0.5981364  0.61210908
```

```
## [1] 0.9114892 0.9114892 0.6319554 0.4759403 0.4759403 0.3246995
```

3.5 Block exogeneity and Granger causality

3.5.1 Block exogeneity

Let's decompose y_t into two subvectors $y_t^{(1)}$ ($n_1 \times 1$) and $y_t^{(2)}$ ($n_2 \times 1$), with $y_t' = [{y_t^{(1)}}', {y_t^{(2)}}']$ (and therefore $n = n_1 + n_2$), such that:

$$\begin{bmatrix} y_t^{(1)} \\ y_t^{(2)} \end{bmatrix} = \begin{bmatrix} \Phi^{(1,1)} & \Phi^{(1,2)} \\ \Phi^{(2,1)} & \Phi^{(2,2)} \end{bmatrix} \begin{bmatrix} y_{t-1}^{(1)} \\ y_{t-1}^{(2)} \end{bmatrix} + \varepsilon_t.$$

Using, e.g., a likelihood ratio test, one can easily test for block exogeneity of $y_t^{(2)}$ (say). The null assumption can be expressed as $\Phi^{(2,1)} = 0$.

3.5.2 Granger Causality

Granger (1969) developed a method to explore **causal relationships** among variables. The approach consists in determining whether the past values of $y_{1,t}$ can help explain the current $y_{2,t}$ (beyond the information already included in the past values of $y_{2,t}$).

Formally, let us denote three information sets:

$$\begin{aligned} \mathcal{J}_{1,t} &= \left\{y_{1,t}, y_{1,t-1}, \ldots\right\} \\ \mathcal{J}_{2,t} &= \left\{y_{2,t}, y_{2,t-1}, \ldots\right\} \\ \mathcal{J}_{t} &= \left\{y_{1,t}, y_{1,t-1}, \ldots y_{2,t}, y_{2,t-1}, \ldots\right\}. \end{aligned}$$

We say that $y_{1,t}$ Granger-causes $y_{2,t}$ if

$$\mathbb{E}\left[y_{2,t} \mid \mathcal{J}_{2,t-1}\right] \neq \mathbb{E}\left[y_{2,t} \mid \mathcal{J}_{t-1}\right].$$

To get the intuition behind the testing procedure, consider the following bivariate VAR(p) process:

$$\begin{aligned} y_{1,t} &= c_1 + \Sigma_{i=1}^{p} \Phi_i^{(11)} y_{1,t-i} + \Sigma_{i=1}^{p} \Phi_i^{(12)} y_{2,t-i} + \varepsilon_{1,t} \\ y_{2,t} &= c_2 + \Sigma_{i=1}^{p} \Phi_i^{(21)} y_{1,t-i} + \Sigma_{i=1}^{p} \Phi_i^{(22)} y_{2,t-i} + \varepsilon_{2,t}, \end{aligned}$$

where $\Phi_k^{(ij)}$ denotes the element (i,j) of Φ_k. Then, $y_{1,t}$ is said not to Granger-cause $y_{2,t}$ if

$$\Phi_1^{(21)} = \Phi_2^{(21)} = \ldots = \Phi_p^{(21)} = 0.$$

The null and alternative hypotheses therefore are:

$$\begin{cases} H_0: & \Phi_1^{(21)} = \Phi_2^{(21)} = ... = \Phi_p^{(21)} = 0 \\ H_1: & \Phi_1^{(21)} \neq 0 \text{ or } \Phi_2^{(21)} \neq 0 \text{ or } ... \Phi_p^{(21)} \neq 0. \end{cases}$$

Loosely speaking, we reject H_0 if some of the coefficients on the lagged $y_{1,t}$'s are statistically significant. Formally, this can be tested using the F-test or asymptotic chi-square test. The F-statistic is

$$F = \frac{(RSS - USS)/p}{USS/(T-2p-1)},$$

where RSS is the Restricted sum of squared residuals and USS is the Unrestricted sum of squared residuals. Under H_0, the F-statistic is distributed as $\mathcal{F}(p, T-2p-1)$ (See Table 8.4).[2]

According to the following lines of code, the output gap Granger-causes inflation, but the reverse is not true:

```
## Granger causality test
##
## Model 1: infl ~ Lags(infl, 1:3) + Lags(y.gdp.gap, 1:3)
## Model 2: infl ~ Lags(infl, 1:3)
##   Res.Df Df      F   Pr(>F)
## 1    214
## 2    217 -3 3.9761 0.008745 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
## Granger causality test
##
## Model 1: y.gdp.gap ~ Lags(y.gdp.gap, 1:3) + Lags(infl, 1:3)
## Model 2: y.gdp.gap ~ Lags(y.gdp.gap, 1:3)
##   Res.Df Df      F Pr(>F)
## 1    214
## 2    217 -3 1.5451 0.2038
```

3.6 Standard identification techniques

3.6.1 The identification problem

In Section 3.4, we have seen how to estimate $\mathbb{V}ar(\varepsilon_t) = \Omega$ and the Φ_k matrices in the context of a VAR model. But the IRFs are functions of B and of the Φ_k's, not of Ω the Φ_k's (see Section 3.2). We have $\Omega = BB'$, which provides some restrictions on the components of B, but this is not sufficient to fully identify B. Indeed, seen a system of equations whose unknowns are the $b_{i,j}$'s (components of B), the system $\Omega = BB'$ contains only $n(n+1)/2$ linearly independent equations. For instance, for $n = 2$:

$$\begin{bmatrix} \omega_{11} & \omega_{12} \\ \omega_{12} & \omega_{22} \end{bmatrix} = \begin{bmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{bmatrix} \begin{bmatrix} b_{11} & b_{21} \\ b_{12} & b_{22} \end{bmatrix}$$

$$\Leftrightarrow \begin{bmatrix} \omega_{11} & \omega_{12} \\ \omega_{12} & \omega_{22} \end{bmatrix} = \begin{bmatrix} b_{11}^2 + b_{12}^2 & b_{11}b_{21} + b_{12}b_{22} \\ b_{11}b_{21} + b_{12}b_{22} & b_{22}^2 + b_{21}^2 \end{bmatrix}.$$

[2] We have $pF \underset{T\to\infty}{\to} \chi^2(p)$.

We then have 3 linearly independent equations but 4 unknowns. Therefore, B is not identified based on second-order moments. Additional restrictions are required to identify B. This section covers two standard identification schemes: **short-run** and **long-run** restrictions.

1. A **short-run restriction (SRR)** prevents a structural shock from affecting an endogenous variable contemporaneously.
   - It is easy to implement: the appropriate entries of B are set to 0.
   - A particular (popular) case is that of the **Cholesky, or recursive approach**.
   - Examples include Bernanke (1986), Sims (1986), Galí (1992), Ruibio-Ramírez et al. (2010).
2. A **long-run restriction (LRR)** prevents a structural shock from having a cumulative impact on one of the endogenous variables.
   - Additional computations are required to implement this. One needs to compute the cumulative effect of one of the structural shocks u_t on one of the endogenous variable.
   - Examples include Blanchard and Quah (1989), Faust and Leeper (1997), Galí (1999), Erceg et al. (2005), Christiano et al. (2007).

The two approaches can be combined (see, e.g., Gerlach and Smets (1995)).

3.6.2 A stylized example motivating short-run restrictions

Let us consider a simple example that could motivate short-run restrictions. Consider the following stylized macro model:

$$\begin{aligned} g_t &= \bar{g} - \lambda(i_{t-1} - \mathbb{E}_{t-1}\pi_t) + \underbrace{\sigma_d \eta_{d,t}}_{\text{demand shock}} && \text{(IS curve)} \\ \Delta\pi_t &= \beta(g_t - \bar{g}) + \underbrace{\sigma_\pi \eta_{\pi,t}}_{\text{cost push shock}} && \text{(Phillips curve)} \\ i_t &= \rho i_{t-1} + \left[\gamma_\pi \mathbb{E}_t \pi_{t+1} + \gamma_g (g_t - \bar{g})\right] \\ &\quad + \underbrace{\sigma_{mp}\eta_{mp,t}}_{\text{Mon. Pol. shock}} && \text{(Taylor rule)}, \end{aligned} \tag{3.17}$$

where:

$$\eta_t = \begin{bmatrix} \eta_{\pi,t} \\ \eta_{d,t} \\ \eta_{mp,t} \end{bmatrix} \sim i.i.d.\, \mathcal{N}(0, I). \tag{3.18}$$

Vector η_t is assumed to be a vector of structural shocks, mutually and serially independent. On date t:

- g_t is contemporaneously affected by $\eta_{d,t}$ only;
- π_t is contemporaneously affected by $\eta_{\pi,t}$ and $\eta_{d,t}$;
- i_t is contemporaneously affected by $\eta_{mp,t}$, $\eta_{\pi,t}$ and $\eta_{d,t}$.

System (3.17) could be rewritten as follows:

$$\begin{bmatrix} g_t \\ \pi_t \\ i_t \end{bmatrix} = \Phi(L) \begin{bmatrix} g_{t-1} \\ \pi_{t-1} \\ i_{t-1} + \end{bmatrix} + \underbrace{\underbrace{\begin{bmatrix} 0 & \bullet & 0 \\ \bullet & \bullet & 0 \\ \bullet & \bullet & \bullet \end{bmatrix}}_{=B} \eta_t}_{=\varepsilon_t}. \tag{3.19}$$

This is the **reduced-form** of the model. This representation suggests three additional restrictions on the entries of B; the latter matrix is therefore identified as soon as $\Omega = BB'$ is known (up to the signs of its columns).

3.6.3 Cholesky: a specific short-run-restriction situation

There are particular cases in which some well-known matrix decomposition of $\Omega = \mathbb{V}ar(\varepsilon_t)$ can be used to easily estimate some specific SVAR. This is the case for the so-called Cholesky decomposition. Consider the following context:

- A first shock (say, $\eta_{n_1,t}$) can affect instantaneously (i.e., on date t) only one of the endogenous variable (say, $y_{n_1,t}$);
- A second shock (say, $\eta_{n_2,t}$) can affect instantaneously (i.e., on date t) two endogenous variables, $y_{n_1,t}$ (the same as before) and $y_{n_2,t}$;
- ...

This implies

1. that column n_1 of B has only 1 non-zero entry (this is the n_1^{th} entry),
2. that column n_2 of B has 2 non-zero entries (the n_1^{th} and the n_2^{th} ones), etc.

Without loss of generality, we can set $n_1 = n$, $n_2 = n - 1$, etc. In this context, matrix B is lower triangular. The Cholesky decomposition of Ω_ε then provides an appropriate estimate of B, since this matrix decomposition yields to a lower triangular matrix satisfying:

$$\Omega_\varepsilon = BB'.$$

For instance, Dedola and Lippi (2005) estimate 5 structural VAR models for the US, the UK, Germany, France and Italy to analyse the monetary-policy transmission mechanisms. They estimate SVAR(5) models over the period 1975-1997. The shock-identification scheme is based on Cholesky decompositions, the ordering of the endogenous variables being: the industrial production, the consumer price index, a commodity price index, the short-term rate, monetary aggregate and the effective exchange rate (except for the US). This ordering implies that monetary policy reacts to the shocks affecting the first three variables but that the latter react to monetary policy shocks with a one-period lag only.

The Cholesky approach can be employed when one is interested in one specific structural shock. This is the case, e.g., of Christiano et al. (1996). Their identification is based on the following relationship between ε_t and η_t:

$$\begin{bmatrix} \varepsilon_{S,t} \\ \varepsilon_{r,t} \\ \varepsilon_{F,t} \end{bmatrix} = \begin{bmatrix} B_{SS} & 0 & 0 \\ B_{rS} & B_{rr} & 0 \\ B_{FS} & B_{Fr} & B_{FF} \end{bmatrix} \begin{bmatrix} \eta_{S,t} \\ \eta_{r,t} \\ \eta_{F,t} \end{bmatrix},$$

where S, r and F respectively correspond to *slow-moving variables*, the policy variable (short-term rate) and *fast-moving variables*. While $\eta_{r,t}$ is scalar, $\eta_{S,t}$ and $\eta_{F,t}$ may be vectors. The space spanned by $\varepsilon_{S,t}$ is the same as that spanned by $\eta_{S,t}$. As a result, because $\varepsilon_{r,t}$ is a linear combination of $\eta_{r,t}$ and $\eta_{S,t}$ (which are $\perp$), it comes that the $B_{rr}\eta_{r,t}$'s are the (population) residuals in the regression of $\varepsilon_{r,t}$ on $\varepsilon_{S,t}$. Because $\mathbb{V}ar(\eta_{r,t}) = 1$, B_{rr} is given by the square root of the variance of $B_{rr}\eta_{r,t}$. $B_{F,r}$ is finally obtained by regressing the components of $\varepsilon_{F,t}$ on the estimates of $\eta_{r,t}$.

An equivalent approach consists in computing the Cholesky decomposition of BB' and the contemporaneous impacts of the monetary policy shock (on the n endogenous variables) are the components of the column of B corresponding to the policy variable.

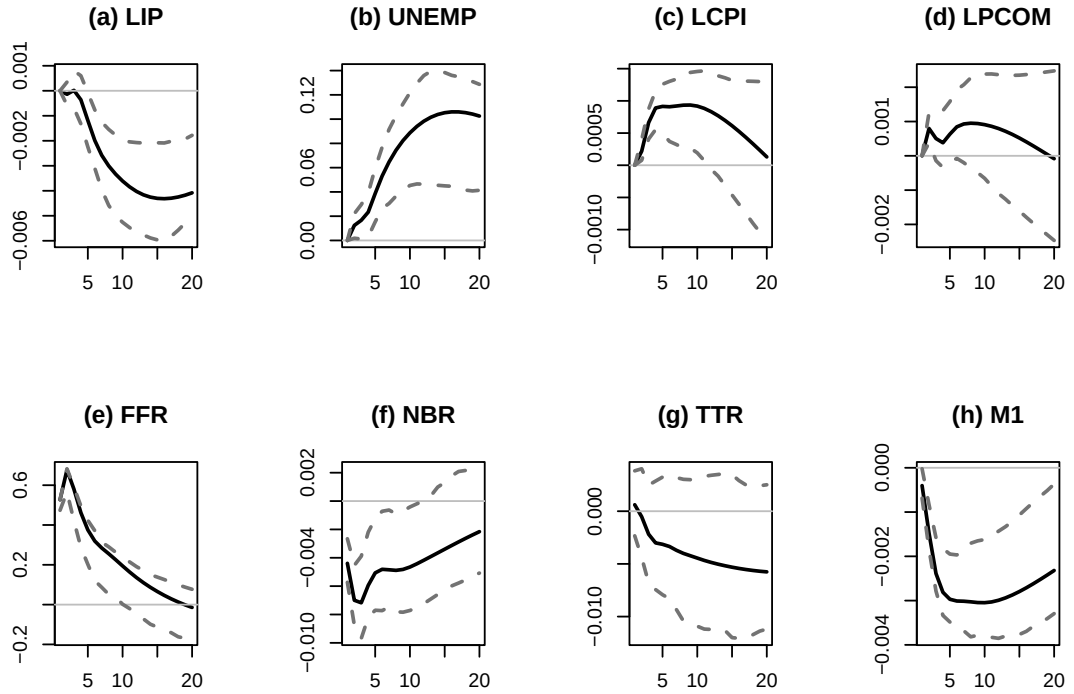


Figure 3.3: Responses of U.S. macroeconomic variables to a contractionary monetary-policy shock identified recursively as in Christiano, Eichenbaum, and Evans (1996). Solid lines are point estimates; dashed lines are 90% pointwise confidence bands obtained by bootstrapping the estimated VAR(3). Monthly sample: January 1965–June 1995.

3.6.4 Long-run restrictions

Let us now turn to **Long-run restrictions**. Such a restriction concerns the long-run influence of a shock on an endogenous variable. Let us consider for instance a structural shock that is assumed to have no "long-run influence" on GDP. How to express this? The long-run change in GDP can be expressed as $GDP_{t+h} - GDP_t$, with h large. Note further that:

$$GDP_{t+h} - GDP_t = \Delta GDP_{t+h} + \Delta GDP_{t+h-1} + \cdots + \Delta GDP_{t+1}.$$

Hence, the fact that a given structural shock ($\eta_{i,t}$, say) has no long-run influence on GDP means that

$$\lim_{h\to\infty} \frac{\partial GDP_{t+h}}{\partial \eta_{i,t}} = \lim_{h\to\infty} \frac{\partial}{\partial \eta_{i,t}} \left(\sum_{k=1}^{h} \Delta GDP_{t+k} \right) = 0.$$

This long-run effect can be formulated as a function of B and of the matrices Φ_i when y_t (including ΔGDP_t) follows a VAR process.

Without loss of generality, we will only consider the VAR(1) case. Indeed, one can always write a VAR(p) as a VAR(1): for that, stack the last p values of vector y_t in vector $y_t^* =$

$[y_t', ..., y_{t-p+1}']'$; Eq. (3.1) can then be rewritten in its **companion form**:

$$y_t^* = \underbrace{\begin{bmatrix} c \\ 0 \\ \vdots \\ 0 \end{bmatrix}}_{=c^*} + \underbrace{\begin{bmatrix} \Phi_1 & \Phi_2 & \cdots & \Phi_p \\ I & 0 & \cdots & 0 \\ 0 & \ddots & 0 & 0 \\ 0 & 0 & I & 0 \end{bmatrix}}_{=\Phi} y_{t-1}^* + \underbrace{\begin{bmatrix} \varepsilon_t \\ 0 \\ \vdots \\ 0 \end{bmatrix}}_{\varepsilon_t^*}, \tag{3.20}$$

where matrices Φ and $\Omega^* = \mathbb{V}ar(\varepsilon_t^*)$ are of dimension $np \times np$; Ω^* is filled with zeros, except the $n \times n$ upper-left block that is equal to $\Omega = \mathbb{V}ar(\varepsilon_t)$. (Matrix Φ had been introduced in Eq. (3.8).)

Let us then focus on the VAR(1) case:

$$\begin{aligned} y_t &= c + \Phi y_{t-1} + \varepsilon_t \\ &= c + \varepsilon_t + \Phi(c + \varepsilon_{t-1}) + ... + \Phi^k(c + \varepsilon_{t-k}) + ... \\ &= \mu + \varepsilon_t + \Phi\varepsilon_{t-1} + ... + \Phi^k\varepsilon_{t-k} + ... \\ &= \mu + B\eta_t + \Phi B\eta_{t-1} + ... + \Phi^k B\eta_{t-k} + ..., \end{aligned}$$

which is the Wold representation of y_t.

The sequence of shocks $\{\eta_t\}$ determines the sequence $\{y_t\}$. What if $\{\eta_t\}$ is replaced with $\{\tilde{\eta}_t\}$, where $\tilde{\eta}_t = \eta_t$ if $t \neq s$ and $\tilde{\eta}_s = \eta_s + \gamma$? Assume $\{\tilde{y}_t\}$ is the associated "perturbated" sequence. We have $\tilde{y}_t = y_t$ if $t < s$. For $t \geq s$, the Wold decomposition of $\{\tilde{y}_t\}$ implies:

$$\tilde{y}_t = y_t + \Phi^{t-s} B\gamma.$$

Therefore, the cumulative impact of γ on $\tilde{y}_t$ will be (for $t \geq s$):

$$\begin{aligned} (\tilde{y}_t - y_t) + (\tilde{y}_{t-1} - y_{t-1}) + \cdots + (\tilde{y}_s - y_s) &= \\ (Id + \Phi + \Phi^2 + \cdots + \Phi^{t-s})B\gamma. \end{aligned} \tag{3.21}$$

Consider a shock on $\eta_{1,t}$, with a magnitude of 1. This shock corresponds to $\gamma = [1, 0, ..., 0]'$. Given Eq. (3.21), the long-run cumulative effect of this shock on the endogenous variables is given by:

$$\underbrace{(Id + \Phi + ... + \Phi^k + ...)}_{=(Id-\Phi)^{-1}} B \begin{bmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix},$$

that is the first column of the $n \times n$ matrix $\Theta \equiv (Id - \Phi)^{-1}B$.

In this context, consider the following long-run restriction: *"the jth structural shock has no cumulative impact on the ith endogenous variable"*. It is equivalent to

$$\Theta_{ij} = 0,$$

where Θ_{ij} is the element (i, j) of Θ.

Blanchard and Quah (1989) have implemented such long-run restrictions in a small-scale VAR. Two variables are considered: GDP and unemployment. Consequently, the VAR is affected by two types of shocks. Specifically, authors want to identify **supply shocks** (that can have

a permanent effect on output) and **demand shocks** (that cannot have a permanent effect on output).[3]

The original dataset of Blanchard and Quah (1989) is quarterly and their VAR features eight lags. The application below uses the open `IdSS` reconstruction based on public BEA real-GNP and BLS unemployment series, transformed in the same spirit as the original variables.

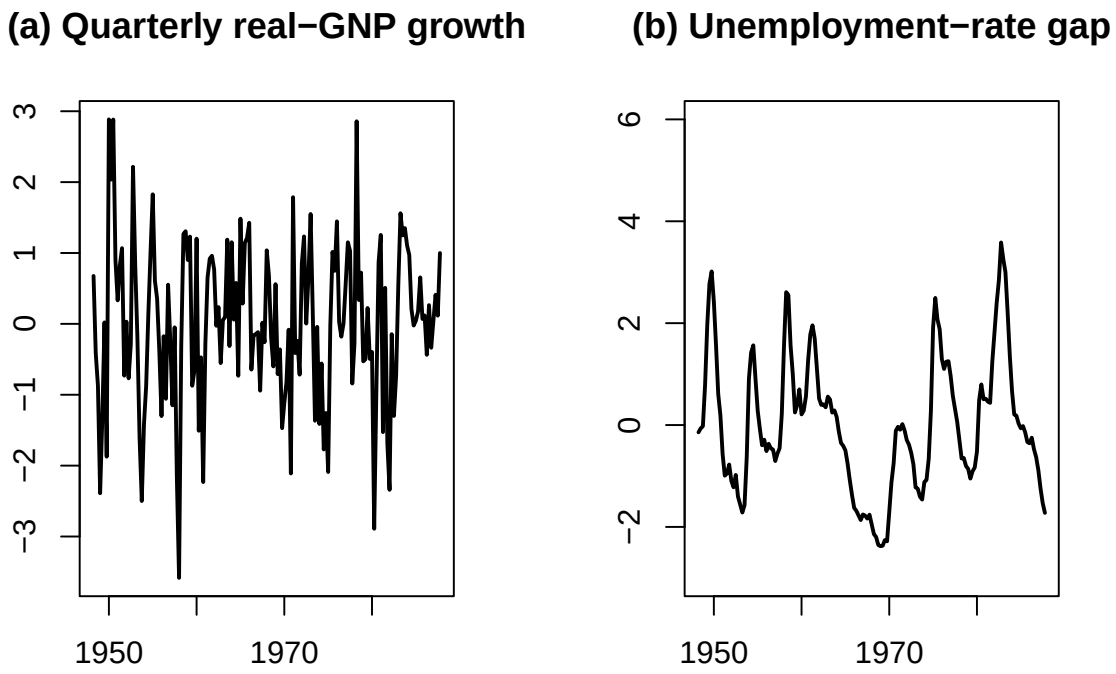


Figure 3.4: Open reconstruction of the quarterly U.S. variables used in the Blanchard–Quah long-run-restriction example: (a) demeaned real-GNP log growth and (b) linearly detrended unemployment. Sample: 1948 Q2–1987 Q4. Sources: U.S. Bureau of Economic Analysis and Bureau of Labor Statistics via FRED.

Estimate a reduced-form VAR(8) model:

Now, let us define a loss function (`loss`) that is equal to zero if (a) $BB' = \Omega$ and (b) the element (1,1) of $\Theta = (Id - \Phi)^{-1}B$ is equal to zero:

```
## [1]  0.8790657 -0.2270598  0.1606735  0.1765915
```

(Note: one can use that type of approach, based on a loss function, to mix short- and long-run restrictions.)

Figure 3.5 displays the resulting IRFs. Note that, for GDP, we cumulate the GDP growth IRF, so as to have the response of the GDP in level.

```
##              Dgdp       unemp
## Dgdp    0.7985735 -0.17122692  0.7985725 -0.17122692
## unemp -0.1712269  0.08274171 -0.1712269  0.08274071
```

[3]The motivation of the authors regarding their long-run restrictions can be obtained from a traditional Keynesian view of fluctuations. The authors propose a variant of a model from Fischer (1977).

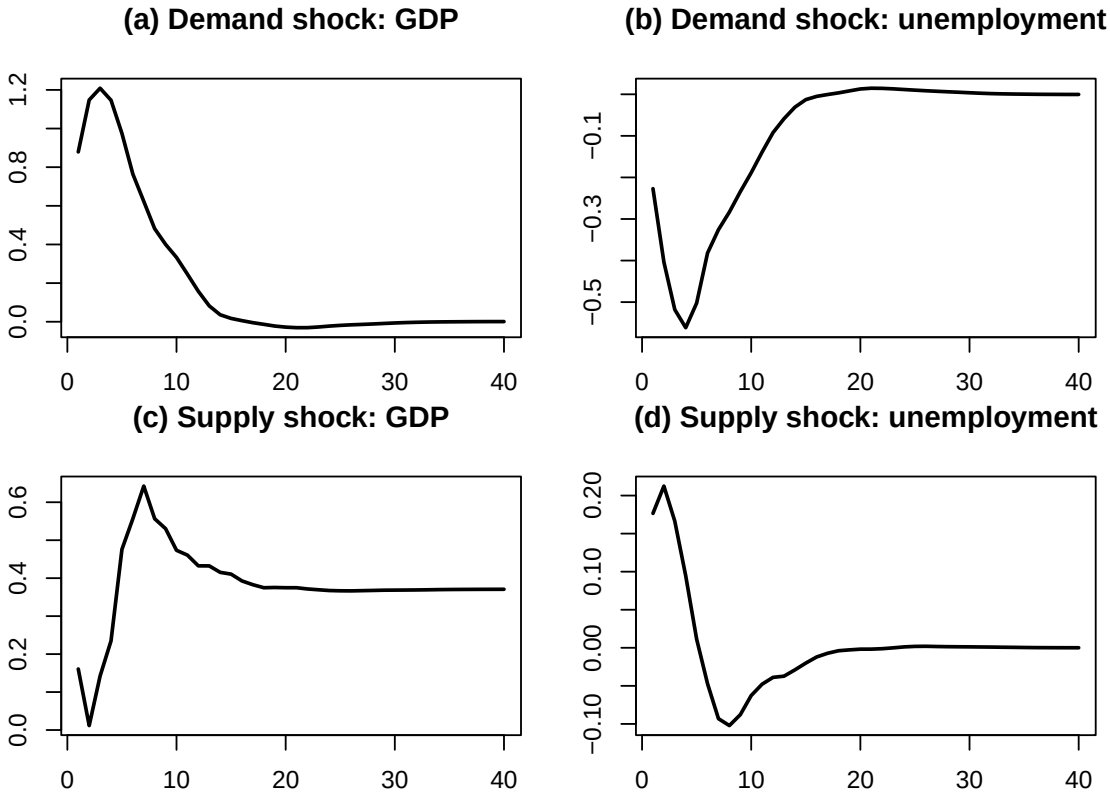


Figure 3.5: Impulse responses in the Blanchard-Quah model: (a) GDP level to a demand shock; (b) unemployment to a demand shock; (c) GDP level to a supply shock; and (d) unemployment to a supply shock. GDP responses are cumulated from the estimated growth-rate responses.

3.7 Exercise

Exercise 3.1 (VAR lag selection, stability, and recursive identification)**. PS3 | Advanced | Multivariate analysis**
Review Section 3.4.

A quarterly VAR contains inflation, the output gap, and the short-term interest rate ($T = 224$). The criteria for lags 1–10 attain their minima at lag 6 for AIC, lag 3 for HQ, and lag 2 for BIC.

1. State the lag choice implied by each criterion.
2. If BIC is followed, explain conditional Gaussian maximum-likelihood estimation of the VAR(2).
3. Explain how the 6×6 companion matrix is formed and how its eigenvalues determine covariance stationarity.

The fitted coefficient matrices are

$$\widehat{\Phi}_1 = \begin{bmatrix} .627 & .054 & .077 \\ .034 & 1.255 & -.087 \\ .143 & .560 & .687 \end{bmatrix}, \qquad \widehat{\Phi}_2 = \begin{bmatrix} .271 & .010 & -.065 \\ -.031 & -.298 & .053 \\ .027 & -.465 & .203 \end{bmatrix}.$$

We seek supply, demand, and monetary-policy shocks. A monetary-policy shock has no contemporaneous effect on activity or inflation, and a demand shock has no contemporaneous effect on activity.

4. Are these three zero restrictions sufficient for exact identification under the stated recursive structure?
5. Give the ordering of variables and shocks that makes the impact matrix lower triangular, and explain it economically.

3.7.1 Exercise from the 2024 exam

Exercise 3.2 (VAR forecasts and structural impacts)**. EXAM-2024 | Exam | True/false**
Review Section 3.

Consider the three-variable VAR(3)

$$y_t = c + \Phi_1 y_{t-1} + \Phi_2 y_{t-2} + \Phi_3 y_{t-3} + B\eta_t,$$

where $\eta_t \overset{i.i.d.}{\sim} \mathcal{N}(0, I_3)$.

State whether each assertion is true or false and justify your answer.

1. $\mathbb{E}_t(y_{t+2}) = c + \Phi_1 c$.
2. $\mathbb{V}ar_t(y_{t+1}) = BB'$.
3. Knowing $\Omega = BB'$ is sufficient to recover an unrestricted B.
4. The contemporaneous impacts of a unit $\eta_{2,t}$ shock on the three variables are $B_{1,2}$, $B_{2,2}$, and $B_{3,2}$.
5. If B is lower triangular, an $\eta_{3,t}$ shock has no contemporaneous effect on $y_{1,t}$ or $y_{2,t}$.

3.7.2 Exercises from the 2025 exam

Exercise 3.3 (A VAR(2) in companion form)**. EXAM-2025 | Exam | True/false**
Review Section 3.

Consider the covariance-stationary, zero-intercept VAR(2)

$$y_t = \Phi_1 y_{t-1} + \Phi_2 y_{t-2} + \varepsilon_t, \qquad \varepsilon_t \overset{i.i.d.}{\sim} \mathcal{N}(\mathbf{0}_{3\times 1}, \Omega),$$

where y_t is 3×1. Define

$$y_t^* = \begin{pmatrix} y_t \\ y_{t-1} \end{pmatrix}, \qquad \Phi = \begin{pmatrix} \Phi_1 & \Phi_2 \\ I_3 & 0_3 \end{pmatrix}, \qquad \Omega^* = \begin{pmatrix} \Omega & \mathbf{0}_{3\times 3} \\ \mathbf{0}_{3\times 3} & \mathbf{0}_{3\times 3} \end{pmatrix}.$$

State whether each assertion is true or false and justify your answer.

1. $\mathbb{E}(y_t) = \mathbf{0}_{3\times 1}$.
2. $\mathbb{V}ar(y_t)$ is the upper-left 3×3 block of $\sum_{j=0}^{\infty} \Phi^j \Omega^* (\Phi^j)'$.
3. $\mathbb{E}_t(y_{t+2}) = \Phi_1 y_t + \Phi_2 y_{t-2}$.
4. $\mathbb{E}_t(y_{t+h})$ is the upper 3×1 block of $\Phi^h y_t^*$.
5. The eigenvalues of Φ have modulus strictly larger than one.

Exercise 3.4 (Granger causality and forecasts in a VAR(1))**. EXAM-2025 | Exam | True/false**
Review Section 3.5.

Consider the VAR(1)

$$y_t = \Phi y_{t-1} + \varepsilon_t, \qquad \Phi = \begin{pmatrix} 0.9 & 0 \\ 0.6 & 0.8 \end{pmatrix}, \qquad \varepsilon_t \overset{i.i.d.}{\sim} \mathcal{N}(\mathbf{0}, I_2),$$

where $y_t = (y_{1,t}, y_{2,t})'$. Conditional moments at date t use current and past information.

State whether each assertion is true or false and justify your answer.

1. $y_{1,t}$ is covariance stationary.
2. $y_{2,t}$ Granger-causes $y_{1,t}$.
3. $y_{1,t}$ Granger-causes $y_{2,t}$.
4. $\mathbb{E}_t(y_{t+h}) = \Phi^h y_t$ for every $h \geq 0$.
5. $\mathbb{V}ar_t(y_{t+2}) = I_2 + \Phi\Phi' + \Phi^2(\Phi^2)'$.

Chapter 4

Forecasting

Forecasting has always been an important part of the time series field (De Gooijer and Hyndman (2006)). Macroeconomic forecasts are done in many places: Public Administration (notably Treasuries), Central Banks, International Institutions (e.g. IMF, OECD), banks, big firms. These institutions are interested in the **point estimates** ($\sim$ most likely value) of the variable of interest. They also sometimes need to measure the **uncertainty** ($\sim$ dispersion of likely outcomes) associated to the point estimates.[1]

Forecasts produced by professional forecasters are available on these web pages:

- Philly Fed Survey of Professional Forecasters.
- ECB Survey of Professional Forecasters.
- IMF World Economic Outlook.
- OECD Global Economic Outlook.
- European Commission Economic Forecasts.

How to formalize the forecasting problem? Assume the current date is t. We want to forecast the value that variable y_t will take on date $t+1$ (i.e., y_{t+1}) based on the observation of a set of variables gathered in vector x_t (x_t may contain lagged values of y_t).

The forecaster aims at minimizing (a function of) the forecast error. It is usal to consider the following (quadratic) loss function:

$$\underbrace{\mathbb{E}([y_{t+1} - y_{t+1}^*]^2)}_{\text{Mean square error (MSE)}}$$

where y_{t+1}^* is the forecast of y_{t+1} (function of x_t).

Proposition 4.1 (Smallest MSE)**.** *The smallest MSE is obtained with the expectation of y_{t+1} conditional on x_t.*

Proof. See Appendix 8.4. □

[1] In its inflation report, the Bank of England displays charts showing the conditional distribution of future inflation, called fan charts. This fan charts show the uncertainty associated with future inflation. See this page.

Proposition 4.2. *Among the class of linear forecasts, the smallest MSE is obtained with the linear projection of y_{t+1} on x_t. This projection, denoted by $\widehat{P}(y_{t+1}|x_t) := \alpha' x_t$, satisfies:*

$$\mathbb{E}\left([y_{t+1} - \alpha' x_t]x_t\right) = \mathbf{0}. \tag{4.1}$$

Proof. Consider the function $f : \alpha \to \mathbb{E}\left([y_{t+1} - \alpha' x_t]^2\right)$. We have:

$$f(\alpha) = \mathbb{E}\left(y_{t+1}^2 - 2y_t x_t'\alpha + \alpha' x_t x_t'\alpha]\right).$$

We have $\partial f(\alpha)/\partial\alpha = \mathbb{E}(-2y_{t+1}x_t + 2x_t x_t'\alpha)$. The function is minimised for $\partial f(\alpha)/\partial\alpha = 0$. □

Eq. (4.1) implies that $\mathbb{E}\left(y_{t+1}x_t\right) = \mathbb{E}\left(x_t x_t'\right)\alpha$. (Note that $x_t x_t'\alpha = x_t(x_t'\alpha) = (\alpha' x_t)x_t'$.) Hence, if $\mathbb{E}\left(x_t x_t'\right)$ is nonsingular,

$$\alpha = \left[\mathbb{E}\left(x_t x_t'\right)\right]^{-1}\mathbb{E}\left(y_{t+1}x_t\right). \tag{4.2}$$

The MSE then is:

$$\mathbb{E}([y_{t+1} - \alpha' x_t]^2) = \mathbb{E}(y_{t+1}^2) - \mathbb{E}\left(y_{t+1}x_t'\right)\left[\mathbb{E}\left(x_t x_t'\right)\right]^{-1}\mathbb{E}\left(x_t y_{t+1}\right).$$

Consider the regression $y_{t+1} = \beta' \mathbf{x}_t + \varepsilon_{t+1}$. The OLS estimate is:

$$\mathbf{b} = \underbrace{\left[\frac{1}{T}\sum_{i=1}^{T}\mathbf{x}_t\mathbf{x}_t'\right]}_{\mathbf{m}_1}^{-1}\underbrace{\left[\frac{1}{T}\sum_{i=1}^{T}\mathbf{x}_t' y_{t+1}\right]}_{\mathbf{m}_2}.$$

If $\{x_t, y_t\}$ is covariance-stationary and ergodic for the second moments then the sample moments ($\mathbf{m}_1$ and $\mathbf{m}_2$) converges in probability to the associated population moments and $\mathbf{b} \xrightarrow{p} \alpha$ (where α is defined in Eq. (4.2)).

Example 4.1 (Forecasting an MA(q) process)**.** Consider the MA(q) process:

$$y_t = \mu + \varepsilon_t + \theta_1\varepsilon_{t-1} + \cdots + \theta_q\varepsilon_{t-q},$$

where $\{\varepsilon_t\}$ is a white noise sequence (Def. 1.1).

We have:[a]

$$\mathbb{E}(y_{t+h}|\varepsilon_t, \varepsilon_{t-1}, \dots) = \begin{cases} \mu + \theta_h\varepsilon_t + \cdots + \theta_q\varepsilon_{t-q+h} & for \quad h \in [1, q] \\ \mu & for \quad h > q \end{cases}$$

and

$$\mathbb{V}ar(y_{t+h}|\varepsilon_t, \varepsilon_{t-1}, \dots) = \mathbb{E}\left([y_{t+h} - \mathbb{E}(y_{t+h}|\varepsilon_t, \varepsilon_{t-1}, \dots)]^2\right) = \begin{cases} \sigma^2(1 + \theta_1^2 + \cdots + \theta_{h-1}^2) & for \quad h \in [1, q] \\ \sigma^2(1 + \theta_1^2 + \cdots + \theta_q^2) & for \quad h > q. \end{cases}$$

[a] The present reasoning relies on the assumption that the ε_t's are observed. But this is generally not the case in practice, where the ε_t's generally have to be estimated.

Example 4.2 (Forecasting an AR(p) process)**.** (See this web interface.) Consider the AR(p) process:

$$y_t = c + \phi_1 y_{t-1} + \phi_2 y_{t-2} + \cdots + \phi_p y_{t-p} + \varepsilon_t,$$

where $\{\varepsilon_t\}$ is a white noise sequence (Def. 1.1).

Using the notation of Eq. (2.4), we have:

$$\mathbf{y}_t - \mu = F(\mathbf{y}_{t-1} - \mu) + \xi_t,$$

with $\mu = [\mu, \dots, \mu]'$ (μ is defined in Eq. (2.9)). Hence:

$$\mathbf{y}_{t+h} - \mu = \xi_{t+h} + F\xi_{t+h-1} + \cdots + F^{h-1}\xi_{t+1} + F^h(\mathbf{y}_t - \mu).$$

Therefore:

$$\begin{aligned} \mathbb{E}(\mathbf{y}_{t+h}|y_t, y_{t-1}, \dots) &= \mu + F^h(\mathbf{y}_t - \mu) \\ \mathbb{V}ar\left([\mathbf{y}_{t+h} - \mathbb{E}(\mathbf{y}_{t+h}|y_t, y_{t-1}, \dots)]\right) &= \Sigma + F\Sigma F' + \cdots + F^{h-1}\Sigma(F^{h-1})', \end{aligned}$$

where:

$$\Sigma = \begin{bmatrix} \sigma^2 & 0 & \dots \\ 0 & 0 & \\ \vdots & & \ddots \end{bmatrix}.$$

Alternative approach: Taking the (conditional) expectations of both sides of

$$y_{t+h} - \mu = \phi_1(y_{t+h-1} - \mu) + \phi_2(y_{t+h-2} - \mu) + \cdots + \phi_p(y_{t-p} - \mu) + \varepsilon_{t+h},$$

we obtain:

$$\begin{aligned} \mathbb{E}(y_{t+h}|y_t, y_{t-1}, \dots) = \mu &+ \phi_1\left(\mathbb{E}[y_{t+h-1}|y_t, y_{t-1}, \dots] - \mu\right) + \\ &\phi_2\left(\mathbb{E}[y_{t+h-2}|y_t, y_{t-1}, \dots] - \mu\right) + \cdots + \\ &\phi_p\left(\mathbb{E}[y_{t+h-p}|y_t, y_{t-1}, \dots] - \mu\right), \end{aligned}$$

which can be exploited recursively.

The recursion begins with $\mathbb{E}(y_{t-k}|y_t, y_{t-1}, \dots) = y_{t-k}$ (for any $k \geq 0$).

Example 4.3 (Forecasting an ARMA(p,q) process)**.** Consider the process:

$$y_t = c + \phi_1 y_{t-1} + \cdots + \phi_p y_{t-p} + \varepsilon_t + \theta_1 \varepsilon_{t-1} + \cdots + \theta_q \varepsilon_{t-q}, \tag{4.3}$$

where $\{\varepsilon_t\}$ is a white noise sequence (Def. 1.1). We assume that the MA part of the process is invertible (see Eq. (2.27)), which implies that the information contained in $\{y_t, y_{t-1}, y_{t-2}, \dots\}$ is identical to that in $\{\varepsilon_t, \varepsilon_{t-1}, \varepsilon_{t-2}, \dots\}$.

While one could use a recursive algorithm to compute the conditional mean (as in Example 4.2), it is convenient to employ the Wold decomposition of this process (see Theorem 2.2 and Prop. 2.7 for the computation of the ψ_i's in the context of ARMA processes):

$$y_t = \mu + \sum_{i=0}^{+\infty} \psi_i \varepsilon_{t-i}.$$

This implies:

$$\begin{aligned} y_{t+h} &= \mu + \sum_{i=0}^{h-1} \psi_i \varepsilon_{t+h-i} + \sum_{i=h}^{+\infty} \psi_i \varepsilon_{t+h-i} \\ &= \mu + \sum_{i=0}^{h-1} \psi_i \varepsilon_{t+h-i} + \sum_{i=0}^{+\infty} \psi_{i+h} \varepsilon_{t-i}. \end{aligned}$$

Since $\mathbb{E}(y_{t+h}|y_t, y_{t-1}, \dots) = \mu + \sum_{i=0}^{+\infty} \psi_{i+h}\varepsilon_{t-i}$, we get:

$$\mathbb{V}ar(y_{t+h}|y_t, y_{t-1}, \dots) = \mathbb{V}ar\left(\sum_{i=0}^{h-1} \psi_i \varepsilon_{t+h-i}\right) = \sigma^2 \sum_{i=0}^{h-1} \psi_i^2.$$

How to use the previous formulas in practice?

One has first to select a specification and to estimate the model. Two methods to determine relevant specifications:

a. Information criteria (see Definition 2.11).
b. Box-Jenkins approach.

Box and Jenkins (1976) have proposed an approach that is now widely used.

1. Data transformation. The data should be transformed to "make them stationary". To do so, one can e.g. take logarithms, take changes in the considered series, remove (deterministic) trends.
2. Select p and q. This can be based on the PACF approach (see Section 2.4), or on selection criteria (see Definition 2.11).
3. Estimate the model parameters. See Section 2.8.
4. Check that the estimated model is consistent with the data. See below.

Assessing the performances of a forecasting model

Once one has fitted a model on a given dataset (of length T, say), one compute MSE (mean square errors) to evaluate the performance of the model. But this MSE is the **in-sample** one. It is

easy to reduce in-sample MSE. Typically, if the model is estimated by OLS, adding covariates mechanically reduces the MSE. That is, even if additional data are irrelevant, the R^2 of the regression increases. Adding irrelevant variables increases the (in-sample) R^2 but is bound to increase the **out-of-sample** MSE.

Therefore, it is important to analyse **out-of-sample** performances of the forecasting model:

a. Estimate a model on a sample of reduced size ($1, \dots, T^*$, with $T^* < T$)
b. Use the remaining available periods ($T^* + 1, \dots, T$) to compute **out-of-sample** forecasting errors (and compute their MSE). In an out-of-sample exercise, it is important to make sure that the data used to produce a forecasts (as of date T^*) where indeed available on date T^*.

Diebold-Mariano test

How to compare different forecasting approaches? Diebold and Mariano (1995) have proposed a simple test to address this question.

Assume that you want to compare approaches A and B. You have historical data sets and you have implemented both approaches in the past, providing you with two sets of forecasting errors: $\{e_t^A\}_{t=1,\dots,T}$ and $\{e_t^B\}_{t=1,\dots,T}$.

It may be the case that your forecasts serve a specific purpose and that, for instance, you dislike positive forecasting errors and you care less about negative errors. We assume you are able to formalise this by means of a **loss function** $L(e)$. For instance:

- If you dislike large positive errors, you may set $L(e) = \exp(e)$.
- If you are concerned about both positive and negative errors (indifferently), you may set $L(e) = e^2$ (standard approach).

Let us define the sequence $\{d_t\}_{t=1,\dots,T} \equiv \{L(e_t^A) - L(e_t^B)\}_{t=1,\dots,T}$ and assume that this sequence is covariance stationary. We consider the following null hypothesis: $H_0 : \bar{d} = 0$, where $\bar{d}$ denotes the population mean of the d_ts. Under H_0 and under the assumption of covariance-stationarity of d_t, we have (Theorem 1.1):

$$\sqrt{T}\bar{d}_T \xrightarrow{d} \mathcal{N}\left(0, \sum_{j=-\infty}^{+\infty} \gamma_j\right),$$

where the γ_js are the autocovariances of d_t.

Hence, assuming that $\widehat{\sigma}^2$ is a consistent estimate of $\sum_{j=-\infty}^{+\infty} \gamma_j$ (for instance the one given by the Newey-West formula, see Def. (1.6)), we have, under H_0:

$$DM_T := \sqrt{T}\frac{\bar{d}_T}{\sqrt{\widehat{\sigma}^2}} \xrightarrow{d} \mathcal{N}(0,1).$$

DM_T is the test statistics. For a test of size α, the critical region is:[2]

$$]-\infty, -\Phi^{-1}(1-\alpha/2)] \cup [\Phi^{-1}(1-\alpha/2), +\infty[,$$

where Φ is the c.d.f. of the standard normal distribution.

[2] This ShinyApp application illustrates the notion of statistical test (illustrating the p-value and the cirtical region, in particular).

Example 4.4 (Forecasting Swiss GDP growth)**.** We use a long historical time series of the Swiss GDP growth taken from the Jordà et al. (2017) dataset (see Figure 1.3, and Example 2.4).

We want to forecast this GDP growth. We envision two specifications : an AR(1) specification (the one advocated by the AIC criteria, see Example 2.4), and an ARMA(2,2) specification. We are interested in 2-year-ahead forecasts (i.e., $h = 2$ since the data are yearly).

```
## Registered S3 method overwritten by 'quantmod':
##   method            from
##   as.zoo.data.frame zoo
```

```
##
##  Diebold-Mariano Test
##
## data:  e1e2
## DM = -0.82989, Forecast horizon = 2, Loss function power = 2, p-value =
## 0.7946
## alternative hypothesis: greater
```

With `alternative = "greater"` The alternative hypothesis is that method 2 is more accurate than method 1. Since we do not reject the null (the p-value being of 0.795), we are not led to use the more sophisticated model (ARMA(2,2)) and we keep the simple AR(1) model.

Assume now that we want to compare the AR(1) process to a VAR model (see Def. 3.1). We consider a bivariate VAR, where GDP growth is complemented with CPI-based inflation rate.

```
##
##  Diebold-Mariano Test
##
## data:  e1e3
## DM = -0.11555, Forecast horizon = 2, Loss function power = 2, p-value =
## 0.5458
## alternative hypothesis: greater
```

Again, we do not find that the alternative model (here the VAR(1) model) is better than the AR(1) model to forecast GDP growth.

4.1 Exercise

Exercise 4.1 (Forecast comparison and the Diebold–Mariano test)**. PS3 | Advanced | Forecast evaluation**
Review Section 4.

Three one-step forecasting models are compared on 200 observations: A, a random walk; B, an AR(1); and C, an ARMA(2,2). The out-of-sample exercise uses 100 rolling estimation windows of length 100.

	Model A	Model B	Model C
In-sample RMSE	1.03	0.96	0.94
Out-of-sample RMSE	1.11	1.02	1.09
Mean loss differential $\bar{d}$ versus A	—	-0.191	-0.044
$\widehat{\gamma}_0$	—	0.090	0.132
$\widehat{\gamma}_1$	—	0.061	0.086
$\widehat{\gamma}_2$	—	0.020	0.047
$\widehat{\gamma}_3$	—	0.003	0.011

Here $d_t = e_{j,t}^2 - e_{A,t}^2$, for $j = B, C$.

1. Discuss the in-sample and out-of-sample rankings.
2. Using a Newey–West long-run variance with maximum lag 3 and Bartlett weights, estimate $\mathbb{V}ar(\bar{d})$.
3. Test at 5% whether B and C have equal predictive accuracy to A, and interpret the sign of the test statistic.

4.1.1 Exercise from the 2024 exam

Exercise 4.2 (Inflation forecasts from an MA(1))**. EXAM-2024 | Exam | True/false**
Review Section 4.

Let annual inflation be $\pi_t = 0.02 + \varepsilon_t + 0.4\varepsilon_{t-1}$, where $\varepsilon_t \overset{i.i.d.}{\sim} \mathcal{N}(0, \sigma^2)$ and $\sigma = 0.01$.

State whether each assertion is true or false and justify your answer.

1. For $h > 0$, $\mathbb{E}(\pi_{t+h} + \cdots + \pi_{t+1} \mid \varepsilon_t, \varepsilon_{t-1}, ...) = 0.02h + 0.4\varepsilon_t$.
2. $\mathbb{V}ar(\pi_t) = 0.01$.
3. Conditional on $\varepsilon_t = 0$, a 95% interval for π_{t+1} is $[-1.96\sigma, 1.96\sigma]$.
4. π_t is correlated with π_{t-1} but uncorrelated with π_{t-2}.
5. $\mathbb{V}ar_t(\pi_{t+3}) = \mathbb{V}ar_t(\pi_{t+2})$.

4.1.2 Exercise from the 2025 exam

Exercise 4.3 (GDP forecasts from an MA(2) growth model)**. EXAM-2025 | Exam | True/false**
Review Section 4.

Let $y_t = \log(GDP_t/GDP_{t-1})$ and suppose

$$y_t = \mu + \varepsilon_t + 0.5\varepsilon_{t-1} + 0.2\varepsilon_{t-2}, \qquad \varepsilon_t \overset{i.i.d.}{\sim} \mathcal{N}(0, \sigma^2).$$

Recall that $\log(GDP_{t+h}) = \log(GDP_t) + y_{t+h} + \cdots + y_{t+1}$. Conditional moments at date t use current and past information.

State whether each assertion is true or false and justify your answer.

1. $\mathbb{E}_t[\log(GDP_{t+2})] = \log(GDP_t) + 2\mu + \varepsilon_{t+2} + 1.5\varepsilon_{t+1} + 0.7\varepsilon_t$.
2. $\mathbb{V}ar(y_t) = 0.29\sigma^2$.
3. Conditional on date-t information, a 95% interval for y_{t+2} is $[\mu - 1.96\sigma, \mu + 1.96\sigma]$.
4. The first-order autocorrelation of y_t is $0.6/1.29$.
5. $\mathbb{E}[\log(GDP_{t+h}/GDP_t)] = h\mu$.

Chapter 5

Non-stationary processes

In time series analysis, nonstationarity has several crucial implications. Indeed, various time-series regression procedures are not reliable anymore when processes are nonstationary.

There are different reasons why a process can be nonstationary. Two examples are trends and breaks. Generally speaking, a trend is a persistent long-term movement of a variable over time. A linear trend is a simple example of (deterministic) trend. We say that process y_t is stationary around a linear trend if it is given by:

$$y_t = a + bt + x_t,$$

where x_t is a stationary process. But "trends" may also be stochastic. A typical example of stochastic trend is a random walk:

$$w_t = w_{t-1} + \varepsilon_t,$$

where ε_t is a sequence of i.i.d. mean-zero shocks with variance σ^2.

As we will see, if $y_t = w_t + x_t$, where w_t is a random walk, then the use of y_t in econometric specifications will have to be operated carefully. Typically, as we shall see, standard inference in linear regression models may no longer be valid since y_t features no unconditional moments.

We have

$$\mathbb{V}ar_t(y_{t+h}) = \mathbb{V}ar_t(\varepsilon_{t+h}) + \cdots + \mathbb{V}ar_t(\varepsilon_{t+1}) = h\sigma^2.$$

Using the law of total variance (and assuming that $\mathbb{V}ar(y_t)$ exists), we have, for any h:

$$\mathbb{V}ar(y_t) = \mathbb{E}[\mathbb{V}ar_{t-h}(y_t)] + \mathbb{V}ar[\mathbb{E}_{t-h}(y_t)]. \tag{5.1}$$

We also have $\mathbb{E}_{t-h}(y_t) = y_{t-h}$ and $\mathbb{V}ar_t(y_{t+h}) = \mathbb{V}ar_t(\varepsilon_{t+h}) + \cdots + \mathbb{V}ar_t(\varepsilon_{t+1}) = h\sigma^2$. Therefore, Eq. (5.1) gives:

$$\mathbb{V}ar(y_t) = h\sigma^2 + \mathbb{V}ar(y_{t-h}).$$

Since the previous equation needs to be satisfied for any h (even infintely large ones), and since $\mathbb{V}ar(y_{t-h}) > 0$, $\mathbb{V}ar(y_t)$ can not be finite. None of the population moments of a random walk is actually defined.

5.1 Issues when working with nonstationary time series

Let us discuss three issues associated with nonstationary processes: 1. Bias of autoregressive coefficient towards 0 (context: autoregressions). 2. Even in large samples, the t-statistics cannot be approximated by the normal distribution (context: OLS regressions). 3. Spurious regressions: OLS-based regression analysis tends to indicate relationships between *independent* nonstationary (or unit-root) series.

5.1.1 The bias of autoregressive coefficient towards zero

Consider a non-stationary y_t process (e.g., y_t follows a random walk). The estimate of ϕ in the (OLS) regression $y_t = c + \phi y_{t-1} + \varepsilon_t$ is then biased toward zero. In particular, we have the approximation $\mathbb{E}(\phi) = 1 - 5.3/T$, where T is the sample size (see, e.g., Abadir (1993)). The 5th percentile of the distribution of ϕ is approximately $1 - 14.1/T$, e.g., 0.824 for $T = 80$. This notably poses problems for forecasts. Indeed, while we have $\mathbb{E}_t(y_{t+h}) = y_t$ if y_t follows a random walk, someone who would fit an AR(1) process and would obtain for instance $\hat{\phi} = 0.824$ would obtain that $\mathbb{E}_t(y_{t+10}) = 0.144 y_t$.

This is illustrated by Figure 5.1. The figure shows the simulated distributions of $\hat{\phi}$, obtained by OLS in the linear model $y_t = c + \phi y_{t-1} + \varepsilon_t$. We consider two sample sizes ($T = 50$, left plot, and $T = 200$, right plot). The dashed vertical lines indicate the simulation means; the solid vertical lines give the approximations $1 - 5.3/T$.

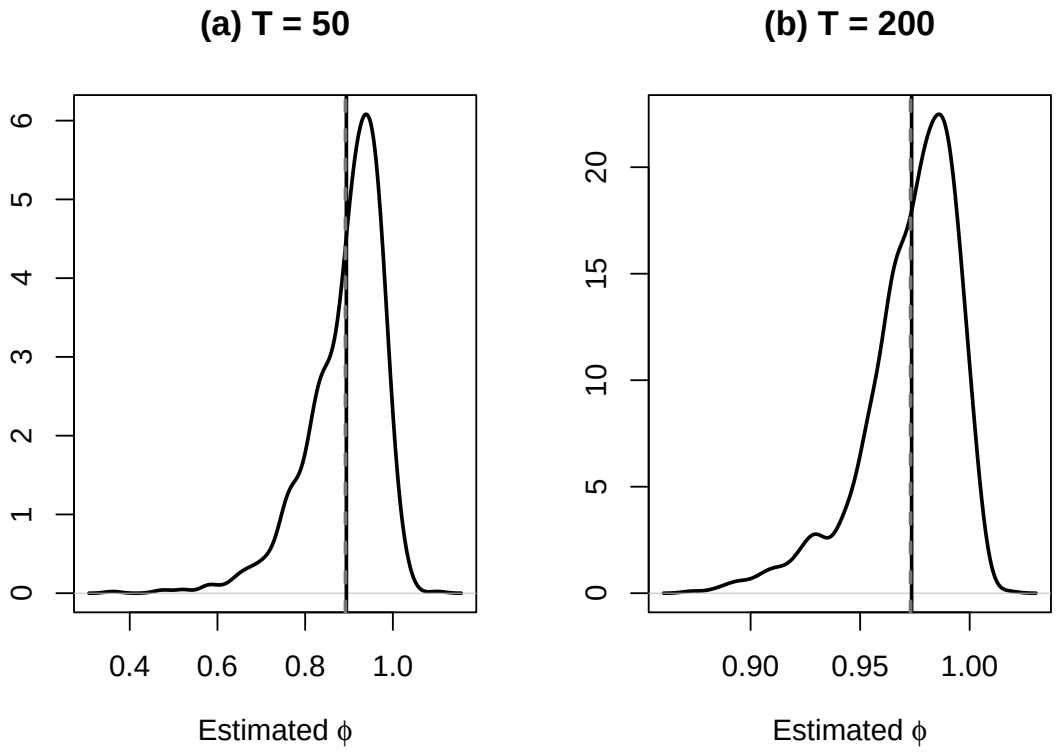


Figure 5.1: Kernel-density estimates of the OLS estimator $\hat{\phi}$ for a unit-root process, based on 1,000 Monte Carlo samples: (a) $T = 50$ and (b) $T = 200$. The dashed grey line marks the simulation mean; the solid black line marks the approximation $1 - 5.3/T$.

5.1.2 Spurious regressions

Consider two independent non-stationary (unit roots) variables. If we regress one on the other, and if we use the standard OLS formulas to compute the standard deviation of the regression parameter, it can be shown that we will tend to underestimate this standard deviation. As a result, if we use standard t-statistics to test for the existence of a relationship between the two variables, we will often have false positive, i.e., we will often reject the null hypothesis of no relationship (H_0: regression coefficient = 0). This phenomenon is called *spurious regressions* (see, e.g., these examples).

This situation is illustrated by Figure 5.2. We simulate two independent random walks, x_t and y_t, and we regress y_t on x_t by OLS. (We consider two different sample sizes, $T = 50$ and $T = 200$.) The solid lines represent the densities of the estimated β's, that are the slope coefficients of the regressions. The dashed densities represent the asymptotic distribution of β based on the standard OLS formula (normal distribution of mean zero and of variance $\hat{\sigma}^2 / \widehat{\mathbb{V}ar}(x_t)$). The fact that the latter distribution features a smaller variance than the former implies that using standard OLS inference is misleading, as this distribution overestimates the accuracy of the estimator.

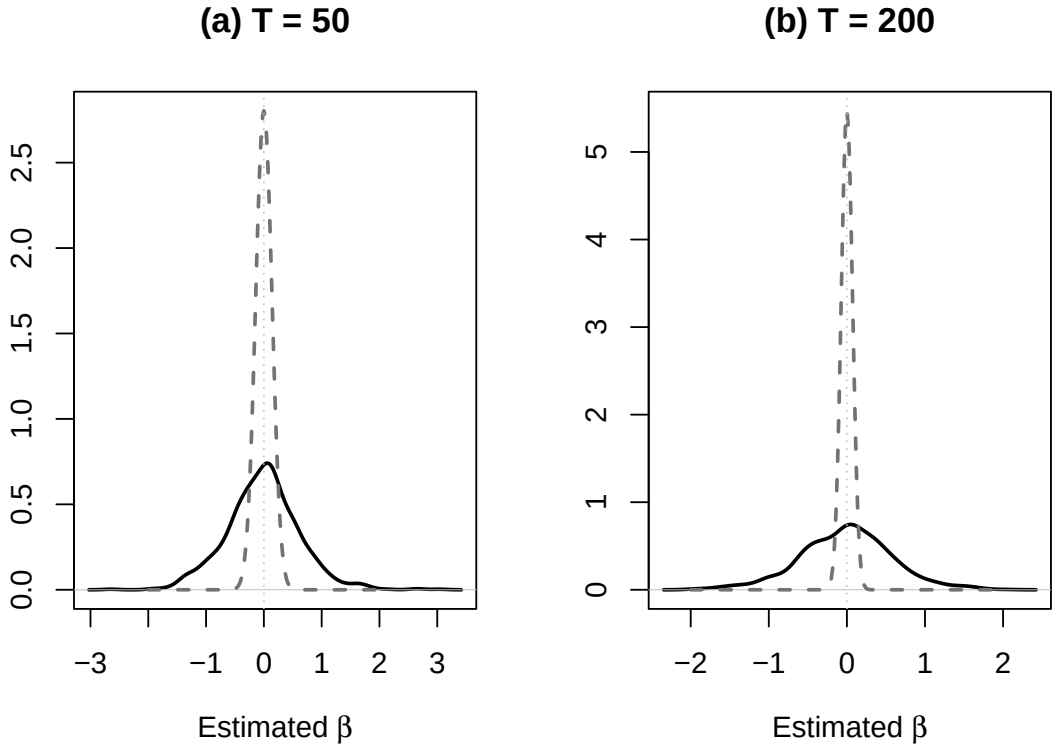


Figure 5.2: Distribution of the OLS slope in regressions between two independent random walks, based on 1,000 Monte Carlo samples: (a) $T = 50$ and (b) $T = 200$. Solid black lines are kernel-density estimates of the simulated slopes; dashed grey lines are the normal approximations implied by conventional OLS standard errors.

5.2 Nonstationarity tests

Hence, employing (OLS) regressions with non-stationary variables may be misleading. It is therefore important, before employing these techniques (OLS), to check that the variables are stationary. To do so, specific tests are used. These tests are called stationarity and non-stationarity (or unit-root) tests. In the context of a stationarity test, the null hypothesis that y_t is stationary or trend-stationary (i.e. equal to the sum of a linear trend and a stationary component). In the context of a non-stationarity (or unit root) tests, the null hypothesis that y_t is not (trend) stationary.

To illustrate, consider the AR(1) case:

$$y_t = \phi y_{t-1} + \varepsilon_t, \quad \varepsilon_t \sim \mathcal{N}(0, 1). \tag{5.2}$$

If y_t is stationary (i.e. if $|\phi| < 1$, Prop. 2.3), it can be shown (Hamilton (1994), p.216) that:

$$\sqrt{T}(\phi^{OLS} - \phi) \xrightarrow{d} \mathcal{N}(0, 1 - \phi^2).$$

The previous equation does not make sense for $\phi = 1$.

Even if standard OLS results are not valid any more in the non-stationary case, many (non)stationary tests make use of statistics that were used in the standard OLS analysis. Even if the t-statistic does not admit the Student-t distribution any more, one can still compute this statistic. The idea behind severeal unit-root tests is simply to determine the distribution of these OLS-based statistics under the null hypothesis of non-stationarity. Let us define H_0 as follows:

$$H_0 : \quad \phi = 1 \quad and \quad H_1 : \quad |\phi| < 1.$$

Test statistic:

$$t_{\phi=1} = \frac{\phi^{OLS} - 1}{\sigma_\phi^{OLS}},$$

where ϕ^{OLS} is the OLS estimate of ϕ and σ_ϕ^{OLS} is the usual OLS-based standard error estimate of ϕ^{OLS}.

Importantly, it has to be noted that, *under the null hypothesis, $t_{\phi=1}$ is not distributed as a Student-t variable* (see Def. 8.10).

Theorem 5.1 (Convergence results)**.** *If y_t follows Eq.* (5.2) *(with $\mathbb{V}ar(\varepsilon_t) = \sigma^2$) and $\phi = 1$, then:*

$$T^{-3/2} \sum_{t=1}^{T} y_t \xrightarrow{d} \sigma \int_0^1 W(r)dr \tag{5.3}$$

$$T^{-2} \sum_{t=1}^{T} y_t^2 \xrightarrow{d} \sigma^2 \int_0^1 W(r)^2 dr \tag{5.4}$$

$$T^{-1} \sum_{t=1}^{T} y_{t-1}\varepsilon_t \xrightarrow{d} \sigma^2 \int_0^1 W(r)dW(r), \tag{5.5}$$

where $W(r)$ denotes a standard Brownian motion (Wiener process) defined on the unit interval.

Proof. See Phillips (1987) or Hamilton (1994) (Subsection 17.4). □

Theorem 5.1 notably implies that, if $\phi = 1$, then:

$$T(\phi^{OLS} - 1) \xrightarrow{d} \frac{\int_0^1 W(r)dW(r)}{\int_0^1 W(r)^2 dr} = \frac{\frac{1}{2}(W(1)^2 - 1)}{\int_0^1 W(r)^2 dr} \tag{5.6}$$

$$t_{\phi=1} \xrightarrow{d} \frac{\int_0^1 W(r)dW(r)}{\left(\int_0^1 W(r)^2 dr\right)^{1/2}} = \frac{\frac{1}{2}(W(1)^2 - 1)}{\left(\int_0^1 W(r)^2 dr\right)^{1/2}} \tag{5.7}$$

The previous theorem notably implies that the convergence rate of ϕ^{OLS} (in T) is faster than if y_t was stationary (in which case it is in $\sqrt{T}$). It also implies that, asymptotically, (a) ϕ^{OLS} is not normally distributed and (b) $t_{\phi=1}$ is not standard normal.

The limiting distribution of $t_{\phi=1}$ has no closed form; it is called the **Dickey-Fuller distribution**.

Although not available in closed form (it has to be evaluated numerically), the distribution of $T(\phi^{OLS}-1)$ can be used to test for the null hypothesis $\phi = 1$.

Nonstationarity tests: About trends again

The right consideration for potential trends in y_t's specification is crucial. We focus on two types of null hypotheses:

- Constant only. Specification:

$$y_t = c + \phi y_{t-1} + \varepsilon_t.$$

  $|\phi| < 1$: y_t is stationary (we note $I(0)$) with non-zero mean.
- Constant + Linear trend. Specification:

$$y_t = c + \delta t + \phi y_{t-1} + \varepsilon_t.$$

  $|\phi| < 1$: y_t is stationary around a deterministic time trend.

So far, we focused on AR(1) processes. But many time series have a more complicated dynamic structure.

Dickey-Fuller (DF) test

The specification underlying the Dickey and Fuller (1979) test is:

$$y_t = \beta' D_t + \phi y_{t-1} + \sum_{i=1}^{p} \psi_i \Delta y_{t-i} + \varepsilon_t, \tag{5.8}$$

where D_t is a vector of deterministic trends ($D_t = 1$ or $D_t = [1, t]'$) and p should be such that the estimated ε_t are serially uncorrelated.

The null hypothesis of the DF test is: $\phi = 1$. Test statistics:

$$\text{ADF bias statistic:} \qquad ADF_\pi = T(\phi^{OLS} - 1)$$

$$\text{ADF } t \text{ statistic:} \qquad ADF_t = t_{\phi=1} = \frac{\phi^{OLS} - 1}{\sigma_\phi^{OLS}}$$

Under the null hypothesis ($\phi = 1$) and if the regression is as in Eq. (5.2) the limiting distributions of the statistics are, respectively, as in Eq. (5.6) and Eq. (5.7).

Alternative formulation of Eq. (5.8):

$$\Delta y_t = \beta' D_t + \pi y_{t-1} + \sum_{i=1}^{p} \psi_i \Delta y_{t-i} + \varepsilon_t,$$

with $\pi = \phi - 1$. Under the null hypothesis $\pi = 0$. In this case:

$$\text{ADF bias statistic:} \qquad ADF_\pi = T\pi^{OLS}$$

$$\text{ADF } t \text{ statistic:} \qquad ADF_t = t_{\pi=0} = \frac{\pi^{OLS}}{\sigma_\pi^{OLS}}$$

Under the null hypothesis ($\phi = 1$) and if the regression is as in Eq. (5.2) the limiting distributions of the statistics are, respectively, as in Eq. (5.6) and Eq. (5.7).

The selection of p can rely on information criteria (see Def. 2.11). Alternatively, Schwert (1989) proposes to use:

$$p = \left[12 \times \left(\frac{T}{100} \right)^{1/4} \right].$$

Importantly, this test is **one-sided left-tailed** test: one rejects the null if the test statistics are sufficiently negative; we are therefore interested in the first quantiles of the limit distribution.

Phillips-Perron (PP) test

The regression underyling the Phillips and Perron (1988) test is as follows:

$$\Delta y_t = \beta' D_t + \pi y_{t-1} + \varepsilon_t.$$

The issue of serial correlation (and heteroskedasticity) in the residual is handled by adjusting the test statistics $t_{\pi=0}$ and $T\pi$:[1]

$$\text{PP } t \text{ stat.:} \qquad Z_t = \sqrt{\frac{\hat{\gamma}_{0,T}}{\hat{\lambda}_T^2}} t_{\pi=0,T} - \frac{\hat{\lambda}_T^2 - \hat{\gamma}_{0,T}}{2\hat{\lambda}_T} \left(\frac{T\sigma_{\pi,T}^{OLS}}{s_T} \right)$$

$$\text{PP bias stat.:} \qquad Z_\pi = T\pi_T^{OLS} - \frac{1}{2}(\hat{\lambda}^2 - \hat{\gamma}_{0,T}^2) \left(\frac{T\sigma_{\pi,T}^{OLS}}{s_T^2} \right)^2$$

where

$$\begin{aligned}
\hat{\gamma}_{j,T} &= \frac{1}{T}\Sigma_{t=j+1}^{T} \hat{\varepsilon}_t \hat{\varepsilon}_{t-j} \\
\hat{\varepsilon}_t &= \text{OLS residuals} \\
\hat{\lambda}_T^2 &= \hat{\gamma}_{0,T} + 2\Sigma_{j=1}^{q} \left(1 - \frac{j}{q+1} \right) \hat{\gamma}_{j,T} \quad \text{(Newey-West formula)} \\
s_T^2 &= \frac{1}{T-k}\Sigma_{t=1}^{T} \hat{\varepsilon}_t^2 \quad (k\text{: number of param. estim. in the regression}) \\
\sigma_{\pi,T}^{OLS} &= \text{OLS standard error of } \pi.
\end{aligned}$$

When the underlying regression is: $y_t = \alpha + \phi y_{t-1} + \varepsilon_t$, and under the null that $\alpha = 0$ and $\phi = 1$, we have that:

- the limiting distribution of Z_π is that of (see, e.g., Hamilton (1994) 17.6.8):

$$\frac{\frac{1}{2}\{W(1)^2 - 1\} - W(1)\int_0^1 W(r)dr}{\int_0^1 W(r)^2 dr - \left[\int_0^1 W(r)dr\right]^2};$$

[1] See Hamilton (1994), Table 17.2 p.514.

- the limiting distribution of Z_t is that of (see, e.g., Hamilton (1994) 17.6.12):

$$\frac{\frac{1}{2}\{W(1)^2 - 1\} - W(1)\int_0^1 W(r)dr}{\left(\int_0^1 W(r)^2 dr - \left[\int_0^1 W(r)dr\right]^2\right)^2}.$$

If $|\phi| < 1$, the OLS estimate ϕ_T^{OLS} is not consistent if the ε_ts are serially correlated (the true residuals and the regressors are correlated). When $\phi = 1$, the rate of convergence of ϕ_T^{OLS} is T (*super-consistency*), which ensures that $\phi_T^{OLS} \xrightarrow{p} 1$ even if the ε_ts are serially correlated.

As the ADF test, this test is **one-sided left-tailed** (reject the null if the test statistics are sufficiently negative). The critical values are obtained by simulation; they can for instance be found here.

5.3 Stationarity test: KPSS

The test proposed by Kwiatkowski et al. (1992) is a stationarity test, i.e., under the null hypothesis, the process is stationary. The underlying specificaiton is the following:

$$y_t = \beta' D_t + \mu_t + \varepsilon_t,$$

with $\mu_t = \mu_{t-1} + \eta_t$, $\mathbb{V}ar(\eta_t) = \sigma_\eta^2$, where $D_t = 1$ or $D_t = [1, t]'$ and where $\{\varepsilon_t\}$ is a covariance-stationary sequence.

The KPSS statistic corresponds to the Lagrange Multiplier test statistic associated with the hypothesis $\sigma_\eta^2 = 0$:

$$\boxed{\xi_T^{KPSS} = \left(\frac{1}{\hat{\lambda}_T^2 T^2}\sum_{t=1}^{T}\hat{S}_t^2\right),}$$

with $\hat{S}_t = \Sigma_{i=1}^t \hat{\varepsilon}_i$, where the $\hat{\varepsilon}_t$s are the residuals of the OLS regression of y_t on D_t, and where $\hat{\lambda}^2$ is a consistent estimate of the long-run variance of $\hat{\varepsilon}_t$ (see Def. 1.6 and Newey-West approach, see Eq. (1.6)).

KPSS show that, under the null hypothesis, ξ_T^{KPSS} converges in distribution towards a distribution that does not depend on β but on the form of D_t. Specifically:

- If $D_t = 1$:

$$\xi_T^{KPSS} \xrightarrow{d} \int_0^1 (W(r) - rW(1))dr.$$

- If $D_t = [1, t]'$:

$$\xi_T^{KPSS} \xrightarrow{d} \int_0^1 \left\{W(r) + r(2-3r)W(1) + 6r(r^2-1)\int_0^1 W(s)ds\right\} dr.$$

This test is a **one-sided right-tailed** test: one rejects the null if ξ_T^{KPSS} is above the $(1-\alpha)$ quantile of the limit distribution. The critical values car be found, e.g., here.

Example 5.1 (Stationarity of inflation and interest rates)**.** Let us use quarterly US data and test the stationarity of inflation and the 3-month short-term rate. Let us first plot the data:

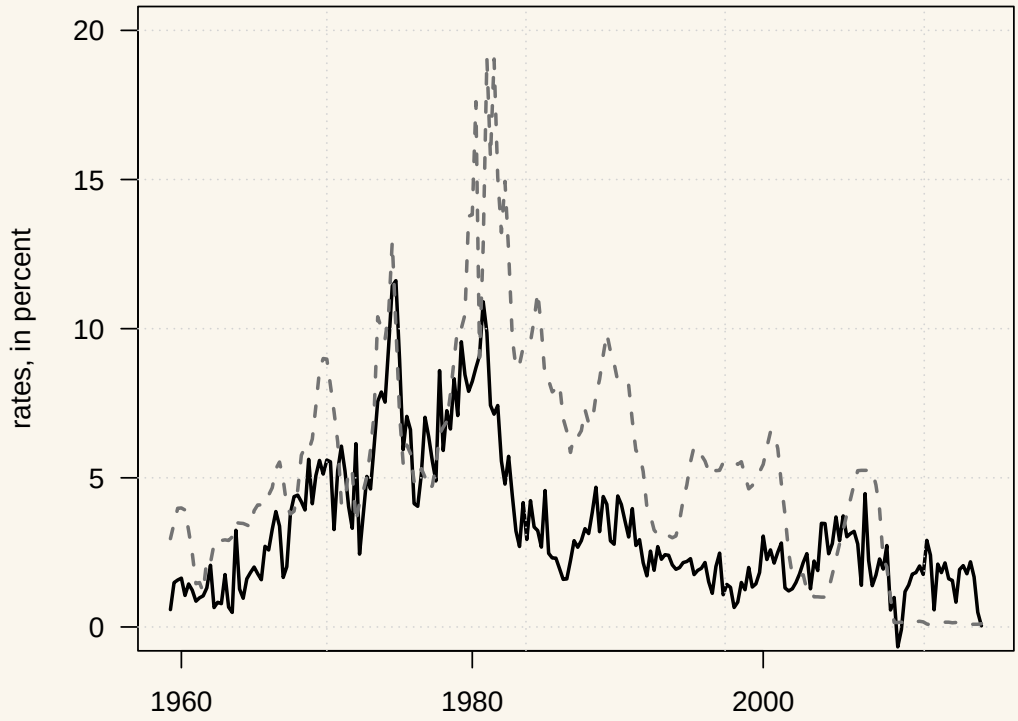


Figure 5.3: Quarterly U.S. inflation and short-term interest rates, in percent. The solid black line is CPI inflation; the dashed grey line is the three-month nominal interest rate. Source: U.S. government series retrieved through FRED and assembled in the IdSS US3var dataset.

Let us now run the tests. Note that the default alternative hypothesis of function `adf.test` of package `tseries` is that the process is trend-stationary. Note that when `kpss.test` returns a p-value of `0.01`, it means that the true p-value is lower then that.

```
## [1] 0.29230878 0.04978275 0.01000000
```

```
## [1] 0.3837577 0.3614365 0.0100000
```

5.4 Exercises

Exercise 5.1 (Orders of integration)**. PS4 | Core | Analytical**
Review Section 5.

Find the order of integration of each process. Innovations ε_t and η_t are independent white noises.

1. $x_t = \rho x_{t-1} + \varepsilon_t$, $|\rho| < 1$.
2. $x_t = x_{t-1} + \varepsilon_t + \theta_1 \varepsilon_{t-1} + \cdots + \theta_q \varepsilon_{t-q}$, assuming the MA polynomial does not cancel the unit root.
3. $x_t = \mu_t + x_{t-1} + \varepsilon_t$, with $\mu_t = \mu_{t-1} + \eta_t$.
4. $x_t = 1.5 x_{t-1} - 0.5 x_{t-2} + \varepsilon_t$.

Exercise 5.2 (True or false: integrated processes)**. PS4 | Core | Conceptual**
Review Section 5.

State whether each assertion is true or false and briefly justify your answer. Let $x_t = x_{t-1} + \varepsilon_t$, where $\varepsilon_t \overset{i.i.d.}{\sim} \mathcal{N}(0,1)$, when the assertion refers to "the random walk."

1. $\mathbb{V}ar_t(x_{t+h}) \to \infty$ as $h \to \infty$.
2. x_t is not covariance stationary, but Δx_t is.
3. If $\Delta^2 x_t$ is $I(0)$ and Δx_t is $I(1)$, then x_t is $I(2)$.
4. The random walk is $I(1)$.
5. The random walk necessarily has unconditional mean zero.
6. In a regression of one unrelated $I(1)$ variable on another, usual stationary OLS standard errors give valid confidence intervals.
7. The OLS slope in a finite-sample regression of the random walk on its lag and a constant is typically biased downward.
8. If x_t is $I(1)$ and y_t is $I(2)$, then $x_t + y_t$ is generally $I(2)$.
9. If x_t is $I(2)$ and y_t is $I(1)$, the residual from regressing x_t on y_t is necessarily $I(1)$.

Exercise 5.3 (Interpreting ADF and Phillips–Perron tests)**. PS4 | Advanced | Empirical interpretation**
Review Section 5.

For samples of 100 observations, ADF and Phillips–Perron tests are computed with a constant and no linear trend:

Statistic	x_t	y_t	z_t
ADF_t	0.14	-3.51	-2.81
Z_t	0.34	-3.82	-3.01
ADF_ρ	-0.96	-18.4	-15.3
Z_ρ	-1.12	-20.2	-11.2

At the 5% level for this specification and sample size, use the approximate critical values -2.89 for the t-type statistics and -14.1 for the normalized-coefficient statistics.

1. State the conclusion of each of the twelve tests.
2. Summarize the evidence for the integration order of each series, noting any disagreement between tests.
3. For which pairs is a conventional cointegration analysis potentially meaningful? Explain why equal integration orders are required.

5.4.1 Exercise from the 2025 exam

Exercise 5.4 (Lag polynomials, stationarity, and integration)**. EXAM-2025 | Exam | True/false**
Review Section 5.

Let $\varepsilon_t \overset{i.i.d.}{\sim} \mathcal{N}(0, \sigma^2)$ and consider

$$(1 - 0.9L)(1 + 0.5L)y_t = \varepsilon_t,$$
$$(1 - L)x_t = -0.9x_{t-1} + \varepsilon_t,$$
$$(1 - 0.5L)w_t = 0.5w_{t-1} + \varepsilon_t.$$

State whether each assertion is true or false and justify your answer.

1. y_t is integrated of order one.
2. x_t is covariance stationary.
3. w_t is covariance stationary.
4. Δy_t is covariance stationary.
5. Δw_t is covariance stationary.

Chapter 6

Introduction to cointegration

6.1 Intuition

Many statistical procedures are well-defined only when the processes of interest are stationary. As a result, especially when one wants to investigate the joint dynamics of different variables, one often begins by making the data stationary (by, e.g., taking first differences or removing deterministic trends). However, doing so may remove information from the data. Heuristically, removing trends amounts to filtering out the long-run variations of the series. However, it may be the case that the different variables interact in the short run *and* in the long run.

For instance, the left plot of Figure 6.1 suggests that the trends of x_t and y_t are positively correlated. However, the right plot shows that, for low values of h, the correlation between $x_t - x_{t-h}$ and $y_t - y_{t-h}$ is negative. This is notably the case for $h = 1$, which means that the first differences of the two variables (i.e., Δx_t and Δy_t) are negatively correlated. Hence, focusing on the first differences would lead the researcher to think that the relationship between x_t and y_t is a negative one (while it is only the case when one focuses on the high-frequency comovements between the two variables).

Definition 6.1 (Integrated variables)**.** A univariate process $\{y_t\}$ is said to be $I(d)$ if its dth difference is stationary (but not its $(d-1)^{th}$ difference).

For instance:

- If y_t is not stationary but $\Delta y_t = y_t - y_{t-1}$ is, then y_t is $I(1)$.
- If Δy_t is not stationary but $\Delta^2 y_t = \Delta(\Delta y_t)$ is, then y_t is $I(2)$.
- If y_t is stationary then y_t is $I(0)$.

6.2 The bivariate case

If we regress an $I(1)$ variable y_t on another independent $I(1)$ variable x_t, the usual (OLS-based) t-tests on regression coefficients often (misleadingly) show statistically significant coefficients (we then speak of spurious regressions, see Section 5). A solution is to regress Δy_t (that is $I(0)$) on Δx_t and then inference will be correct. However, as stated above, the economic interpretation of the regression then changes, as doing so amounts to focusing on the high-frequency movements of the variables.

Let us now consider the case where y_t and x_t are still $I(1)$, but where they satisfy:

$$y_t = \beta x_t + \varepsilon_t, \tag{6.1}$$

with $\varepsilon_t \sim I(0)$. That is, there is a linear combination of the two $I(1)$ variables that is $I(0)$.

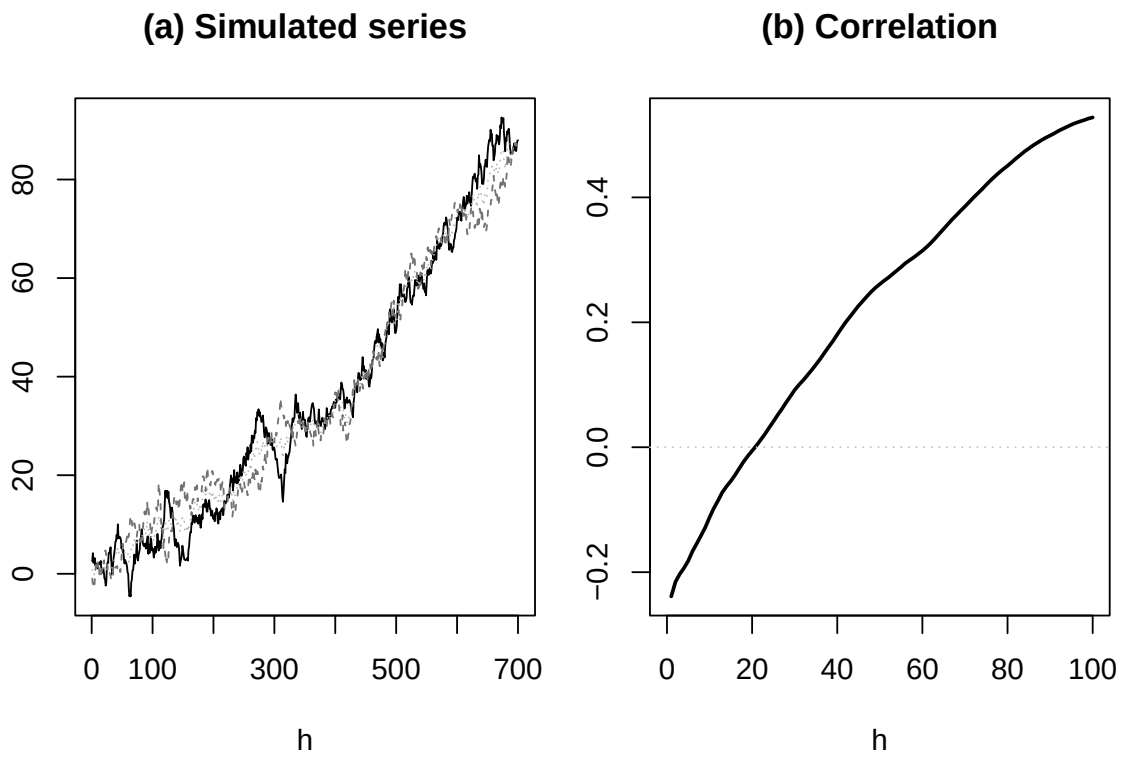


Figure 6.1: Simulated illustration of frequency-dependent comovement: (a) x_t (solid black), y_t (dashed grey), and their common stochastic trend w_t (dotted light grey); (b) correlation between $x_t - x_{t-h}$ and $y_t - y_{t-h}$ as a function of the differencing horizon h. Although x_t and y_t share a positive long-run trend, their short-run changes are negatively correlated.

In that case, the convergence of b, the OLS estimate of β, is fast. Indeed, the convergence rate is in $1/T$ (versus $1/\sqrt{T}$ in the purely stationary case). This stems from the properties of non-statinary processes that are stated in Prop. 6.1. We have:

$$b_T = \frac{\sum_t x_t y_t}{\sum_t x_t^2} = \frac{\sum_t x_t(\beta x_t + \varepsilon_t)}{\sum_t x_t^2} = \beta + \frac{\sum_t x_t \varepsilon_t}{\sum_t x_t^2}.$$

In particular, if ε_t was a white noise, using properties (ii) and (iv) of Prop. 6.1, we would get:

$$b_T \approx \beta + \frac{1}{T}\frac{\int_0^1 x_\infty(r) dW_r^\varepsilon}{\int_0^1 x_\infty^2(r) dr}. \tag{6.2}$$

where the random variables x_∞^2 and W_r^ε are defined in Prop. 6.1.

Proposition 6.1 (Properties of an I(1) process)**.** *If $\{y_t\}$ is $I(1)$ and such that $y_t - y_{t-1} = H(L)\varepsilon_t$ where $H(L)\varepsilon_t$ is $I(0)$, then:*

i. $\frac{1}{\sqrt{T}}\bar{y}_T \xrightarrow{d} \int_0^1 y_\infty(r)dr$,

ii. $\frac{1}{T^2}\sum_{t=1}^T y_t^2 \xrightarrow{d} \int_0^1 y_\infty^2(r)dr$,

iii. $\frac{1}{T}\sum_{t=1}^T y_t(y_t - y_{t-1}) \xrightarrow{d} \frac{1}{2}y_\infty^2(1) + \frac{1}{2}\mathbb{V}ar(y_t - y_{t-1})$,

iv. $\frac{1}{T}\sum_{t=1}^T y_t\eta_t \xrightarrow{d} \sigma_\eta \int_0^1 y_\infty(r)dW_r^\eta$ *where η_t is a white noise of variance σ_η^2,*

where $y_\infty(r)$ is of the form ωW_r (W_r being a Brownian motion (see extra material), with $\omega = \sum_{k=-\infty}^{\infty}\gamma_k$, the γ_ks being the autocovariances of $H(L)\varepsilon_t$, and where $\eta_\infty(r)$ is of the form $\sigma_\eta W_r^\eta$ (W_r^η Brownian motion "associated" to η_t).

Proof. (of 1) We have:

$$\frac{1}{\sqrt{T}}\bar{y}_T = \frac{1}{T}\sum_{t=1}^T \frac{y_t}{\sqrt{T}} = \frac{1}{T}\sum_{t=1}^T \tilde{y}_T(t/T),$$

where $\tilde{y}_T(r) = (1/\sqrt{T})y_{[rT]}$. Now $\tilde{y}_T(r) = r\sqrt{T}\left(\frac{1}{Tr}\sum_{t=1}^{[Tr]} H(L)\varepsilon_t\right)$. By Eq. (1.4) in Theorem 1.1, we have $\sqrt{r}\sqrt{Tr}\left(\frac{1}{Tr}\sum_{t=1}^{[Tr]} H(L)\varepsilon_t\right) \to \omega W_r$. Therefore, for large T, $\frac{1}{T}\sum_{t=1}^T \tilde{y}_T(t/T)$ approximates $\frac{1}{T}\sum_{t=1}^T \omega W_{t/T}$, which is a Riemann sum that converges to $\int_0^1 y_\infty(r)dr$. □

6.3 Multivariate case and VECM

In the following, we consider an n-dimensional vector y_t. Moreover, ε_t is an n-dimensional white noise process. The notion of integration (Def. 6.1) can also be defined in the multivariate case:

Definition 6.2 (Order of integration (multivariate case)). $\{y_t\}$ is $I(d)$ if

$$(1-L)^d y_t = \mu + H(L)\varepsilon_t, \tag{6.3}$$

where $H(L)\varepsilon_t$ is a stationary process (but $(1-L)^{d-1}y_t$ is not).

Definition 6.3 (Cointegration). If y_t is integrated of order d, then its components are said to be cointegrated if there exists a linear combination of the components of y_t that is integrated of an order equal to, or lower than, $d-1$.

For instance, Eq. (6.1) implies that $[1, -\beta]'$ is a **cointegrating vector** and that $[y_t, x_t]'$ is cointegrated.

Consider an $I(1)$ process, y_t, that is such that the Wold representation of Δy_t is Eq. (6.3). We have:

$$y_t = \mu + H(L)\varepsilon_t + y_{t-1} = t\mu + H(L)(\varepsilon_t + \varepsilon_{t-1} + \cdots + \varepsilon_1) + y_0.$$

It can be shown that:

$$y_t = t\mu + H(1)(\varepsilon_t + \varepsilon_{t-1} + \cdots + \varepsilon_1) + \xi_t,$$

where ξ_t is a stationary process.

Assume that y_t possesses a cointegrating vector π such that $\pi' y_t$ is a (univariate) stationary process.

Necessarily, we must have $\pi'\mu = 0$ and $\pi' H(1) = 0$. Reciprocally, if $\pi'\mu = 0$ and $\pi' H(1) = 0$, then π is a cointegrating vector of y_t. This proves the following proposition:

Proposition 6.2 (Necessary and sufficient conditions of cointegration). *If y_t is $I(1)$ and admits the Wold representation Eq.* (6.3)*, with $d = 1$, then π is a cointegrating vector if and only if (iff):*

$$\pi'\mu = 0 \textit{ (scalar equation) and } \pi' H(1) = 0 \textit{ (vectorial equation)}.$$

Consider the following VAR(p) model, where y_t is $I(1)$:

$$y_t = c + \Phi_1 y_{t-1} + \cdots + \Phi_p y_{t-p} + \varepsilon_t, \tag{6.4}$$

or $\Phi(L) y_t = c + \varepsilon_t$ where $\Phi(L) = I - \Phi_1 L - \cdots - \Phi_p L^p$.

Suppose that the Wold representation of Δy_t is Eq. (6.3). Premultiplying Eq. (6.3) by $\Phi(L)$ gives:

$$(1-L)\Phi(L) y_t = \Phi(1)\mu + \Phi(L)H(L)\varepsilon_t,$$

or

$$(1-L)\varepsilon_t = \Phi(1)\mu + \Phi(L)H(L)\varepsilon_t.$$

Taking the expectation on both sides gives $\Phi(1)\mu = 0$. Therefore, for the previous equation to hold for any ε_t, we must have

$$(1-L)Id = \Phi(L)H(L).$$

The previous equality implies that $(1-z)Id = \Phi(z)H(z)$ for all z. In particular, for $z = 1$:

$$0 = \Phi(1)H(1).$$

Take any row π' of $\Phi(1)$. Since $\Phi(1)\mu = 0$, we have $\pi'\mu = 0$ (scalar equation). Since $\Phi(1)H(1) = 0$, we have $\pi' H(1) = 0$ (vectorial equation). Prop. 6.2 then implies that the rows of $\Phi(1)$ are cointegrating vectors of y_t. Therefore, if $\{a_1, ..., a_h\}$ constitutes a basis for the space of cointegrating vectors, then π can be expressed as a linear combination of the a_i's. That is, we must have:

$$\pi = [a_1 \quad a_2 \quad ... \quad a_h]b = \underset{n\times h}{\underbrace{A}} \; \underset{h\times 1}{\underbrace{b}},$$

where $A = [a_1 \quad a_2 \quad ... \quad a_h]$.

Since this is true for all rows of $\Phi(1)$, it comes that this matrix is of the form:

$$\underset{n\times n}{\underbrace{\Phi(1)}} = \underset{n\times h}{\underbrace{B}} \; \underset{h\times n}{\underbrace{A'}}. \tag{6.5}$$

This shows that the number of independent cointegrating vectors—he order of cointegration of y_t—is the rank of $\Phi(1)$. This has important implications for the dynamics of Δy_t.

Consider a process y_t whose VAR representation is as in Eq. (6.4). This VAR representation can be rewritten:

$$y_t = (c + \rho y_{t-1}) + \zeta_1 \Delta y_{t-1} + \cdots + \zeta_{p-1}\Delta y_{t-p+1} + \varepsilon_t, \tag{6.6}$$

where $\zeta_k = -\Phi_{k+1} - \cdots - \Phi_p$ and $\rho = \Phi_1 + \cdots + \Phi_p$.

Example 6.1 (VAR(2)). For a VAR(2) with $y_t = c + \Phi_1 y_{t-1} + \Phi_2 y_{t-2} + \varepsilon_t$, we have:

$$y_t = c + \{\Phi_1 + \Phi_2\}y_{t-1} + \{-\Phi_2\}\Delta y_{t-1} + \varepsilon_t,$$

Subtracting y_{t-1} from both sides of Eq. (6.6) and remarking that $-\Phi(1) = \rho - Id$ (recall that $\Phi(L) = I - \Phi_1 L - \cdots - \Phi_p L^p$), we get:

$$\Delta y_t = \{c - \underbrace{\Phi(1)y_{t-1}}_{=BA'y_{t-1}}\} + \zeta_1 \Delta y_{t-1} + \cdots + \zeta_{p-1}\Delta y_{t-p+1} + \varepsilon_t.$$

Using Eq. (6.5), and denoting by z_t the h-dimensional vector $A'y_t$, we obtain the **error correction representation** of the cointegrated variable y_t:

$$\boxed{\Delta y_t = c - Bz_{t-1} + \zeta_1 \Delta y_{t-1} + \cdots + \zeta_{p-1}\Delta y_{t-p+1} + \varepsilon_t.} \tag{6.7}$$

This type of model is called **Vector Error Correction Model** (VECM). This is because the components of z_t can be considered as errors that, multiplied by the components of B generate correction forces that imply that, in the long run, the congregation relationships are satisfied. Example 6.2 illustrates this.

Example 6.2 (VECM example)**.** Consider the VAR(1) process y_t that follows:

$$y_t = \Phi_1 y_{t-1} + \varepsilon_t = \begin{bmatrix} 0.5 & 0.5 \\ 0.2 & 0.8 \end{bmatrix} y_{t-1} + \varepsilon_t, \quad \varepsilon_t \sim i.i.d. \mathcal{N}(0, \Sigma)$$

1 is an eigenvalue of Φ_1 and y_t is $I(1)$. We have:

$$\Phi(1) = Id - \begin{bmatrix} 0.5 & 0.5 \\ 0.2 & 0.8 \end{bmatrix} = \begin{bmatrix} 0.5 & -0.5 \\ -0.2 & 0.2 \end{bmatrix},$$

which is of rank 1. Therefore y_t is cointegrated of order 1.

For that process, Eq. (6.7) writes:

$$\Delta y_t = -\begin{bmatrix} 0.5 & -0.5 \\ -0.2 & 0.2 \end{bmatrix} y_{t-1} + \varepsilon_t = \begin{bmatrix} -0.5 \\ 0.2 \end{bmatrix} z_{t-1} + \varepsilon_t,$$

where $z_t = y_{1,t} - y_{2,t}$.

The process z_t is stationary (we have $z_t = 0.3 z_{t-1} + \varepsilon_{1,t} - \varepsilon_{2,t}$).

We say that there is a *long-run relationship* between $y_{1,t}$ and $y_{2,t}$:

When $y_{1,t}$ is substantially above $y_{2,t}$, z_t is large, the influence of $-0.5 z_t$ on $\Delta y_{1,t+1}$ is negative, which tends to "correct" $y_{1,t}$ and brings it closer to $y_{2,t}$.

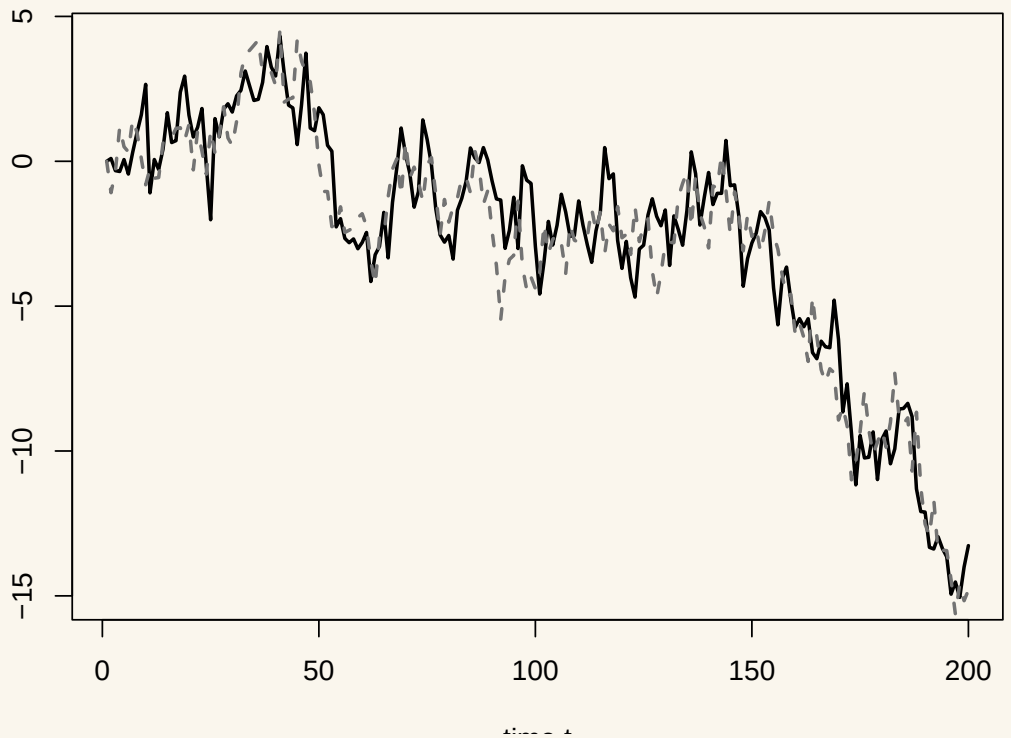


Figure 6.2: Simulated cointegrated series $y_{1,t}$ and $y_{2,t}$. The solid black line represents $y_{1,t}$; the dashed grey line represents $y_{2,t}$. Their difference follows the stationary process $z_t = 0.3 z_{t-1} + \varepsilon_{1,t} - \varepsilon_{2,t}$.

6.4 Cointegration in practice

Assume that we have a vector of variables y_t and that we want to investigate the joint dynamics of its components. If the $y_{i,t}$s are $I(1)$, we may need to use a VECM. Therefore, in the first place, one has to test for the stationarity of y_t's components (see Section 5).

If the components of y_t are $I(1)$, one has to determine the existence of cointegrating vectors. There are two general possibilities to do this:

1. The theory provides us with relevant cointegration relationships, for instance:
   - The Purchasing Power Parity (PPP) suggests the existence of a long-run relationship between domestic prices, foreign prices, and the exchange rate.
   - If real rates are stationary, the Fisher equation ($r = i - \pi$) implies a cointegration relatinsip between the nominal interest rate (i) and inflation π.
2. We have no a priori regarding the cointegration relationship. We have to estimate (and test) the potential cointegration relationship.

In the first case, we proceed in two steps: a. If π^* is suspected to be a cointegrating vector, we can then use unit-root or stationarity tests on $z_t = {\pi^*}' y_t$ (see Section 5). b. Estimate Eq. (6.7) by OLS.

In the following, we focus on the second case, when there is at most one cointegration relationship. (In the general case, one can for instance implement the Johansen (1991) approach.)

Engle and Granger (1987) propose a two-step estimation procedure to estimate error-correction models. This method proceeds under the assumption that there is a single cointegration equation, i.e. $A' = [\alpha_1, ..., \alpha_n]$. (Recall that $\Phi(1) = BA'$, see Eq. (6.5).) Without loss of generality, one can set $\alpha_1 = 1$. In that case, the cointegration relationship, if it exists, is of the form:

$$y_{1,t} = -\alpha_2 y_{2,t} - \cdots - \alpha_n y_{n,t} + z_t,$$

where $z_t \sim I(0)$.

The first step consists in estimating the previous equation by OLS (regression of $y_{1,t}$ on $y_{2,t}, ..., y_{2,t}$). The second step consists in estimating Eq. (6.7) also by OLS, after having replaced z_{t-1} by the (lagged) residuals of the first step OLS regression ($\widehat{z}_{t-1}$). Because of high speed convergence of the first-step regression (the convergence is in $1/T$, see Eq. (6.2)), the asymptotic properties of the second-step estimates are the standard ones. That is, one can use the standard t-statistic to assess the statistical significativity of the parameters.

It remains to explain how to test for the existence of a cointegration relationship. This amounts to testing whether z_t is stationary. Note however that we do not observe the "true" z_t, but only OLS-based estimates $\widehat{z}_t$. Therefore, the critical values of the usual unit-root tests are not the same. The appropriate critical values are given by Phillips and Ouliaris (1990).

Example 6.3 (VECM for US inflation and nominal interest rate)**.** The data are as in Example 5.1. In the first step, we compute z_t and use Phillips and Ouliaris (1990)'s test to see whether the two variables are cointegrated:

```
##
##  Phillips-Ouliaris Cointegration Test
##
## data:  cbind(i, infl)
## Phillips-Ouliaris demeaned = -29.484, Truncation lag parameter = 2,
## p-value = 0.01
```

We reject the null hypothesis of unit root in the residuals. That is, the results are in favor of cointegration. Let us then esitmate the VECM model; this amounts to running two OLS regressions:

Table 6.1 reports the estimated coefficients in `eq1` and `eq2`, together with their p-values:

Table 6.1: Results of the second-stage OLS regressions (Engle-Granger approach). The first variable is the short-term nominal interest rate; the second variable is inflation.

	Coef Y1	p-values Y1	Coef Y2	p-values Y2
(Intercept)	-0.0189	0.8093	-0.0047	0.9457
z_1	-0.1040	0.0014	-0.0034	0.9046
dinfl_1	0.0224	0.7852	-0.3845	0.0000
di_1	-0.0841	0.2225	0.1196	0.0473
dinfl_2	0.0322	0.6864	-0.2238	0.0014
di_2	-0.0393	0.5625	0.0828	0.1626

Imporantly, one of the parameters associated with z_{t-1} (parameters that are called speeds of adjustement) is significant and with the expected signs: if the short term nominal interest rate i_t is high then z_t becomes positive (since, up to an intercept, $z_t = i_t - \beta\pi_t$, where β comes from the first-step regression), but since $B_{1,1}$ is negative (it is equal to -0.104), a positive z_t will generate a negative correction (i.e., $B_{1,1}z_t$) for i_{t+1} on date $t+1$, thereby "correcting" the high level of i_t.

6.5 Exercise

Exercise 6.1 (A bivariate cointegrated VAR)**. PS4 | Core | Analytical**
Review Section 6.

Consider

$$Y_t = \begin{bmatrix} .7 & -.6 \\ -.2 & .6 \end{bmatrix} Y_{t-1} + \varepsilon_t,$$

where ε_t is i.i.d. Gaussian with mean zero and nonsingular covariance matrix. The coefficient matrix has eigenvalues 1 and 0.3.

1. Express ΔY_t in terms of Y_{t-1}.
2. Show that $\Delta Y_t = B z_{t-1} + \varepsilon_t$, where $z_t = \gamma' Y_t$, $\gamma' = (1, 2)$, and identify B.
3. Derive the dynamics and unconditional mean of z_t.
4. What does this imply about Y_t and its cointegration rank?
5. Compute $\mathbb{E}(\Delta y_{1,t} \mid z_{t-1} = 1)$ and $\mathbb{E}(\Delta y_{2,t} \mid z_{t-1} = 1)$, and interpret the adjustment.

Chapter 7

ARCH and GARCH models

7.1 Conditional heteroskedasticity

Many financial and macroeconomic variables are hit by shocks whose variance is not constant through time, i.e. by *heteroskedatic* shocks. Often, the conditional variance of shocks features a persistent behavior (volatility clustering). The observation that large shocks (in absolute value) tend to be followed by other important shocks has notably been established by Mandelbrot (1963). Such a situation is illustrated by Figure 7.1.

Autoregressive Conditional Heteroskedasticity (ARCH) and its generalized version (GARCH) constitute useful tools to model such time series.

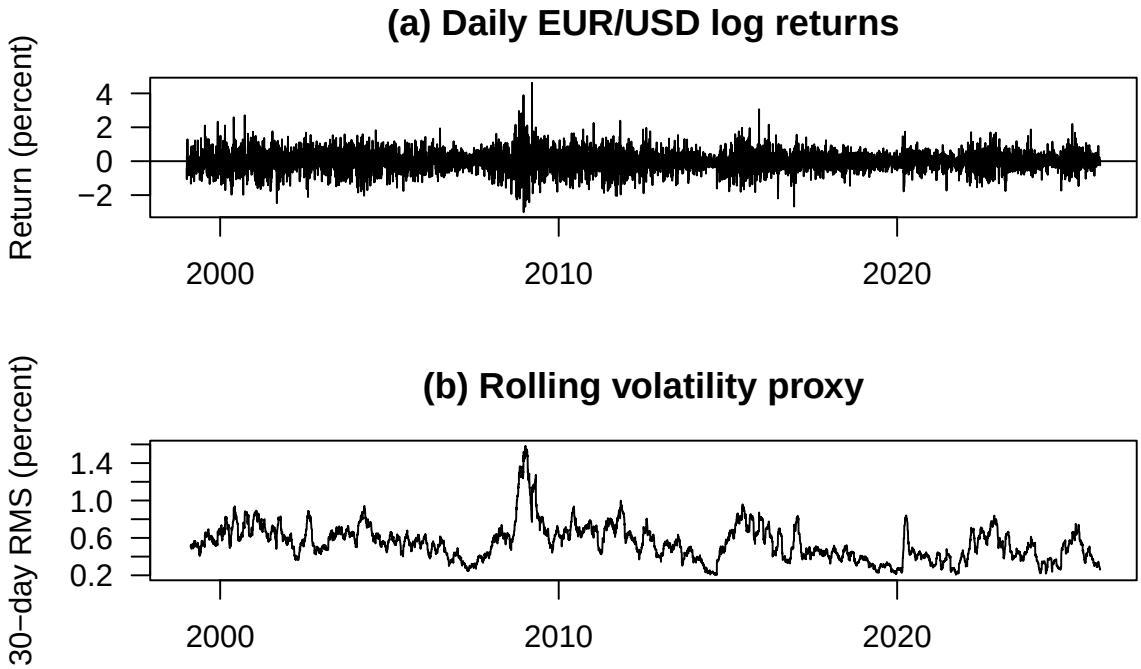


Figure 7.1: EUR/USD returns and volatility clustering: (a) daily log returns, $100\Delta \log(S_t)$, in percent; (b) 30-day rolling root mean square of daily returns, used as a descriptive volatility measure. Periods of large absolute returns coincide with sustained increases in the rolling measure. Source: Board of Governors of the Federal Reserve System, H.10 series DEXUSEU, retrieved from FRED; 5 January 1999–31 December 2025.

In order to test whether a times series of shocks $\{u_1, \dots, u_T\}$ features persistent conditional heteroskedasticity, a simple test has been designed by Engle (1982). This test consists in regressing u_t^2 on it last m lags (i.e., on $u_{t-1}^2, \dots, u_{t-m}^2$) by OLS.Under the null hypothesis that

u_t is i.i.d., we have:

$$T \times R^2 \xrightarrow{d} \chi^2(m),$$

where R^2 is the centered R^2 of the OLS regression.

Let us employ this test on the return series plotted in panel (a) of Figure 7.1:

The p-value is extremely close to zero. Hence we strongly reject the null of i.i.d. EUR/USD returns.

7.2 The ARCH model

7.2.1 The two ARCH specifications

Consider the auto-regressive process following:

$$y_t = c + \phi_1 y_{t-1} + \cdots + \phi_p y_{t-p} + u_t, \tag{7.1}$$

where u_t is a white noise (Def. 1.1), i.e. $\mathbb{E}(u_t) = 0$, $\mathbb{E}(u_t^2) = \sigma^2$ and $\mathbb{E}(u_t u_s) = 0$ if $s \neq t$. Importantly, note that while the *unconditional variance* of u_t is σ^2, its *conditional variance* can be time-varying. This is the case in the context of (G)ARCH models.

In the ARCH(m) model, u_t follows:

$$u_t^2 = \zeta + \alpha_1 u_{t-1}^2 + \cdots + \alpha_m u_{t-m}^2 + w_t, \tag{7.2}$$

where w_t is a non-autocorrelated white noise process that is exogenous to u_t, in the sense that:

$$\mathbb{E}(w_t | u_{t-1}, u_{t-2}, \dots) = 0.$$

In this case:

$$\mathbb{E}(u_t^2 | u_{t-1}, u_{t-2}, \dots) = \zeta + \alpha_1 u_{t-1}^2 + \cdots + \alpha_m u_{t-m}^2. \tag{7.3}$$

For Eq. (7.2) to make sense, it has to be the case that

$$\zeta + \alpha_1 u_{t-1}^2 + \cdots + \alpha_m u_{t-m}^2 + w_t \geq 0$$

for all realisations of $\{u_t\}$. This is the case if $w_t > -\zeta$, with $\zeta > 0$, and if $\alpha_i \geq 0$ for $i \in \{1, \dots, m\}$. Assuming this is the case, u_t^2 is covariance-stationary if the roots of:

$$g(z) = 1 - \alpha_1 z - \cdots - \alpha_m z^m = 0$$

lie outside the unit circle (Prop. 2.5). A necessary condition for not having a root between 0 and 1 is that $\sum_i \alpha_i < 1$.[1] This is also a sufficient condition.[2]

If $w_t > -\zeta$, with $\zeta > 0$, $\alpha_i \geq 0$ for $i \in \{1, \dots, m\}$ and $\sum_i \alpha_i < 1$, then the unconditional variance of u_t is:

$$\sigma^2 = \mathbb{E}(u_t^2) = \frac{\zeta}{1 - \sum_{i=1}^m \alpha_i}.$$

[1] If we have $1 - \sum_i \alpha_i \leq 0$, then $g(1) \leq 0$. We would then have $g(0) = 1 > 0$ and $g(1) \leq 0$. Since g is continuous, this would imply that $\exists z \in [0, 1]$ s.t. $g(z) = 0$.

[2] Indeed, for $z \in [-1, 1]$, we have $1 - \alpha_1 z - \cdots - \alpha_m z^m \geq 1 - \sum_i |\alpha_i z^i| \geq 1 - \sum_i \alpha_i > 0$.

An alternative representation of an ARCH(m) process is as follows:

$$u_t = \sqrt{h_t} v_t, \tag{7.4}$$

where v_t is an i.i.d. sequence with zero mean and unit variance, i.e.:

$$\mathbb{E}(v_t) = 0, \quad \mathbb{E}(v_t^2) = 1.$$

If h_t follows:

$$h_t = \zeta + \alpha_1 u_{t-1}^2 + \cdots + \alpha_m u_{t-m}^2, \tag{7.5}$$

then Eq. (7.3) is also true. Since $u_t^2 = h_t v_t^2$, Eq. (7.2) holds with $w_t = h_t(v_t^2 - 1)$. This alternative representation is convenient because v_t is not necessarily bounded.

7.2.2 Maximum Likelihood Estimation of an ARCH process

Assume the complete model is:

$$y_t = \mathbf{x}_t' \beta + u_t,$$

where $\mathbf{x}_t$ is a $k \times 1$ vector of explanatory variables and u_t is as specified in Eq. (7.4).

To write the likelihood function, it is convenient to condition on the first m observations. Let us denote by $\mathcal{J}_t$ the following information set:

$$\mathcal{J}_t = (y_t, y_{t-1}, \ldots, y_0, y_{-1}, \ldots, y_{-m+1}, \mathbf{x}_t, \mathbf{x}_{t-1}, \ldots, \mathbf{x}_0, \mathbf{x}_{-1}, \ldots, \mathbf{x}_{-m+1}).$$

If $v_t \sim \mathcal{N}(0,1)$, we have, for $t \geq 1$:

$$f(y_t | \mathbf{x_t}, \mathcal{J}_{t-1}) = \frac{1}{\sqrt{2\pi h_t}} \exp\left(\frac{-(y_t - \mathbf{x}_t'\beta)^2}{2h_t}\right), \tag{7.6}$$

where h_t is given by:

$$h_t = \zeta + \alpha_1 (y_{t-1} - \mathbf{x}_{t-1}'\beta)^2 + \cdots + \alpha_m (y_{t-m} - \mathbf{x}_{t-m}'\beta)^2.$$

The log likelihood function is given by:

$$\log \mathcal{L}(\theta) = \sum_{t=1}^{T} \log(f(y_t | \mathbf{x_t}, \mathcal{J}_{t-1})),$$

where θ, the vector of unknown parameters, is $[\beta', \zeta, \alpha']'$, where $\alpha = [\alpha_1, \ldots, \alpha_m]'$. The maximisation of the log likelihood is performed numerically.

Note that one can also use non-Gaussian distributions for v_t. (For that, one has to replace the normal distribution in Eq. (7.6).)

Let us fit an ARCH(2) model on the EUR/USD return data (panel (a) of Figure 7.1). For this, we make use of function `compute.garch` of package `IdSS`. This function takes four arguments:

- vector `theta` contains the model parameterization: first, ζ, then the α_i's, then for GARCH models (see Subsection 7.3 below), the δ_i's.
- vector `x` contains the observations of the process.
- `m` is the number of lags in the ARCH specification.
- `r` is the number of lags in the GARCH specification (see Subsection 7.3 below).

Table 7.1: ARCH(2) maximum-likelihood estimates. Data: daily EUR/USD log returns, in percent.

	Estim. param.	Std dev.	t-stat
zeta	-0.2676	0.0069	-38.8331
alpha1	-0.0805	0.0141	-5.7278
alpha2	-0.1224	0.0153	-8.0204

Function `compute.garch` returns a list, one entry of each is the log-likelihood associated with a parameterization $[\zeta, \alpha']'$, the second being the resulting sequence of h_t's (see Eq. (7.5)). Let us create a function that returns only the log-likelihood:

Now, let us maximize the log-likelihood:

Table 7.1 reports the estimated parameters and their standard deviation. Figure 7.2 displays the resulting 95% confidence intervals (i.e., $\pm 2\sqrt{h_t}$).

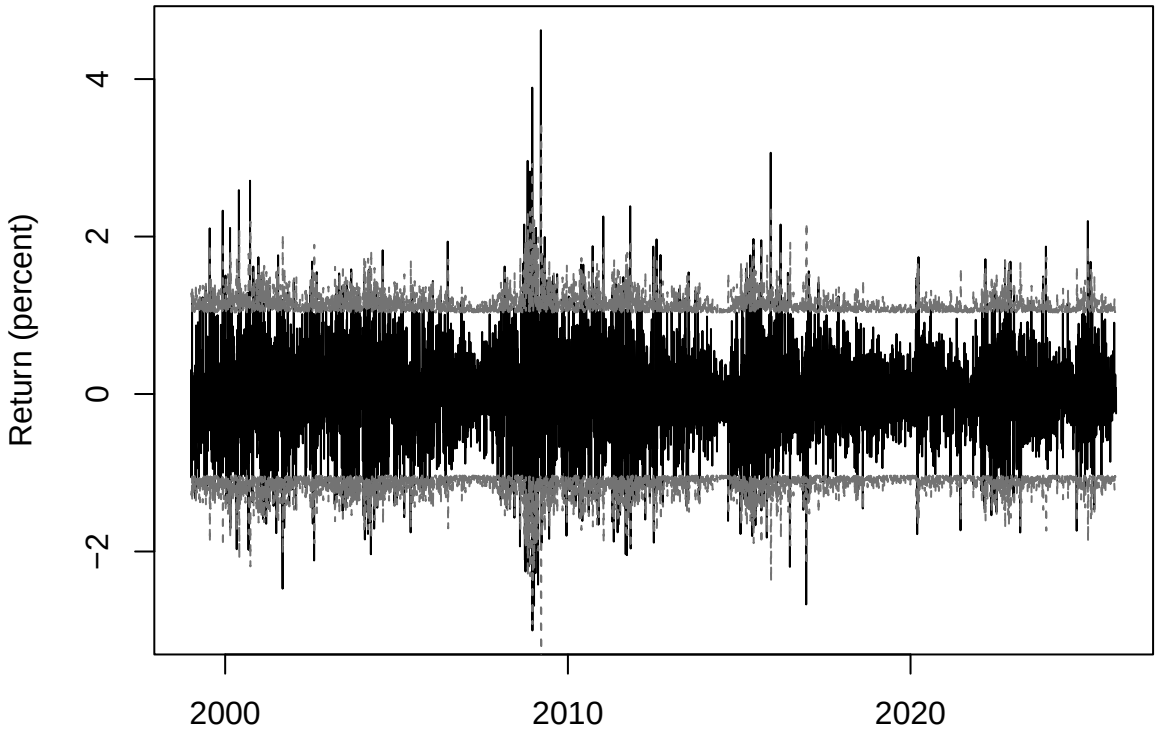


Figure 7.2: Daily EUR/USD log returns and conditional-volatility bands from the estimated ARCH(2) model. The solid black line shows returns; dashed grey lines show $\pm 2\sqrt{h_t}$. The estimation sample runs from 5 January 1999 through 31 December 2025.

7.3 The GARCH model

One can generalize the model and replace Eq. (7.5) with:

$$\begin{aligned} h_t &= (1 - \delta_1 - \delta_2 - \cdots - \delta_r)\zeta + \\ &\quad \delta_1 h_{t-1} + \delta_2 h_{t-2} + \cdots + \delta_r h_{t-r} + \\ &\quad \alpha_1 u_{t-1}^2 + \cdots + \alpha_m u_{t-m}^2. \end{aligned} \tag{7.7}$$

Table 7.2: GARCH(1,1) maximum-likelihood estimates. Data: daily EUR/USD log returns, in percent.

	Estim. param.	Std dev.	t-stat
zeta	-0.2128	NaN	NaN
alpha1	-0.0591	NaN	NaN
delta1	0.8600	NaN	NaN

This generalised autoregressive conditional heteroskedasticity model is denoted by GARCH(r,m). Non-negativity is satisfied as soon as $\kappa > 0$, $\alpha_j \geq 0$, $\delta_j \geq 0$ for $j \leq p$.

Denoting $u_t^2 - h_t$ by w_t, and $(1 - \delta_1 - \cdots - \delta_r)\zeta$ by κ, it can be checked that:[3]

$$\begin{aligned} u_t^2 &= \kappa + (\delta_1 + \alpha_1)u_{t-1}^2 + (\delta_2 + \alpha_2)u_{t-2}^2 + ... \\ &\quad +(\delta_p + \alpha_p)u_{t-p}^2 + w_t - \delta_1 w_{t-1} - \cdots - \delta_r w_{t-r}, \end{aligned} \tag{7.8}$$

where $p = max(m, r)$, $\alpha_j = 0$ for $j > m$ and $\delta_j = 0$ for $j > r$.

We have $w_t = h_t(v_t^2 - 1)$. Under regularity assumptions, $\{w_t\}$ is a white noise. Hence, u_t^2 follows an ARMA(p,r) process. Accordingly, it comes that u_t^2 is covariance stationary if the roots of:

$$1 - (\delta_1 + \alpha_1)z - \cdots - (\delta_1 + \alpha_1)z^p = 0$$

lie outside the unit circle (Prop. 2.5). When the $\delta_i + \alpha_i$ are nonnegative, and using the same reasoning as for ARCH models, this is the case iff:

$$\sum_i \delta_i + \sum_i \alpha_i < 1.$$

In that case, the unconditional variance of u_t, i.e. the unconditional mean of u_t^2, is:

$$\mathbb{E}(u_t^2) = \sigma^2 = \frac{\kappa}{1 - \sum_i \delta_i - \sum_i \alpha_i}.$$

GARCH models can also be estimated by the ML approach.Table 7.2 reports the estimated parameters when fitting an GARCH(1,1) model on the EUR/USD return dataset. Figure

Figure 7.3 compare the condiitonal standard deviations ($\sqrt{h_t}$) resulting from the ARCH(2) and the GARCH(1,1) specifications.

```
## initial  value 6589.631790
## iter  10 value 5610.514412
## final  value 5610.514412
## stopped after 10 iterations
```

```
## Warning in sqrt(diag(solve(res.opt$hessian))): NaNs produced
```

7.4 ARCH-in-mean

Another extension of the ARCH model is the **ARCH-in-Mean**, or ARCH-M model. That model is close to that specified by Eqs. (7.1), (7.4) and (7.5), but it also allows for a potential effect of

[3]Note that $w_t = h_t(v_t^2 - 1)$ is a martingale difference sequence (see Def. 1.2) because v_t is a zero-mean unit-root i.i.d. sequence.

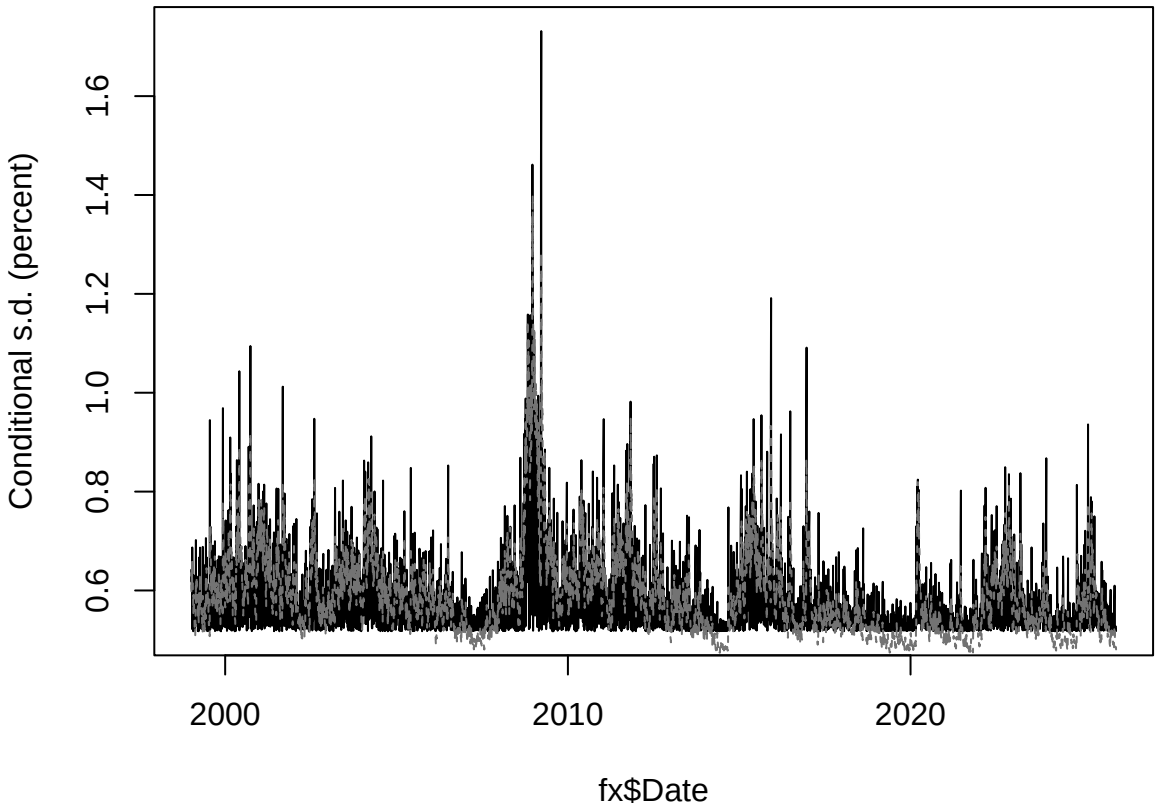


Figure 7.3: Estimated conditional standard deviation of daily EUR/USD returns. The solid black line is the ARCH(2) estimate; the dashed grey line is the GARCH(1,1) estimate. Both models are estimated by Gaussian maximum likelihood over 5 January 1999–31 December 2025.

h_t on $\mathbb{E}_{t-1}(y_t)$:

$$\begin{aligned} y_t &= c + \phi_1 y_{t-1} + \cdots + \phi_p y_{t-p} + \delta h_t + u_t \\ u_t &= \sqrt{h_t} v_t \\ h_t &= \zeta + \alpha_1 u_{t-1}^2 + \cdots + \alpha_m u_{t-m}^2, \end{aligned}$$

where v_t is a zero-mean unit-variance i.i.d. sequence.

Chapter 8

Appendix

8.1 Principal component analysis (PCA)

Principal component analysis (PCA) is a classical and easy-to-use statistical method to reduce the dimension of large datasets containing variables that are linearly driven by a relatively small number of factors. This approach is widely used in data analysis and image compression.

Suppose that we have T observations of a n-dimensional random vector x, denoted by $x_1, x_2, \dots, x_T$. We suppose that each component of x is of mean zero.

Let denote with X the matrix given by $\begin{bmatrix} x_1 & x_2 & \dots & x_T \end{bmatrix}'$. Denote the jth column of X by X_j.

We want to find the linear combination of the x_i's ($x.u$), with $\|u\| = 1$, with "maximum variance." That is, we want to solve:

$$\begin{array}{ll} \arg\max\limits_{u} & u'X'Xu. \\ \text{s.t.} & \|u\| = 1 \end{array} \tag{8.1}$$

Since $X'X$ is a positive definite matrix, it admits the following decomposition:

$$\begin{aligned} X'X &= PDP' \\ &= P \begin{bmatrix} \lambda_1 & & \\ & \ddots & \\ & & \lambda_n \end{bmatrix} P', \end{aligned}$$

where P is an orthogonal matrix whose columns are the eigenvectors of $X'X$.

We can order the eigenvalues such that $\lambda_1 \geq \dots \geq \lambda_n$. (Since $X'X$ is positive definite, all these eigenvalues are positive.)

Since P is orthogonal, we have $u'X'Xu = u'PDP'u = y'Dy$ where $\|y\| = 1$. Therefore, we have $y_i^2 \leq 1$ for any $i \leq n$.

As a consequence:

$$y'Dy = \sum_{i=1}^{n} y_i^2 \lambda_i \leq \lambda_1 \sum_{i=1}^{n} y_i^2 = \lambda_1.$$

It is easily seen that the maximum is reached for $y = [1, 0, \cdots, 0]'$. Therefore, the maximum of the optimization program (Eq. (8.1)) is obtained for $u = P\,[1, 0, \cdots, 0]'$. That is, u is the eigenvector of $X'X$ that is associated with its larger eigenvalue (first column of P).

Let us denote with F the vector that is given by the matrix product XP. The columns of F, denoted by F_j, are called **factors**. We have:

$$F'F = P'X'XP = D.$$

Therefore, in particular, the F_j's are orthogonal.

Since $X = FP'$, the X_j's are linear combinations of the factors. Let us then denote with $\hat{X}_{i,j}$ the part of X_i that is explained by factor F_j, we have:

$$\begin{aligned} \hat{X}_{i,j} &= p_{ij} F_j \\ X_i &= \sum_j \hat{X}_{i,j} = \sum_j p_{ij} F_j. \end{aligned}$$

Consider the share of variance that is explained—through the n variables $(X_1, \dots, X_n)$—by the first factor F_1:

$$\frac{\sum_i \hat{X}'_{i,1} \hat{X}_{i,1}}{\sum_i X'_i X_i} = \frac{\sum_i p_{i1} F'_1 F_1 p_{i1}}{tr(X'X)} = \frac{\sum_i p_{i1}^2 \lambda_1}{tr(X'X)} = \frac{\lambda_1}{\sum_i \lambda_i}.$$

Intuitively, if the first eigenvalue is large, it means that the first factor captures a large share of the fluctutaions of the n X_i's.

By the same token, it is easily seen that the fraction of the variance of the n variables that is explained by factor j is given by:

$$\frac{\sum_i \hat{X}'_{i,j} \hat{X}_{i,j}}{\sum_i X'_i X_i} = \frac{\lambda_j}{\sum_i \lambda_i}.$$

Let us illustrate PCA on the term structure of yields. The term structure of yields (or yield curve) is known to be driven by only a small number of factors (e.g., Litterman and Scheinkman (1991)). One can employ PCA to recover such factors. The data used in the example below are taken from the FRED database. The second panel shows the factor loadings: the first factor (solid black line) is a level factor, the second (dashed dark-grey line) is a slope factor, and the third (dotted light-grey line) is a curvature factor.

To run a PCA, one simply has to apply function `prcomp` to a matrix of data:

Let us know visualize some results. The first plot of Figure 8.1 shows the share of total variance explained by the different principal components (PCs). The second plot shows the factor loadings. The two bottom plots show how yields (in black) are fitted by linear combinations of the first two PCs only.

8.2 Linear algebra: definitions and results

Definition 8.1 (Eigenvalues)**.** The eigenvalues of of a matrix M are the numbers λ for which:

$$|M - \lambda I| = 0,$$

where $|\bullet|$ is the determinant operator.

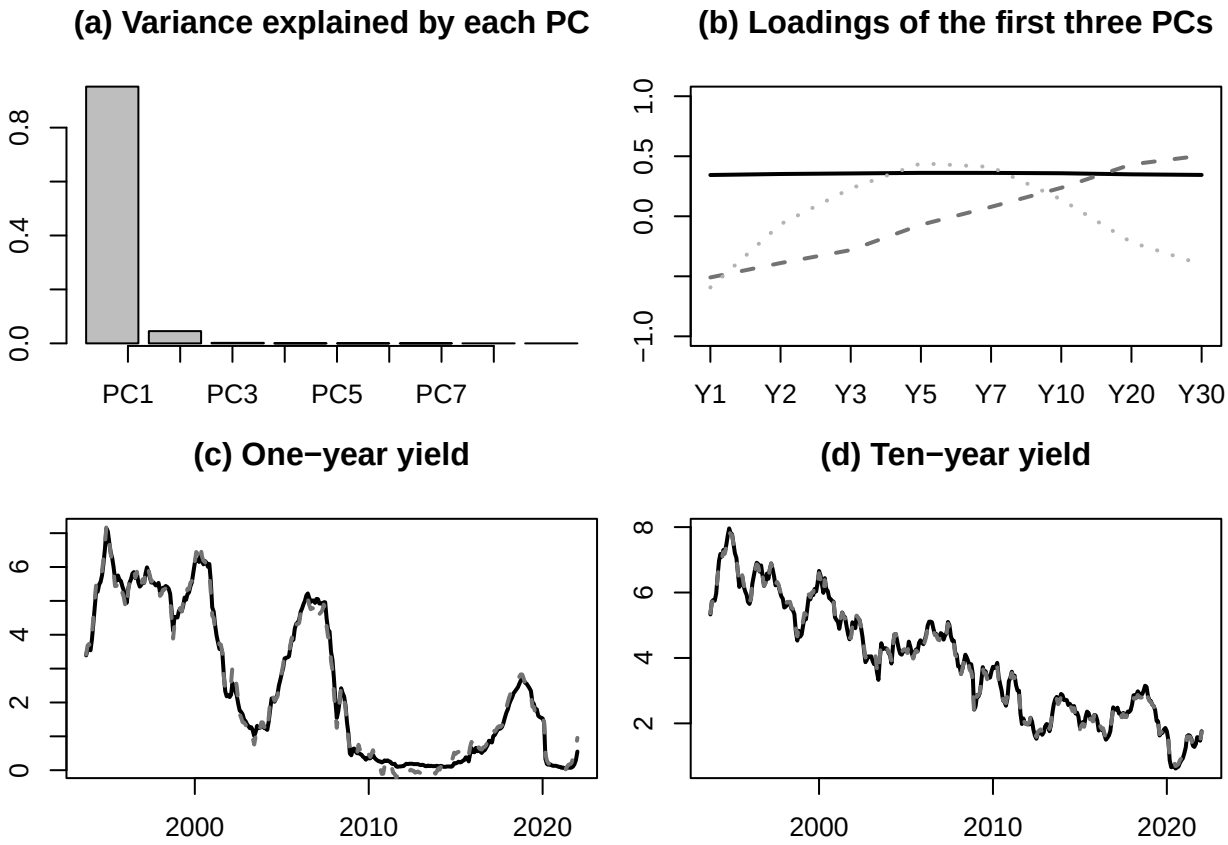


Figure 8.1: Principal-component analysis of eight monthly U.S. Treasury yields: (a) share of standardized yield variance explained by each component; (b) loadings of the first three components, shown respectively as solid black, dashed dark-grey, and dotted light-grey lines; (c) observed and two-component fitted one-year yield; and (d) observed and two-component fitted ten-year yield. In panels (c) and (d), observed yields are solid black and fitted yields are dashed grey. Source: Federal Reserve Board H.15 series retrieved through FRED.

Proposition 8.1 (Properties of the determinant)**.** *We have:*

- $|MN| = |M| \times |N|$.
- $|M^{-1}| = |M|^{-1}$.
- *If M admits the diagonal representation $M = TDT^{-1}$, where D is a diagonal matrix whose diagonal entries are $\{\lambda_i\}_{i=1,\dots,n}$, then:*

$$|M - \lambda I| = \prod_{i=1}^{n} (\lambda_i - \lambda).$$

Definition 8.2 (Moore-Penrose inverse)**.** If $M \in \mathbb{R}^{m \times n}$, then its Moore-Penrose pseudo inverse (exists and) is the unique matrix $M^* \in \mathbb{R}^{n \times m}$ that satisfies:

i. $MM^*M = M$
ii. $M^*MM^* = M^*$
iii. $(MM^*)' = MM^*$ .iv $(M^*M)' = M^*M$.

Proposition 8.2 (Properties of the Moore-Penrose inverse)**.**

- *If M is invertible then $M^* = M^{-1}$.*
- *The pseudo-inverse of a zero matrix is its transpose.* *
- *
  The pseudo-inverse of the pseudo-inverse is the original matrix.

Definition 8.3 (Idempotent matrix)**.** Matrix M is idempotent if $M^2 = M$.

If M is a symmetric idempotent matrix, then $M'M = M$.

Proposition 8.3 (Roots of an idempotent matrix)**.** *The eigenvalues of an idempotent matrix are either 1 or 0.*

Proof. If λ is an eigenvalue of an idempotent matrix M then $\exists x \neq 0$ s.t. $Mx = \lambda x$. Hence $M^2x = \lambda Mx \Rightarrow (1-\lambda)Mx = 0$. Either all element of Mx are zero, in which case $\lambda = 0$ or at least one element of Mx is nonzero, in which case $\lambda = 1$. □

Proposition 8.4 (Idempotent matrix and chi-square distribution)**.** *The rank of a symmetric idempotent matrix is equal to its trace.*

Proof. The result follows from Prop. 8.3, combined with the fact that the rank of a symmetric matrix is equal to the number of its nonzero eigenvalues. □

Proposition 8.5 (Constrained least squares)**.** *The solution of the following optimisation problem:*

$$\min_{\beta} \quad ||\mathbf{y} - \mathbf{X}\beta||^2$$
$$\text{subject to } \mathbf{R}\beta = \mathbf{q}$$

is given by:

$$\boxed{\beta^r = \beta_0 - (\mathbf{X}'\mathbf{X})^{-1}\mathbf{R}'\{\mathbf{R}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{R}'\}^{-1}(\mathbf{R}\beta_0 - \mathbf{q}),}$$

where $\beta_0 = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y}$.

Proof. See for instance Jackman, 2007. □

Proposition 8.6 (Inverse of a partitioned matrix)**.** *We have:*

$$\begin{bmatrix} \mathbf{A}_{11} & \mathbf{A}_{12} \\ \mathbf{A}_{21} & \mathbf{A}_{22} \end{bmatrix}^{-1} =$$
$$\begin{bmatrix} (\mathbf{A}_{11} - \mathbf{A}_{12}\mathbf{A}_{22}^{-1}\mathbf{A}_{21})^{-1} & -\mathbf{A}_{11}^{-1}\mathbf{A}_{12}(\mathbf{A}_{22} - \mathbf{A}_{21}\mathbf{A}_{11}^{-1}\mathbf{A}_{12})^{-1} \\ -(\mathbf{A}_{22} - \mathbf{A}_{21}\mathbf{A}_{11}^{-1}\mathbf{A}_{12})^{-1}\mathbf{A}_{21}\mathbf{A}_{11}^{-1} & (\mathbf{A}_{22} - \mathbf{A}_{21}\mathbf{A}_{11}^{-1}\mathbf{A}_{12})^{-1} \end{bmatrix}.$$

Definition 8.4 (Matrix derivatives)**.** Consider a fonction $f : \mathbb{R}^K \to \mathbb{R}$. Its first-order derivative is:

$$\frac{\partial f}{\partial \mathbf{b}}(\mathbf{b}) = \begin{bmatrix} \frac{\partial f}{\partial b_1}(\mathbf{b}) \\ \vdots \\ \frac{\partial f}{\partial b_K}(\mathbf{b}) \end{bmatrix}.$$

We use the notation:

$$\frac{\partial f}{\partial \mathbf{b}'}(\mathbf{b}) = \left(\frac{\partial f}{\partial \mathbf{b}}(\mathbf{b})\right)'.$$

Proposition 8.7. *We have:*

- *If* $f(\mathbf{b}) = A'\mathbf{b}$ *where* A *is a* $K \times 1$ *vector then* $\frac{\partial f}{\partial \mathbf{b}}(\mathbf{b}) = A$.
- *If* $f(\mathbf{b}) = \mathbf{b}'A\mathbf{b}$ *where* A *is a* $K \times K$ *matrix, then* $\frac{\partial f}{\partial \mathbf{b}}(\mathbf{b}) = 2A\mathbf{b}$.

Proposition 8.8 (Square and absolute summability)**.** *We have:*

$$\underbrace{\sum_{i=0}^{\infty} |\theta_i| < +\infty}_{\text{Absolute summability}} \quad \Rightarrow \quad \underbrace{\sum_{i=0}^{\infty} \theta_i^2 < +\infty}_{\text{Square summability}} \quad .$$

Proof. See Appendix 3.A in Hamilton. Idea: Absolute summability implies that there exist N such that, for $j > N$, $|\theta_j| < 1$ (deduced from Cauchy criterion, Theorem 8.2 and therefore $\theta_j^2 < |\theta_j|$. □

8.3 Statistical analysis: definitions and results

8.3.1 Moments and statistics

Definition 8.5 (Partial correlation)**.** The **partial correlation** between y and z, controlling for some variables **X** is the sample correlation between y^* and z^*, where the latter two variables are the residuals in regressions of y on **X** and of z on **X**, respectively.

This correlation is denoted by $r_{yz}^{\mathbf{x}}$. By definition, we have:

$$r_{yz}^{\mathbf{x}} = \frac{\mathbf{z}^{*\prime}\mathbf{y}^*}{\sqrt{(\mathbf{z}^{*\prime}\mathbf{z}^*)(\mathbf{y}^{*\prime}\mathbf{y}^*)}}. \tag{8.2}$$

Definition 8.6 (Skewness and kurtosis)**.** Let Y be a random variable whose fourth moment exists. The expectation of Y is denoted by μ.

- The skewness of Y is given by:

$$\frac{\mathbb{E}[(Y-\mu)^3]}{\{\mathbb{E}[(Y-\mu)^2]\}^{3/2}}.$$

- The kurtosis of Y is given by:

$$\frac{\mathbb{E}[(Y-\mu)^4]}{\{\mathbb{E}[(Y-\mu)^2]\}^{2}}.$$

Theorem 8.1 (Cauchy-Schwarz inequality)**.** *We have:*

$$|\mathbb{C}ov(X,Y)| \leq \sqrt{\mathbb{V}ar(X)\mathbb{V}ar(Y)}$$

and, if $X \neq=$ and $Y \neq 0$, the equality holds iff X and Y are the same up to an affine transformation.

Proof. If $\mathbb{V}ar(X) = 0$, this is trivial. If this is not the case, then let's define Z as $Z = Y - \frac{\mathbb{C}ov(X,Y)}{\mathbb{V}ar(X)}X$. It is easily seen that $\mathbb{C}ov(X,Z) = 0$. Then, the variance of $Y = Z + \frac{\mathbb{C}ov(X,Y)}{\mathbb{V}ar(X)}X$ is equal to the sum of the variance of Z and of the variance of $\frac{\mathbb{C}ov(X,Y)}{\mathbb{V}ar(X)}X$, that is:

$$\mathbb{V}ar(Y) = \mathbb{V}ar(Z) + \left(\frac{\mathbb{C}ov(X,Y)}{\mathbb{V}ar(X)}\right)^2 \mathbb{V}ar(X) \geq \left(\frac{\mathbb{C}ov(X,Y)}{\mathbb{V}ar(X)}\right)^2 \mathbb{V}ar(X).$$

The equality holds iff $\mathbb{V}ar(Z) = 0$, i.e. iff $Y = \frac{\mathbb{C}ov(X,Y)}{\mathbb{V}ar(X)}X + cst$. □

Definition 8.7 (Mean ergodicity)**.** The covariance-stationary process y_t is ergodic for the mean if:

$$\text{plim}_{T\to+\infty}\frac{1}{T}\sum_{t=1}^{T}y_t=\mathbb{E}(y_t).$$

Definition 8.8 (Second-moment ergodicity)**.** The covariance-stationary process y_t is ergodic for second moments if, for all j:

$$\text{plim}_{T\to+\infty}\frac{1}{T}\sum_{t=1}^{T}(y_t-\mu)(y_{t-j}-\mu)=\gamma_j.$$

It should be noted that ergodicity and stationarity are different properties. Typically if the process $\{x_t\}$ is such that, $\forall t$, $x_t\equiv y$, where $y\sim\mathcal{N}(0,1)$ (say), then $\{x_t\}$ is stationary but not ergodic.

8.3.2 Standard distributions

Definition 8.9 (F distribution)**.** Consider $n=n_1+n_2$ i.i.d. $\mathcal{N}(0,1)$ r.v. X_i. If the r.v. F is defined by:

$$F=\frac{\sum_{i=1}^{n_1}X_i^2}{\sum_{j=n_1+1}^{n_1+n_2}X_j^2}\frac{n_2}{n_1}$$

then $F\sim\mathcal{F}(n_1,n_2)$. (See Table 8.4 for quantiles.)

Definition 8.10 (Student-t distribution)**.** Z follows a Student-t (or t) distribution with ν degrees of freedom (d.f.) if:

$$Z=X_0\Big/\sqrt{\frac{\sum_{i=1}^{\nu}X_i^2}{\nu}},\quad X_i\sim i.i.d.\mathcal{N}(0,1).$$

We have $\mathbb{E}(Z)=0$, and $\mathbb{V}ar(Z)=\frac{\nu}{\nu-2}$ if $\nu>2$. (See Table 8.2 for quantiles.)

Definition 8.11 (Chi-square distribution)**.** Z follows a χ^2 distribution with ν d.f. if $Z=\sum_{i=1}^{\nu}X_i^2$ where $X_i\sim i.i.d.\mathcal{N}(0,1)$. We have $\mathbb{E}(Z)=\nu$. (See Table 8.3 for quantiles.)

Definition 8.12 (Cauchy distribution)**.**

The probability distribution function of the Cauchy distribution defined by a location parameter μ and a scale parameter γ is:

$$f(x) = \frac{1}{\pi\gamma\left(1 + \left[\frac{x-\mu}{\gamma}\right]^2\right)}.$$

The mean and variance of this distribution are undefined.

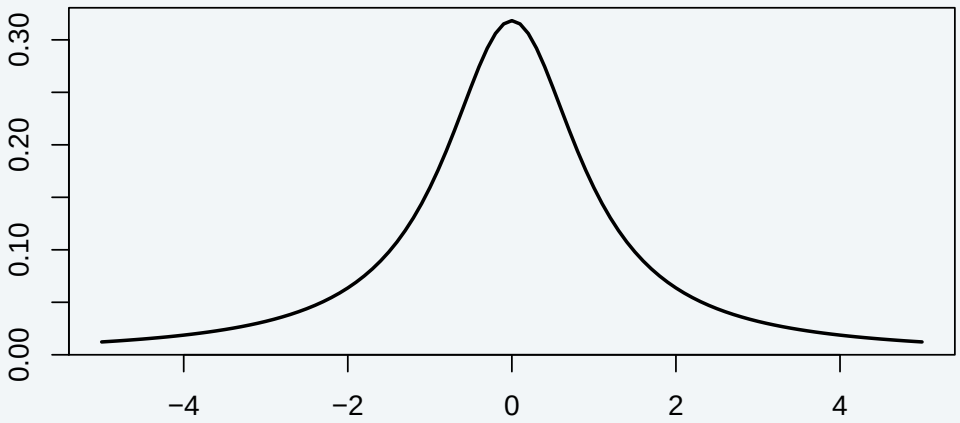


Figure 8.2: Pdf of the Cauchy distribution ($\mu = 0, \gamma = 1$).

Proposition 8.9 (Inner product of a multivariate Gaussian variable)**.** *Let X be a n-dimensional multivariate Gaussian variable: $X \sim \mathcal{N}(0, \Sigma)$. We have:*

$$X'\Sigma^{-1}X \sim \chi^2(n).$$

Proof. Because Σ is a symmetrical definite positive matrix, it admits the spectral decomposition PDP' where P is an orthogonal matrix (i.e. $PP' = Id$) and D is a diagonal matrix with non-negative entries. Denoting by $\sqrt{D^{-1}}$ the diagonal matrix whose diagonal entries are the inverse of those of D, it is easily checked that the covariance matrix of $Y := \sqrt{D^{-1}}P'X$ is Id. Therefore Y is a vector of uncorrelated Gaussian variables. The properties of Gaussian variables imply that the components of Y are then also independent. Hence $Y'Y = \sum_i Y_i^2 \sim \chi^2(n)$.

It remains to note that $Y'Y = X'PD^{-1}P'X = X'\mathbb{V}ar(X)^{-1}X$ to conclude. □

8.3.3 Stochastic convergences

Definition 8.13 (Convergence in probability)**.** The random variable sequence x_n converges in probability to a constant c if $\forall \varepsilon$, $\lim_{n\to\infty} \mathbb{P}(|x_n - c| > \varepsilon) = 0$.

It is denoted as: plim $x_n = c$.

Definition 8.14 (Convergence in the Lr norm)**.** x_n converges in the r-th mean (or in the L^r-norm) towards x, if $\mathbb{E}(|x_n|^r)$ and $\mathbb{E}(|x|^r)$ exist and if

$$\lim_{n\to\infty} \mathbb{E}(|x_n - x|^r) = 0.$$

It is denoted as: $x_n \overset{L^r}{\to} c$.

For $r = 2$, this convergence is called **mean square convergence**.

Definition 8.15 (Almost sure convergence)**.** The random variable sequence x_n converges almost surely to c if $\mathbb{P}(\lim_{n\to\infty} x_n = c) = 1$.

It is denoted as: $x_n \overset{a.s.}{\to} c$.

Definition 8.16 (Convergence in distribution)**.** x_n is said to converge in distribution (or in law) to x if

$$\lim_{n\to\infty} F_{x_n}(s) = F_x(s)$$

for all s at which F_X –the cumulative distribution of X– is continuous.

It is denoted as: $x_n \overset{d}{\to} x$.

Theorem 8.2 (Cauchy criterion (non-stochastic case))**.** *We have that $\sum_{i=0}^{T} a_i$ converges ($T \to \infty$) iff, for any $\eta > 0$, there exists an integer N such that, for all $M \geq N$,*

$$\left| \sum_{i=N+1}^{M} a_i \right| < \eta.$$

Theorem 8.3 (Cauchy criterion (stochastic case))**.** *We have that $\sum_{i=0}^{T} \theta_i \varepsilon_{t-i}$ converges in mean square ($T \to \infty$) to a random variable iff, for any $\eta > 0$, there exists an integer N such that, for all $M \geq N$,*

$$\mathbb{E}\left[\left(\sum_{i=N+1}^{M} \theta_i \varepsilon_{t-i} \right)^2 \right] < \eta.$$

8.3.4 Multivariate Gaussian distribution

Proposition 8.10 (p.d.f. of a multivariate Gaussian variable)**.** *If $Y \sim \mathcal{N}(\mu, \Omega)$ and if Y is a n-dimensional vector, then the density function of Y is:*

$$\frac{1}{(2\pi)^{n/2}|\Omega|^{1/2}} \exp\left[-\frac{1}{2}(Y-\mu)' \Omega^{-1} (Y-\mu)\right].$$

8.3.5 Central limit theorem

Theorem 8.4 (Law of large numbers)**.** *The sample mean is a consistent estimator of the population mean.*

Proof. Let's denote by ϕ_{X_i} the characteristic function of a r.v. X_i. If the mean of X_i is μ then the Talyor expansion of the characteristic function is:

$$\phi_{X_i}(u) = \mathbb{E}(\exp(iuX)) = 1 + iu\mu + o(u).$$

The properties of characteristic functions imply that:

$$\phi_{\frac{1}{n}(X_1+\cdots+X_n)}(u) = \prod_{i=1}^{n} \left(1 + i\frac{u}{n}\mu + o\left(\frac{u}{n}\right)\right) \to e^{iu\mu}.$$

The facts that (a) $e^{iu\mu}$ is the characteristic function of the constant μ and (b) that a characteristic function uniquely characterises a distribution imply that the sample mean converges in distribution to the constant μ, which further implies that it converges in probability to μ. □

Theorem 8.5 (Lindberg-Levy Central limit theorem, CLT)**.** *If x_n is an i.i.d. sequence of random variables with mean μ and variance σ^2 ($\in]0, +\infty[$), then:*

$$\sqrt{n}(\bar{x}_n - \mu) \xrightarrow{d} \mathcal{N}(0, \sigma^2), \quad \textit{where} \quad \bar{x}_n = \frac{1}{n}\sum_{i=1}^{n} x_i.$$

Proof. Let us introduce the r.v. $Y_n := \sqrt{n}(\bar{X}_n - \mu)$. We have $\phi_{Y_n}(u) = \left[\mathbb{E}\left(\exp(i\frac{1}{\sqrt{n}}u(X_1-\mu))\right)\right]^n$. We have:

$$\begin{aligned}
& \left[\mathbb{E}\left(\exp\left(i\frac{1}{\sqrt{n}}u(X_1-\mu)\right)\right)\right]^n \\
= & \left[\mathbb{E}\left(1 + i\frac{1}{\sqrt{n}}u(X_1-\mu) - \frac{1}{2n}u^2(X_1-\mu)^2 + o(u^2)\right)\right]^n \\
= & \left(1 - \frac{1}{2n}u^2\sigma^2 + o(u^2)\right)^n.
\end{aligned}$$

Therefore $\phi_{Y_n}(u) \underset{n\to\infty}{\to} \exp\left(-\frac{1}{2}u^2\sigma^2\right)$, which is the characteristic function of $\mathcal{N}(0, \sigma^2)$. □

8.4 Proofs

Proof of Eq. (1.3)

Proof. We have:

$$\begin{aligned}
& T\mathbb{E}\left[(\bar{y}_T-\mu)^2\right] \\
= \; & T\mathbb{E}\left[\left(\frac{1}{T}\sum_{t=1}^{T}(y_t-\mu)\right)^2\right] = \frac{1}{T}\mathbb{E}\left[\sum_{t=1}^{T}(y_t-\mu)^2+2\sum_{s<t\leq T}(y_t-\mu)(y_s-\mu)\right] \\
= \; & \gamma_0+\frac{2}{T}\left(\sum_{t=2}^{T}\mathbb{E}\left[(y_t-\mu)(y_{t-1}-\mu)\right]\right)+\frac{2}{T}\left(\sum_{t=3}^{T}\mathbb{E}\left[(y_t-\mu)(y_{t-2}-\mu)\right]\right)+\ldots \\
& +\frac{2}{T}\left(\sum_{t=T-1}^{T}\mathbb{E}\left[(y_t-\mu)(y_{t-(T-2)}-\mu)\right]\right)+\frac{2}{T}\mathbb{E}\left[(y_t-\mu)(y_{t-(T-1)}-\mu)\right] \\
= \; & \gamma_0+2\frac{T-1}{T}\gamma_1+\cdots+2\frac{1}{T}\gamma_{T-1}.
\end{aligned}$$

Therefore:

$$\begin{aligned}
& T\mathbb{E}\left[(\bar{y}_T-\mu)^2\right]-\sum_{j=-\infty}^{+\infty}\gamma_j \\
= \; & -2\frac{1}{T}\gamma_1-2\frac{2}{T}\gamma_2-\cdots-2\frac{T-1}{T}\gamma_{T-1}-2\gamma_T-2\gamma_{T+1}+\ldots
\end{aligned}$$

And then:

$$\begin{aligned}
& \left|T\mathbb{E}\left[(\bar{y}_T-\mu)^2\right]-\sum_{j=-\infty}^{+\infty}\gamma_j\right| \\
\leq \; & 2\frac{1}{T}|\gamma_1|+2\frac{2}{T}|\gamma_2|+\cdots+2\frac{T-1}{T}|\gamma_{T-1}|+2|\gamma_T|+2|\gamma_{T+1}|+\ldots
\end{aligned}$$

For any $q\leq T$, we have:

$$\begin{aligned}
\left|T\mathbb{E}\left[(\bar{y}_T-\mu)^2\right]-\sum_{j=-\infty}^{+\infty}\gamma_j\right| \; \leq \; & 2\frac{1}{T}|\gamma_1|+2\frac{2}{T}|\gamma_2|+\cdots+2\frac{q-1}{T}|\gamma_{q-1}|+2\frac{q}{T}|\gamma_q|+ \\
& 2\frac{q+1}{T}|\gamma_{q+1}|+\cdots+2\frac{T-1}{T}|\gamma_{T-1}|+2|\gamma_T|+2|\gamma_{T+1}|+\ldots \\
\leq \; & \frac{2}{T}\left(|\gamma_1|+2|\gamma_2|+\cdots+(q-1)|\gamma_{q-1}|+q|\gamma_q|\right)+ \\
& 2|\gamma_{q+1}|+\cdots+2|\gamma_{T-1}|+2|\gamma_T|+2|\gamma_{T+1}|+\ldots
\end{aligned}$$

Consider $\varepsilon>0$. The fact that the autocovariances are absolutely summable implies that there exists q_0 such that (Cauchy criterion, Theorem 8.2):

$$2|\gamma_{q_0+1}|+2|\gamma_{q_0+2}|+2|\gamma_{q_0+3}|+\cdots<\varepsilon/2.$$

Then, if $T > q_0$, it comes that:

$$\left| T\mathbb{E}\left[(\bar{y}_T - \mu)^2\right] - \sum_{j=-\infty}^{+\infty} \gamma_j \right| \leq \frac{2}{T}\left(|\gamma_1| + 2|\gamma_2| + \cdots + (q_0-1)|\gamma_{q_0-1}| + q_0|\gamma_{q_0}|\right) + \varepsilon/2.$$

If $T \geq 2\left(|\gamma_1| + 2|\gamma_2| + \cdots + (q_0-1)|\gamma_{q_0-1}| + q_0|\gamma_{q_0}|\right)/(\varepsilon/2)$ $(= f(q_0)$, say) then

$$\frac{2}{T}\left(|\gamma_1| + 2|\gamma_2| + \cdots + (q_0-1)|\gamma_{q_0-1}| + q_0|\gamma_{q_0}|\right) \leq \varepsilon/2.$$

Then, if $T > f(q_0)$ and $T > q_0$, i.e. if $T > \max(f(q_0), q_0)$, we have:

$$\left| T\mathbb{E}\left[(\bar{y}_T - \mu)^2\right] - \sum_{j=-\infty}^{+\infty} \gamma_j \right| \leq \varepsilon.$$

□

Proof of Proposition 4.1

Proof. We have:

$$\begin{aligned}
\mathbb{E}([y_{t+1} - y^*_{t+1}]^2) &= \mathbb{E}\left([\{y_{t+1} - \mathbb{E}(y_{t+1}|x_t)\} + \{\mathbb{E}(y_{t+1}|x_t) - y^*_{t+1}\}]^2\right) \\
&= \mathbb{E}\left([y_{t+1} - \mathbb{E}(y_{t+1}|x_t)]^2\right) + \mathbb{E}\left([\mathbb{E}(y_{t+1}|x_t) - y^*_{t+1}]^2\right) \\
&\quad + 2\mathbb{E}\left([y_{t+1} - \mathbb{E}(y_{t+1}|x_t)][\mathbb{E}(y_{t+1}|x_t) - y^*_{t+1}]\right). \qquad (8.3)
\end{aligned}$$

Let us focus on the last term. We have:

$$\begin{aligned}
& \mathbb{E}\left([y_{t+1} - \mathbb{E}(y_{t+1}|x_t)][\mathbb{E}(y_{t+1}|x_t) - y^*_{t+1}]\right) \\
= \; & \mathbb{E}(\mathbb{E}([y_{t+1} - \mathbb{E}(y_{t+1}|x_t)]\underbrace{[\mathbb{E}(y_{t+1}|x_t) - y^*_{t+1}]}_{\text{function of } x_t}|x_t)) \\
= \; & \mathbb{E}([\mathbb{E}(y_{t+1}|x_t) - y^*_{t+1}]\mathbb{E}([y_{t+1} - \mathbb{E}(y_{t+1}|x_t)]|x_t)) \\
= \; & \mathbb{E}([\mathbb{E}(y_{t+1}|x_t) - y^*_{t+1}]\underbrace{[\mathbb{E}(y_{t+1}|x_t) - \mathbb{E}(y_{t+1}|x_t)]}_{=0}) = 0.
\end{aligned}$$

Therefore, Eq. (8.3) becomes:

$$\begin{aligned}
& \mathbb{E}([y_{t+1} - y^*_{t+1}]^2) \\
= \; & \underbrace{\mathbb{E}\left([y_{t+1} - \mathbb{E}(y_{t+1}|x_t)]^2\right)}_{\geq 0 \text{ and does not depend on } y^*_{t+1}} + \underbrace{\mathbb{E}\left([\mathbb{E}(y_{t+1}|x_t) - y^*_{t+1}]^2\right)}_{\geq 0 \text{ and depends on } y^*_{t+1}}.
\end{aligned}$$

This implies that $\mathbb{E}([y_{t+1} - y^*_{t+1}]^2)$ is always larger than $\mathbb{E}([y_{t+1} - \mathbb{E}(y_{t+1}|x_t)]^2)$, and is therefore minimized if the second term is equal to zero, that is if $\mathbb{E}(y_{t+1}|x_t) = y^*_{t+1}$. □

Proof of Proposition 3.2

Proof. Using Proposition 8.10, we obtain that, conditionally on x_1, the log-likelihood is given by

$$\begin{aligned}\log \mathcal{L}(Y_T;\theta) &= -(Tn/2)\log(2\pi) + (T/2)\log\left|\Omega^{-1}\right| \\ &\quad -\frac{1}{2}\sum_{t=1}^{T}\left[(y_t - \Pi' x_t)' \Omega^{-1}(y_t - \Pi' x_t)\right].\end{aligned}$$

Let's rewrite the last term of the log-likelihood:

$$\begin{aligned}\sum_{t=1}^{T}\left[(y_t - \Pi' x_t)' \Omega^{-1}(y_t - \Pi' x_t)\right] &= \\ \sum_{t=1}^{T}\left[\left(y_t - \hat{\Pi}' x_t + \hat{\Pi}' x_t - \Pi' x_t\right)' \Omega^{-1}\left(y_t - \hat{\Pi}' x_t + \hat{\Pi}' x_t - \Pi' x_t\right)\right] &= \\ \sum_{t=1}^{T}\left[\left(\hat{\varepsilon}_t + (\hat{\Pi} - \Pi)' x_t\right)' \Omega^{-1}\left(\hat{\varepsilon}_t + (\hat{\Pi} - \Pi)' x_t\right)\right],\end{aligned}$$

where the jth element of the $(n \times 1)$ vector $\hat{\varepsilon}_t$ is the sample residual, for observation t, from an OLS regression of $y_{j,t}$ on x_t. Expanding the previous equation, we get:

$$\begin{aligned}&\sum_{t=1}^{T}\left[(y_t - \Pi' x_t)' \Omega^{-1}(y_t - \Pi' x_t)\right] = \sum_{t=1}^{T}\hat{\varepsilon}_t' \Omega^{-1}\hat{\varepsilon}_t \\ &+2\sum_{t=1}^{T}\hat{\varepsilon}_t' \Omega^{-1}(\hat{\Pi} - \Pi)' x_t + \sum_{t=1}^{T} x_t'(\hat{\Pi} - \Pi)\Omega^{-1}(\hat{\Pi} - \Pi)' x_t.\end{aligned}$$

Let's apply the trace operator on the second term (that is a scalar):

$$\begin{aligned}\sum_{t=1}^{T}\hat{\varepsilon}_t' \Omega^{-1}(\hat{\Pi} - \Pi)' x_t &= Tr\left(\sum_{t=1}^{T}\hat{\varepsilon}_t' \Omega^{-1}(\hat{\Pi} - \Pi)' x_t\right) \\ = Tr\left(\sum_{t=1}^{T}\Omega^{-1}(\hat{\Pi} - \Pi)' x_t \hat{\varepsilon}_t'\right) &= Tr\left(\Omega^{-1}(\hat{\Pi} - \Pi)' \sum_{t=1}^{T} x_t \hat{\varepsilon}_t'\right).\end{aligned}$$

Given that, by construction (property of OLS estimates), the sample residuals are orthogonal to the explanatory variables, this term is zero. Introducing $\tilde{x}_t = (\hat{\Pi} - \Pi)' x_t$, we have

$$\sum_{t=1}^{T}\left[(y_t - \Pi' x_t)' \Omega^{-1}(y_t - \Pi' x_t)\right] = \sum_{t=1}^{T}\hat{\varepsilon}_t' \Omega^{-1}\hat{\varepsilon}_t + \sum_{t=1}^{T}\tilde{x}_t' \Omega^{-1}\tilde{x}_t.$$

Since Ω is a positive definite matrix, Ω^{-1} is as well. Consequently, the smallest value that the last term can take is obtained for $\tilde{x}_t = 0$, i.e. when $\Pi = \hat{\Pi}$.

The MLE of Ω is the matrix $\hat{\Omega}$ that maximizes $\Omega \xrightarrow{\ell} L(Y_T; \hat{\Pi}, \Omega)$. We have:

$$\log \mathcal{L}(Y_T; \hat{\Pi}, \Omega) = -(Tn/2)\log(2\pi) + (T/2)\log\left|\Omega^{-1}\right| - \frac{1}{2}\sum_{t=1}^{T}\left[\hat{\varepsilon}_t' \Omega^{-1}\hat{\varepsilon}_t\right].$$

Matrix $\widehat{\Omega}$ is a symmetric positive definite. It is easily checked that the (unrestricted) matrix that maximizes the latter expression is symmetric positive definite matrix. Indeed:

$$\frac{\partial \log \mathcal{L}(Y_T; \widehat{\Pi}, \Omega)}{\partial \Omega} = \frac{T}{2}\Omega' - \frac{1}{2}\sum_{t=1}^{T} \widehat{\varepsilon}_t \widehat{\varepsilon}_t' \Rightarrow \widehat{\Omega}' = \frac{1}{T}\sum_{t=1}^{T} \widehat{\varepsilon}_t \widehat{\varepsilon}_t',$$

which leads to the result. □

Proof of Proposition 3.3

Proof. Let us drop the i subscript. Rearranging Eq. (2.23), we have:

$$\sqrt{T}(\mathbf{b} - \beta) = (X'X/T)^{-1}\sqrt{T}(X'\varepsilon/T).$$

Let us consider the autocovariances of $\mathbf{v}_t = x_t\varepsilon_t$, denoted by γ_j^v. Using the fact that x_t is a linear combination of past ε_ts and that ε_t is a white noise, we get that $\mathbb{E}(\varepsilon_t x_t) = 0$. Therefore

$$\gamma_j^v = \mathbb{E}(\varepsilon_t \varepsilon_{t-j} x_t x_{t-j}').$$

If $j > 0$, we have $\mathbb{E}(\varepsilon_t\varepsilon_{t-j}x_t x_{t-j}') = \mathbb{E}(\mathbb{E}[\varepsilon_t\varepsilon_{t-j}x_t x_{t-j}'|\varepsilon_{t-j}, x_t, x_{t-j}]) = \mathbb{E}(\varepsilon_{t-j}x_t x_{t-j}'\mathbb{E}[\varepsilon_t|\varepsilon_{t-j}, x_t, x$ 0. Note that we have $\mathbb{E}[\varepsilon_t|\varepsilon_{t-j}, x_t, x_{t-j}] = 0$ because $\{\varepsilon_t\}$ is an i.i.d. white noise sequence.
If $j = 0$, we have:

$$\gamma_0^v = \mathbb{E}(\varepsilon_t^2 x_t x_t') = \mathbb{E}(\varepsilon_t^2)\mathbb{E}(x_t x_t') = \sigma^2\mathbf{Q}.$$

The convergence in distribution of $\sqrt{T}(X'\varepsilon/T) = \sqrt{T}\frac{1}{T}\sum_{t=1}^{T} v_t$ results from the Central Limit Theorem for covariance-stationary processes, using the γ_j^v computed above. □

8.5 Inference

8.5.1 Monte Carlo method

We use Monte Carlo when we need to approximate the distribution of a variable whose distribution is unknown but which is a function of another variable whose distribution is known. For instance, suppose we know the distribution of a random variable X, which takes values in $\mathbb{R}$, with density function p. Assume we want to compute the mean of $\varphi(X)$. We have:

$$\mathbb{E}(\varphi(X)) = \int_{-\infty}^{+\infty} \varphi(x)p(x)dx$$

Suppose that the above integral does not have a simple expression. We cannot compute $\mathbb{E}(\varphi(X))$ but, by virtue of the law of large numbers, we can approximate it as follows:

$$\mathbb{E}(\varphi(X)) \approx \frac{1}{N}\sum_{i=1}^{N} \varphi(X^{(i)}),$$

where $\{X^{(i)}\}_{i=1,\ldots,N}$ are N independent draws of X. More generally, the distribution of $\varphi(X)$ can be approximated by the empirical distribution of the $\varphi(X^{(i)})$'s. Typically, if 10'000 values of $\varphi(X^{(i)})$ are drawn, the 5th percentile of the p.d.f. of $\varphi(X)$ can be approximated by the 500th value of the 10'000 draws of $\varphi(X^{(i)})$ (after arranging these values in ascending order).

For instance, as regards the computation of confidence intervals around IRFs, one has to think of $\{\widehat{\Phi}_j\}_{j=1,...,p}$, and of $\widehat{\Omega}$ as X and $\{\widehat{\Psi}_j\}_{j=1,...}$ as $\varphi(X)$. (Proposition 3.4 provides us with the asymptotic distribution of the "X.")

To summarize, here are the steps one can implement to derive confidence intervals for the IRFs using the Monte-Carlo approach: For each iteration k,

1. Draw $\{\widehat{\Phi}_j^{(k)}\}_{j=1,...,p}$ and $\widehat{\Omega}^{(k)}$ from their asymptotic distribution (using Proposition 3.4).
2. Compute the matrix $B^{(k)}$ so that $\widehat{\Omega}^{(k)} = B^{(k)}B^{(k)'}$, according to your identification strategy.
3. Compute the associated IRFs $\{\widehat{\Psi}_j\}^{(k)}$.

Perform N replications and report the median impulse response (and its confidence intervals).

8.5.2 Delta method

Suppose β is a vector of parameters and β is an estimator such that

$$\sqrt{T}(\hat{\beta} - \beta) \xrightarrow{d} \mathcal{N}(0, \Sigma_\beta),$$

where d denotes convergence in distribution, $N(0, \Sigma_\beta)$ denotes the multivariate normal distribution with mean vector 0 and covariance matrix Σ_β and T is the sample size used for estimation.

Let $g(\beta) = (g_l(\beta), ..., g_m(\beta))'$ be a continuously differentiable function with values in $\mathbb{R}^m$, and assume that $\partial g_i/\partial\beta' = (\partial g_i/\partial\beta_j)$ is nonzero at β for $i = 1, \dots, m$. Then

$$\sqrt{T}(g(\hat{\beta}) - g(\beta)) \xrightarrow{d} \mathcal{N}\left(0, \frac{\partial g}{\partial \beta'}\Sigma_\beta\frac{\partial g'}{\partial \beta}\right).$$

Using this property, Lütkepohl (1990) provides the asymptotic distributions of the Ψ_j's. The following lines of code can be used to get approximate confidence intervals for IRFs.

A limit of the last two approaches (Monte Carlo and the Delta method) is that they rely on asymptotic results. Boostrapping approaches are more robust in small-sample situations.

8.5.3 Bootstrap

IRFs' confidence intervals are intervals where 90% (or 95%, 75%, …) of the IRFs would lie, if we were to repeat the estimation a large number of times in similar conditions (T observations). We obviously cannot do this, because we have only one sample: $\{y_t\}_{t=1,..,T}$. But we can try to *construct* such samples.

Bootstrapping consists in:

- re-sampling N times, i.e., constructing N samples of T observations, using the estimated VAR coefficients and

a. a sample of residuals from the distribution $N(0, BB')$ (**parametric approach**), or
b. a sample of residuals drawn randomly from the set of the actual estimated residuals $\{\hat{\varepsilon}_t\}_{t=1,..,T}$. (**non-parametric approach**).

- re-estimating the SVAR N times.

Here is the algorithm:

1. Construct a sample

$$y_t^{(k)} = \widehat{\Phi}_1 y_{t-1}^{(k)} + \cdots + \widehat{\Phi}_p y_{t-p}^{(k)} + \widehat{\varepsilon}_t^{(k)},$$

   with $\widehat{\varepsilon}_t^{(k)} = \widehat{\varepsilon}_{s_t^{(k)}}$, where $\{s_1^{(k)}, .., s_T^{(k)}\}$ is a random set from $\{1, .., T\}^T$.
2. Re-estimate the SVAR and compute the IRFs $\{\widehat{\Psi}_j\}^{(k)}$.

Perform N replications and report the median impulse response (and its confidence intervals).

8.6 Statistical Tables

Table 8.1: Quantiles of the $\mathcal{N}(0,1)$ distribution. If a and b are respectively the row and column number; then the corresponding cell gives $\mathbb{P}(0 < X \leq a + b)$, where $X \sim \mathcal{N}(0,1)$.

	0	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0	0.5000	0.6179	0.7257	0.8159	0.8849	0.9332	0.9641	0.9821	0.9918	0.9965
0.1	0.5040	0.6217	0.7291	0.8186	0.8869	0.9345	0.9649	0.9826	0.9920	0.9966
0.2	0.5080	0.6255	0.7324	0.8212	0.8888	0.9357	0.9656	0.9830	0.9922	0.9967
0.3	0.5120	0.6293	0.7357	0.8238	0.8907	0.9370	0.9664	0.9834	0.9925	0.9968
0.4	0.5160	0.6331	0.7389	0.8264	0.8925	0.9382	0.9671	0.9838	0.9927	0.9969
0.5	0.5199	0.6368	0.7422	0.8289	0.8944	0.9394	0.9678	0.9842	0.9929	0.9970
0.6	0.5239	0.6406	0.7454	0.8315	0.8962	0.9406	0.9686	0.9846	0.9931	0.9971
0.7	0.5279	0.6443	0.7486	0.8340	0.8980	0.9418	0.9693	0.9850	0.9932	0.9972
0.8	0.5319	0.6480	0.7517	0.8365	0.8997	0.9429	0.9699	0.9854	0.9934	0.9973
0.9	0.5359	0.6517	0.7549	0.8389	0.9015	0.9441	0.9706	0.9857	0.9936	0.9974
1	0.5398	0.6554	0.7580	0.8413	0.9032	0.9452	0.9713	0.9861	0.9938	0.9974
1.1	0.5438	0.6591	0.7611	0.8438	0.9049	0.9463	0.9719	0.9864	0.9940	0.9975
1.2	0.5478	0.6628	0.7642	0.8461	0.9066	0.9474	0.9726	0.9868	0.9941	0.9976
1.3	0.5517	0.6664	0.7673	0.8485	0.9082	0.9484	0.9732	0.9871	0.9943	0.9977
1.4	0.5557	0.6700	0.7704	0.8508	0.9099	0.9495	0.9738	0.9875	0.9945	0.9977
1.5	0.5596	0.6736	0.7734	0.8531	0.9115	0.9505	0.9744	0.9878	0.9946	0.9978
1.6	0.5636	0.6772	0.7764	0.8554	0.9131	0.9515	0.9750	0.9881	0.9948	0.9979
1.7	0.5675	0.6808	0.7794	0.8577	0.9147	0.9525	0.9756	0.9884	0.9949	0.9979
1.8	0.5714	0.6844	0.7823	0.8599	0.9162	0.9535	0.9761	0.9887	0.9951	0.9980
1.9	0.5753	0.6879	0.7852	0.8621	0.9177	0.9545	0.9767	0.9890	0.9952	0.9981
2	0.5793	0.6915	0.7881	0.8643	0.9192	0.9554	0.9772	0.9893	0.9953	0.9981
2.1	0.5832	0.6950	0.7910	0.8665	0.9207	0.9564	0.9778	0.9896	0.9955	0.9982
2.2	0.5871	0.6985	0.7939	0.8686	0.9222	0.9573	0.9783	0.9898	0.9956	0.9982
2.3	0.5910	0.7019	0.7967	0.8708	0.9236	0.9582	0.9788	0.9901	0.9957	0.9983
2.4	0.5948	0.7054	0.7995	0.8729	0.9251	0.9591	0.9793	0.9904	0.9959	0.9984
2.5	0.5987	0.7088	0.8023	0.8749	0.9265	0.9599	0.9798	0.9906	0.9960	0.9984
2.6	0.6026	0.7123	0.8051	0.8770	0.9279	0.9608	0.9803	0.9909	0.9961	0.9985
2.7	0.6064	0.7157	0.8078	0.8790	0.9292	0.9616	0.9808	0.9911	0.9962	0.9985
2.8	0.6103	0.7190	0.8106	0.8810	0.9306	0.9625	0.9812	0.9913	0.9963	0.9986
2.9	0.6141	0.7224	0.8133	0.8830	0.9319	0.9633	0.9817	0.9916	0.9964	0.9986

Table 8.2: Quantiles of the Student-t distribution. The rows correspond to different degrees of freedom (ν, say); the columns correspond to different probabilities (z, say). The cell gives q that is s.t. $\mathbb{P}(-q < X < q) = z$, with $X \sim t(\nu)$.

	0.05	0.1	0.75	0.9	0.95	0.975	0.99	0.999
1	0.079	0.158	2.414	6.314	12.706	25.452	63.657	636.619
2	0.071	0.142	1.604	2.920	4.303	6.205	9.925	31.599
3	0.068	0.137	1.423	2.353	3.182	4.177	5.841	12.924
4	0.067	0.134	1.344	2.132	2.776	3.495	4.604	8.610
5	0.066	0.132	1.301	2.015	2.571	3.163	4.032	6.869
6	0.065	0.131	1.273	1.943	2.447	2.969	3.707	5.959
7	0.065	0.130	1.254	1.895	2.365	2.841	3.499	5.408
8	0.065	0.130	1.240	1.860	2.306	2.752	3.355	5.041
9	0.064	0.129	1.230	1.833	2.262	2.685	3.250	4.781
10	0.064	0.129	1.221	1.812	2.228	2.634	3.169	4.587
20	0.063	0.127	1.185	1.725	2.086	2.423	2.845	3.850
30	0.063	0.127	1.173	1.697	2.042	2.360	2.750	3.646
40	0.063	0.126	1.167	1.684	2.021	2.329	2.704	3.551
50	0.063	0.126	1.164	1.676	2.009	2.311	2.678	3.496
60	0.063	0.126	1.162	1.671	2.000	2.299	2.660	3.460
70	0.063	0.126	1.160	1.667	1.994	2.291	2.648	3.435
80	0.063	0.126	1.159	1.664	1.990	2.284	2.639	3.416
90	0.063	0.126	1.158	1.662	1.987	2.280	2.632	3.402
100	0.063	0.126	1.157	1.660	1.984	2.276	2.626	3.390
200	0.063	0.126	1.154	1.653	1.972	2.258	2.601	3.340
500	0.063	0.126	1.152	1.648	1.965	2.248	2.586	3.310

Table 8.3: Quantiles of the χ^2 distribution. The rows correspond to different degrees of freedom; the columns correspond to different probabilities.

	0.05	0.1	0.75	0.9	0.95	0.975	0.99	0.999
1	0.004	0.016	1.323	2.706	3.841	5.024	6.635	10.828
2	0.103	0.211	2.773	4.605	5.991	7.378	9.210	13.816
3	0.352	0.584	4.108	6.251	7.815	9.348	11.345	16.266
4	0.711	1.064	5.385	7.779	9.488	11.143	13.277	18.467
5	1.145	1.610	6.626	9.236	11.070	12.833	15.086	20.515
6	1.635	2.204	7.841	10.645	12.592	14.449	16.812	22.458
7	2.167	2.833	9.037	12.017	14.067	16.013	18.475	24.322
8	2.733	3.490	10.219	13.362	15.507	17.535	20.090	26.124
9	3.325	4.168	11.389	14.684	16.919	19.023	21.666	27.877
10	3.940	4.865	12.549	15.987	18.307	20.483	23.209	29.588
20	10.851	12.443	23.828	28.412	31.410	34.170	37.566	45.315
30	18.493	20.599	34.800	40.256	43.773	46.979	50.892	59.703
40	26.509	29.051	45.616	51.805	55.758	59.342	63.691	73.402
50	34.764	37.689	56.334	63.167	67.505	71.420	76.154	86.661
60	43.188	46.459	66.981	74.397	79.082	83.298	88.379	99.607
70	51.739	55.329	77.577	85.527	90.531	95.023	100.425	112.317
80	60.391	64.278	88.130	96.578	101.879	106.629	112.329	124.839
90	69.126	73.291	98.650	107.565	113.145	118.136	124.116	137.208
100	77.929	82.358	109.141	118.498	124.342	129.561	135.807	149.449
200	168.279	174.835	213.102	226.021	233.994	241.058	249.445	267.541
500	449.147	459.926	520.950	540.930	553.127	563.852	576.493	603.446

Table 8.4: Quantiles of the $\mathcal{F}$ distribution. The columns and rows correspond to different degrees of freedom (resp. n_1 and n_2). The different panels correspond to different probabilities (α) The corresponding cell gives z that is s.t. $\mathbb{P}(X \leq z) = \alpha$, with $X \sim \mathcal{F}(n_1, n_2)$.

	1	2	3	4	5	6	7	8	9	10
alpha = 0.9										
5	4.060	3.780	3.619	3.520	3.453	3.405	3.368	3.339	3.316	3.297
10	3.285	2.924	2.728	2.605	2.522	2.461	2.414	2.377	2.347	2.323
15	3.073	2.695	2.490	2.361	2.273	2.208	2.158	2.119	2.086	2.059
20	2.975	2.589	2.380	2.249	2.158	2.091	2.040	1.999	1.965	1.937
50	2.809	2.412	2.197	2.061	1.966	1.895	1.840	1.796	1.760	1.729
100	2.756	2.356	2.139	2.002	1.906	1.834	1.778	1.732	1.695	1.663
500	2.716	2.313	2.095	1.956	1.859	1.786	1.729	1.683	1.644	1.612
alpha = 0.95										
5	6.608	5.786	5.409	5.192	5.050	4.950	4.876	4.818	4.772	4.735
10	4.965	4.103	3.708	3.478	3.326	3.217	3.135	3.072	3.020	2.978
15	4.543	3.682	3.287	3.056	2.901	2.790	2.707	2.641	2.588	2.544
20	4.351	3.493	3.098	2.866	2.711	2.599	2.514	2.447	2.393	2.348
50	4.034	3.183	2.790	2.557	2.400	2.286	2.199	2.130	2.073	2.026
100	3.936	3.087	2.696	2.463	2.305	2.191	2.103	2.032	1.975	1.927
500	3.860	3.014	2.623	2.390	2.232	2.117	2.028	1.957	1.899	1.850
alpha = 0.99										
5	16.258	13.274	12.060	11.392	10.967	10.672	10.456	10.289	10.158	10.051
10	10.044	7.559	6.552	5.994	5.636	5.386	5.200	5.057	4.942	4.849
15	8.683	6.359	5.417	4.893	4.556	4.318	4.142	4.004	3.895	3.805
20	8.096	5.849	4.938	4.431	4.103	3.871	3.699	3.564	3.457	3.368
50	7.171	5.057	4.199	3.720	3.408	3.186	3.020	2.890	2.785	2.698
100	6.895	4.824	3.984	3.513	3.206	2.988	2.823	2.694	2.590	2.503
500	6.686	4.648	3.821	3.357	3.054	2.838	2.675	2.547	2.443	2.356

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